STAT 330 - Mathematical Statistics

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Table of Contents

Contents

1 Probability Model **4**
- 1.1 Random Experiment 4
- 1.2 Conditional Probability 6
- 1.3 Independence of Events 6
- 1.4 Independence vs. Overlap (non-exclusive) 6
- 1.5 Mutual Independence & Pairwise Independence 7

2 Random Variable **7**
- 2.1 Cumulative Distribution Function (CDF) 7

3 Discrete Random Variables **9**
- 3.1 Discrete Random Variables 9
- 3.2 Important Discrete Random Variables 10
  - 3.2.1 Bernoulli 10
  - 3.2.2 Binomial 10
  - 3.2.3 Geometric 11
  - 3.2.4 Negative Binomial 11
  - 3.2.5 Poisson 12

4 Continuous Random Variables **12**
- 4.1 Cumulative Density Function 12
- 4.2 Probability Density Functions 12
- 4.3 Properties of PDFs 13
- 4.4 Gamma Function 14
  - 4.4.1 Properties of the Gamma Function 14
- 4.5 Some Continuous Random Variables 15
  - 4.5.1 Gamma Distribution 15
  - 4.5.2 Weibull Distribution 16
  - 4.5.3 Normal Distribution 17

5 Expectation **18**
- 5.1 Expectation as a Function of a Random Variable 21
- 5.2 Linearity of Expectation 21
- 5.3 Moments 21

6 Moment Generating Functions **23**
- 6.1 Properties of the MGF 24
  - 6.1.1 MGF to Moment 24
  - 6.1.2 MGF vs. Distribution 25

7 Joint Distributions **25**
- 7.1 The Joint CDF 25
- 7.2 Joint Discrete Random Variables 26
- 7.3 Joint Continuous Random Variables 27

8 Joint CDF, PMF/PDFs **28**
- 8.1 Independence 30

9 Joint Expectation, Variance and Conditionals **35**
- 9.1 Joint Expectation 35
- 9.2 Properties of Joint Expectation 35

9.3 Correlation Coefficient . . . 37
9.4 Conditional Distributions . . . 39
10 Conditional Distributions and their Expectation . . . 40
10.1 Conditional Expectation . . . 42
10.2 Double Expectation Theorem . . . 43
11 (Cont.) Conditional Expectation . . . 45
11.1 Joint Moment Generating Function . . . 47
12 Multinomial Distribution . . . 50
12.1 Bivariate Normal Distribution . . . 52
12.2 Function of Random Variables . . . 56
12.2.1 CDF Method . . . 56
12.3 CDF Method . . . 58
12.4 One-to-one Transformation Method . . . 59
12.5 One-to-One Bivariate Transformation Method . . . 60
12.6 Non-one-to-one Transformations . . . 62
12.7 MGF Technique . . . 63
12.8 Normal Distribution . . . 63
12.9 Some Important Distributions . . . 64
12.10 t distribution . . . 65
12.10.1 How to find the pdf ot T? . . . 65
12.11 F Distribution . . . 65
12.12 Distributional Results from Normal Samples . . . 66
12.13 Sample Mean and Sample Variance . . . 66
12.14 Important Distributional Results for Asymptotics . . . 67
13 Introduction to Asymptotics . . . 67
13.1 Convergence in Distribution . . . 68
13.2 Convergence in Probability . . . 72
13.3 Convergence to a Constant . . . 72
13.4 Markov Inequality . . . 74
13.5 Chernoff Bounds . . . 75
13.6 Markov Inequality - WLLN . . . 75
13.7 Central Limit Theorem . . . 77
13.8 Continuous Mapping Theorem . . . 79
13.9 Slutsky's Theorem . . . 80
14 (Cont') Delta Method . . . 82
14.1 Big $O_p(\cdot)$ Small $o_p(\cdot)$ Notations: . . . 82
14.2 Combining Big/small O: . . . 83
14.3 Alternative Proof . . . 83
14.4 Examples . . . 84
15 Point Estimation . . . 84
15.1 Estimator & Estimate . . . 85
15.2 How to evaluate an estimator? . . . 85
15.3 Method of Moments . . . 85
15.4 Examples . . . 86
15.5 Likelihood Function . . . 86
15.6 Comments: . . . 86
15.7 Example . . . 87
15.8 Maximum Likelihood Method . . . 87
15.9 Examples . . . 88
15.10 Inverse Gaussian . . . 89
15.11 Properties of ML Estimator . . . 90
15.12 Preparations . . . 90

16 (Cont') Fisher Information . . . 91
16.1 Examples: . . . 91
16.2 The Cramer-Rao Lower Bound . . . 92
16.3 Comments: . . . 92
16.4 Examples: . . . 93
16.5 ML Estimator Property . . . 94
16.6 Example . . . 94
17 Frequentist vs. Bayesian Statistics . . . 96
17.1 Two notations: $f(x;\theta)$ vs. $f(x \mid \theta)$ . . . 96
18 Bayesian inference . . . 96
18.1 Bayes Estimator . . . 96
18.2 Example 1 . . . 97
18.3 Example 3: Improper Prior . . . 98
18.4 How to choose prior distributions . . . 99
18.5 Example 1 . . . 99
18.6 Example 2 . . . 99
18.7 Invariance . . . 99
18.8 Meaning of Invariance . . . 100
18.9 Proof . . . 100
18.10 Example . . . 100

1 Probability Model

Definition 1.1. A probability model is a mathematical representation of a random experiment.

1.1 Random Experiment

Definition 1.2. Any procedure that can be repeated and has a well-defined set of outcomes.

Example 1.1.

E.g. toss a coin/die, discrete. the lifespan of phone battery, continuous.

How to define a probability model?

1. **Sample Space:** All possible outcomes of a random experiment.

   E.g. toss a coin: $S_1 = \{H, T\}$

   toss two coins: $S_2 = \{(H,H),(H,T),(T,H),(T,T)\}$

   lifespan: $S_3 = \mathbb{R}^+$

2. **Event:** A subset of the sample space.

   E.g. toss a coin: $A_1 = \{H\} \subset S_1,\ A_2 = \{H,T\} \subset S_1,\ A_3 = H$ **NOT an event**

   toss two coins: $B_1 = \{(H,H),(H,T)\} \subset S_2$

   $B_2 = \{$"First and Second toss have different outcomes"$\} = \{(H,T),(T,H)\} \subset S_2$

   Theorem 1.1 (Properties of a Probability Function)**.**

   Probability Function: $\Pr(\cdot)$ *is a function of event.*

   (a) $\Pr(A) \in [0,1],\ \ \forall A \subset S.$

   (b) $\Pr(S) = 1.$

   (c) If $A_1, A_2, A_3, \ldots$ *is a sequence of subsets of* S *and they are pairwise mutually exclusive*

   $$\forall i \neq j,\ A_i \cap A_j = \varnothing,$$

   $$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i).$$

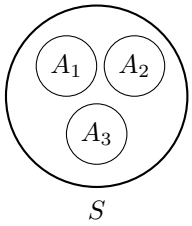


Figure 1: Sample space S with mutually exclusive events A_1, A_2, A_3.

Example 1.2 (Example of Defining $\Pr(A)$).

$$S = \{(H,H),(H,T),(T,H),(T,T)\}$$

$$\Pr(A) = \frac{\#\{\text{outcomes in } A\}}{\#\{\text{outcomes in } S\}}$$

$$\#\{\text{outcomes in } A\} = 0,1,2,3,4, \qquad \#\{\text{outcomes in } S\} = 4$$
$$\Rightarrow \Pr(A) \in [0,1]$$

(a) $\Pr(A) = \dfrac{\#\{\text{outcomes in } A\}}{4}$

(b) $\Pr(S) = \dfrac{4}{4} = 1$

(c) (skip ✓)

Theorem 1.2 (Properties of Probability Functions)**.**

1. $\Pr(\varnothing) = 0$. ***Proof:*** *By (c),* $\Pr(S \cup \varnothing) = \Pr(S) + \Pr(\varnothing)$. *By (b),* $\Pr(S \cup \varnothing) = \Pr(S) = 1$. $\Rightarrow \Pr(\varnothing) = 0$.
2. *Let A^c be the* ***complement*** *of A.*

   $$A \cup A^c = S, \quad A \cap A^c = \varnothing$$

   Then

   $$\Pr(A) + \Pr(A^c) = 1 \quad \textit{(Proof by (c))}$$

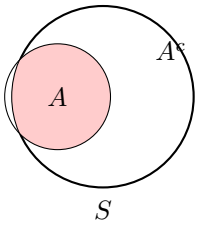


Figure 2: Complement A^c of A.

3. *If A_1 and A_2 are two events,*

   $$\Pr(A_1 \cup A_2) = \Pr(A_1) + \Pr(A_2) - \Pr(A_1 \cap A_2)$$

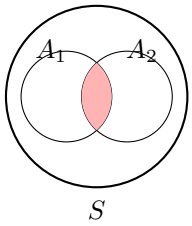


Figure 3: Intersection $A_1 \cap A_2$ highlighted.

4. *If A_1 and A_2 are exclusive,*

$$\Pr(A_1 \cup A_2) = \Pr(A_1) + \Pr(A_2)$$

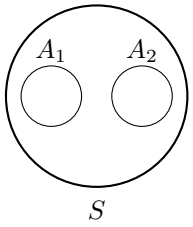


Figure 4: Exclusive events A_1, A_2.

5. *If $A_1 \subset A_2$, then $\Pr(A_1) \leq \Pr(A_2)$.*

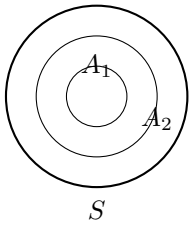


Figure 5: Subset relation $A_1 \subset A_2$.

Probability Model = sample space + **a set of events** + probability function

Definition 1.3.

$$\textit{sigma algebra} = \text{a set of subsets of the sample space}$$

Generally **different** from the full power set of S.

1.2 Conditional Probability

Theorem 1.3. *For two events A and B that satisfy $\Pr(B) > 0$ and $\Pr(A) \geq 0$, the conditional probability of A given B is defined as*

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}.$$

1.3 Independence of Events

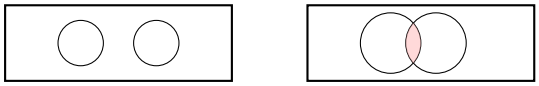
Theorem 1.4. *We say events A and B are independent if*

$$\Pr(A \cap B) = \Pr(A)\Pr(B).$$

Equivalently,

$$\Pr(A \mid B) = \Pr(A), \quad \Pr(B \mid A) = \Pr(B), \quad \textit{given } \Pr(A), \Pr(B) > 0.$$

1.4 Independence vs. Overlap (non-exclusive)



exclusive non-exclusive

Figure 6: Exclusive vs. non-exclusive events.

Example 1.3. Take the sample space $S = \{(H,H),(H,T),(T,H),(T,T)\}$

$$A = \{\text{the first toss is head}\} = \{(H,H),(H,T)\}$$

$$B = \{\text{the second toss is head}\} = \{(H,H),(T,H)\}$$

$$C = \{\text{the first toss is tail}\} = \{(T,H),(T,T)\}$$

Then, we can conclude that,

$$\Pr(A \cap B) = \tfrac{1}{4}, \quad \Pr(A)\Pr(B) = \tfrac{1}{2}\cdot\tfrac{1}{2} = \tfrac{1}{4} \quad \Rightarrow A, B \text{ non-exclusive, independent.}$$

$$\Pr(A \cap C) = 0, \quad \Pr(A)\Pr(C) = \tfrac{1}{2}\cdot\tfrac{1}{2} = \tfrac{1}{4} \quad \Rightarrow A, C \text{ exclusive, dependent.}$$

1.5 Mutual Independence & Pairwise Independence

Events A, B, C are **mutually independent** if

$$\Pr(A \cap B \cap C) = \Pr(A)\Pr(B)\Pr(C).$$

They are **pairwise independent** if

$$\Pr(A \cap B) = \Pr(A)\Pr(B), \quad \Pr(A \cap C) = \Pr(A)\Pr(C), \quad \Pr(B \cap C) = \Pr(B)\Pr(C).$$

- Mutual independence $\Rightarrow$ pairwise independence.
- Pairwise independence $\not\Rightarrow$ mutual independence.

2 Random Variable

We usually use capital letters to denote random variables. For example, X and Y. (At least in this course.)

Definition 2.1.

1. A random variable is a function:
$$X : S \to \mathbb{R}$$
If X is a r.v., it takes in outcomes of a random experiment, not the event.
2. $\{X \le x\}$ is an event.
$$\{X \le 2\} = S, \quad \Pr(X \le 2) = 1$$
$$\{X \le 1\} = \{(H,T),(T,H),(T,T)\}, \quad \Pr(X \le 1) = \tfrac{3}{4}$$
$$\{X \le 0\} = \{(T,T)\}, \quad \Pr(X \le 0) = \tfrac{1}{4}$$

Example 2.1. Example: $S = \{(H,H),(H,T),(T,H),(T,T)\}$

$$X((H,H)) = 2, \quad X((H,T)) = X((T,H)) = 1, \quad X((T,T)) = 0$$

So $X =$ "number of heads".

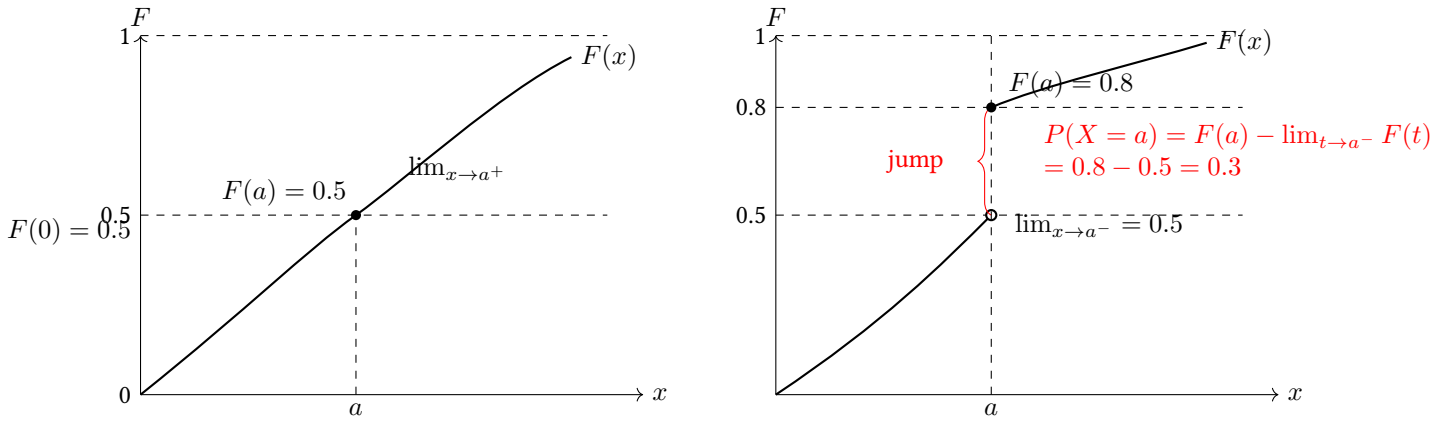
2.1 Cumulative Distribution Function (CDF)

The default notation for a cdf is $F(\cdot)$.

$$F(x) = \Pr(X \le x), \quad x \in \mathbb{R}.$$

Theorem 2.1 (Properties of the Cumulative Distribution Function)**.**

1. $F(x)$ is a non-decreasing function. If $x_1 \le x_2$, then $F(x_1) \le F(x_2)$ since $\{X \le x_1\} \subset \{X \le x_2\}$.
2. $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to+\infty} F(x) = 1$.
3. $F(x)$ is a right-continuous function of x.



Theorem 2.2 (More Properties of CDFs)**.**

4. For $a < b$,
$$\Pr(a < X \le b) = \Pr(X \le b) - \Pr(X \le a) = F(b) - F(a).$$
Using disjoint unions of sets, we can write,
$$\{X \le b\} = \{a < X \le b\} \cup \{X \le a\}, \quad \{a < X \le b\} \cap \{X \le a\} = \emptyset$$
$$\implies P(X \le b) = P(a < X \le b) + P(X \le a)$$

5. $\Pr(X = x) = \Pr(\text{Jump at } x) = F(x) - \lim_{t\uparrow x} F(t)$.
$$\begin{aligned} P(X = x) &= P(X \le x) - P(X < x) \\ &= F(x) - \lim_{t\to x^-} F(t) \end{aligned}$$

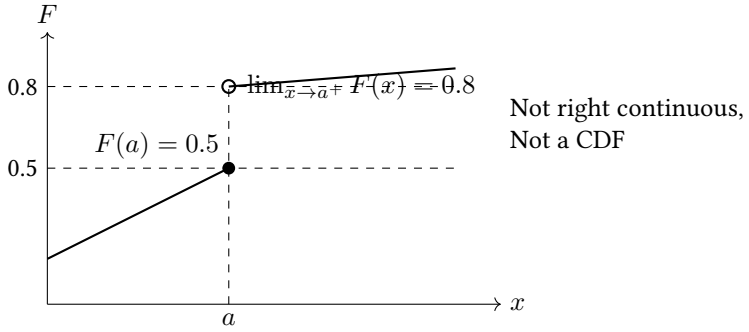


Figure 7: An invalid function: not right continuous, hence not a CDF.

Example 2.2. For X = number of heads in two coin tosses:
$$\Pr(X = 0) = \tfrac{1}{4}, \quad \Pr(X = 1) = \tfrac{1}{2}, \quad \Pr(X = 2) = \tfrac{1}{4}.$$
$$F(x) = \begin{cases} 0 & x < 0, \\ \frac{1}{4} & 0 \le x < 1, \\ \frac{3}{4} & 1 \le x < 2, \\ 1 & x \ge 2. \end{cases}$$
$$\Pr(X = 1) = F(1) - \lim_{t\uparrow 1} F(t) = \tfrac{3}{4} - \tfrac{1}{4} = \tfrac{1}{2}.$$

3 Discrete Random Variables

3.1 Discrete Random Variables

Definition 3.1. If a random variable X can only take a finite or countable number of values, then we say that X is a discrete random variable.

Example 3.1.

$$X \in \{0, 1\} \quad \text{(finite)}$$
$$X \in \{0, 1, 2, \dots\} \quad \text{(countably infinite)}$$
$$X \in \{\text{Any values on all rational numbers}\} \quad \text{(countably infinite)}$$

Remark 3.1. **CDF**: $F(x) = \mathbb{P}(X = x)$ is a right-continuous step function,

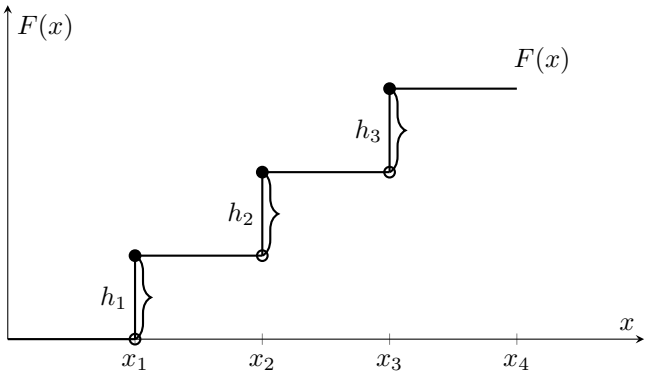


Figure 8: CDF $F(x)$ of a discrete random variable.

Theorem 3.1 (Probability Mass Function)**.**

1. $f(x) = P(X = x) = F(x) - \lim_{t \to x^-} F(t) = \lim_{t \to x^+} F(t) - \lim_{t \to x^-} F(t)$
2. $f(x_1) = h_1, f(x_2) = h_2, f(x_3) = h_3$ and $f(x_4) = 0$ because the slope is 0.
3. **Support** of a discrete random variable X of its pmf is,
$$A : \{x : f(x) > 0\}$$
so for our example in the figure, $x_1, x_2, x_3 \in A$ but $x_4 \notin A$.
4. **Properties of a pmf**
   (a) $f(x) \geq 0$
   (b) $\sum_{x \in A} f(x) = 1$

Remark 3.2. In the **Property (b)** observe that this is just the summation over all possible vaclues of x, where x are values in the support of the probability mass function.

3.2 Important Discrete Random Variables

3.2.1 Bernoulli

1. **Notation**: $X \sim \text{BER}(p)$ where p is an unknown parameter defined $p \in (0,1)$
2. **Motivation**: Toss a coin and the probability of heads is p
3. X is the number of heads you get
4. **Support**: $A = \{0,1\}$
5. **Probability Mass Function**

$$f(x) = \begin{cases} P(X=1) & = p \quad \text{if } x = 1 \\ P(X=0) & = 1-p \quad \text{if } x = 0 \end{cases}$$

The pmf above satisfies $f(x) \geq 0$ and $f(0) + f(1) = 1 - p + p = 1$.

3.2.2 Binomial

1. **Notation**: $X \sim \text{BIN}(n,p)$ where n is known and p is unknown.
2. **Motivation**: Tossing n coins (iid) where p is the probability of getting heads
3. **Support**: $A = \{0,1,\ldots,n\}$ and X is the number of heads you get
4. **Probability Mass Function**

$$f(x) = P(X=x) = \underbrace{\binom{n}{x}}_{\text{\# of outcomes}} \cdot \underbrace{p^x(1-p)^{n-x}}_{\text{proba. of each \{outcome\}}}$$

It's easy to show that $f(x) \geq 0$ and that $\sum_{x=0}^{n} f(x) = 1$,

Proof.

$$f(x) = P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0,1,\ldots,n.$$

$$\geq 0 \quad \text{since } \binom{n}{x} \geq 0,\ p^x \geq 0,\ (1-p)^{n-x} \geq 0.$$

$$\sum_{x=0}^{n} f(x) = \sum_{x=0}^{n} \binom{n}{x} p^x (1-p)^{n-x}$$

$$= (p + (1-p))^n \quad \text{(by the Binomial Theorem)}$$

$$= 1^n = 1.$$

Note:
1. If $n = 1$ then Bin r.v. = Ber r.v
2. If $X \sim \text{BIN}(n,p)$ then $X = \sum_{i=1}^{n} X_i, \quad X_i \sim^{\text{iid}} \text{BER}(p)$

3.2.3 Geometric

1. **Notation**: $X \sim \text{GEO}(p)$ where p is unknown
2. **Motivation**: X is the number of tails you get before you get your first head and p is the probability of getting heads
3. **Support**: $A = \{0, 1, 2, \dots\}$
4. **Probability Mass Function**
$$f(x) = P(X = x) = p(1-p)^x$$

Remark 3.3 (Visual Interpretation of Coin Toss for Geometric).

$$\begin{aligned} x &= 0 \quad H \\ x &= 1 \quad TH \\ x &= 2 \quad TTH \\ &\vdots \quad \underbrace{T \cdots T}_{x} \underbrace{H}_{p} \end{aligned}$$

The intuition here is that, we get x T's before we encounter our first H, where the sequence of T's occurring before the first H each occur with probability $1 - p$

It is easy to show that $f(x) \geq 0$ and $\sum_{x=0}^{\infty} f(x) = 1$ for the probability mass function of the geometric distribution.

Proof.

$$\begin{aligned} \sum_{k=0}^{\infty} p(1-p)^k &= p \cdot \sum_{k=0}^{\infty} (1-p)^k \\ &= p \cdot \frac{1}{1-(1-p)} \\ &= p \cdot \frac{1}{1-1+p} = p \cdot \frac{1}{p} = 1 \end{aligned}$$

3.2.4 Negative Binomial

1. **Notation**: $X \sim$ NEG BIN(r, p) where r is known and p is unknown
2. **Motivation**: X is the number of tails you get when you first reach r heads
3. **Support**: $A = \{0, 1, 2, \dots\}$
4. **Probability Mass Function**
$$f(x) = P(X = x) \binom{x + r - 1}{x} (1-p)^x p^2$$

Remark 3.4. **Geometric and Negative Binomial**

1. $\text{GEO}(p) = \text{NEG BIN}(1, p)$ i.e., the case for the negative binomial where $r = 1$ (1 success)
2. If $X \sim \text{NEG BIN}(r, p)$ then $X = \sum_{i=1}^{r} X_i$ with $X_i \sim^{\text{iid}} \text{GEO}(p)$

3.2.5 Poisson

1. **Notation**: $X \sim \text{POI}(\mu)$ is an unknown parameter
2. **Motivation**: Poisson random variables are the limit of BIN random variables as $n \to \infty$

$$\text{BIN}(n, p) : np = \mu \quad \text{or} \quad p = \frac{\mu}{n}$$

Proof. Given,

$$f_{\text{BIN}}(x) = \binom{n}{x} p^x (1-p)^{n-x}$$
$$f(x) = e^{-\mu} \frac{\mu^x}{x!}$$

we aim to show that $f_{\text{BIN}}(x)$ approximates $f(x)$ as $n \to \infty$.

$$\begin{aligned} f_{\text{BIN}}(x) &= \binom{n}{x} p^x (1-p)^{n-x} \\ &= \frac{n!}{x!(n-x)!} \left(\frac{\mu}{n}\right)^x \left(1 - \frac{\mu}{n}\right)^{n-x} \\ &= \frac{\mu^x}{x!} \cdot \frac{n!}{(n-x)! n^x} \left(1 - \frac{\mu}{n}\right)^{-x} \left(1 - \frac{\mu}{n}\right)^n \end{aligned}$$

As $n \to \infty$,

$$\left(1 - \frac{\mu}{n}\right)^x \to 1$$
$$\left(1 - \frac{\mu}{n}\right)^n \to e^{-\mu} \quad \text{by } \lim_{n\to\infty} \left(1 + \frac{\delta}{n}\right)^n = e^{\delta}$$
$$\frac{n!}{(n-x)! n^x} = \frac{n}{n} \cdot \frac{n-1}{n} \cdots \frac{n-x+1}{n} \to 1$$

So we can say that $f_{\text{BIN}}(x) \to e^{-\mu} \cdot \frac{\mu^x}{x!}$ which is just $f(x)$ which is the probability mass function of the Poisson distribution with support $A = \{0, 1, 2, \dots\}$ also satisfying the properties $f(x) \geq 0$ and $\sum_{x \in A} f(x) = 1$.

4 Continuous Random Variables

Definition 4.1. X is a random variable. If the possible values of X is an interval of $\mathbb{R}$ then X is said to be a continuous random variable.

4.1 Cumulative Density Function

Definition 4.2. Cumulative Density Function $F(x) = \mathbb{P}(X = x)$

1. $F(x)$ is a continuous function
2. $F(x)$ is almost surely (a.s.) differentiable, that is, it is NOT differentiable for at most a countable number of points.

Definition 4.3. Probability Density Function denoted $f(x)$

$$f(x) = \begin{cases} F(x) & \text{if } F \text{ is differetiable at } x \\ 0 & \text{otherwise} \end{cases}$$

with support $A = \{x : f(x) > 0\}$.

4.2 Probability Density Functions

4.3 Properties of PDFs

Theorem 4.1. ***Properties of Probability Density Functions***

1. $f(x) \geq 0$
2. $$\int_{-\infty}^{+\infty} f(x)dx = 1 \text{ since } \int_{-\infty}^{+\infty} f(x)dx = F(x)|_{-\infty}^{+\infty} = F(\infty) - F(-\infty) = 1$$
3. $$F(x) = \int_{-\infty}^{x} f(t)dt$$
4. $f(x) = F'(x)$ *when differetiable*
5. $P(X = x) = 0$
6. $$P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b) = \int_a^b f(x)dx$$

Example 4.1. Suppose that we have the Cumulative Distribution Function,

$$F(x) = \begin{cases} 0 & x \leq a, \\ \dfrac{x-a}{b-a} & a < x < b, \\ 1 & x \geq b. \end{cases}$$

Find the pdf.

Solution. The probability density function is the derivative of the CDF.

$$f(x) = \begin{cases} 0 & x \leq a, \\ \dfrac{1}{b-a} & a < x < b, \\ 0 & x \geq b. \end{cases}$$

Thus, $f(x)$ is the uniform density on the interval (a, b).

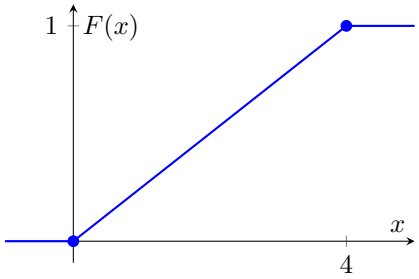


Figure 9: Cumulative distribution function $F(x)$ with $a = 0$, $b = 4$.

■

Example 4.2. Suppose that the pdf is

$$f(x) = \begin{cases} \dfrac{\theta}{x^{\theta+1}} & x \geq 1, \\ 0 & x < 1, \quad \theta > 0 \end{cases}$$

.

(i) Find the cumulative distribution function (CDF) $F(x)$.

Solution. For $x < 1$, clearly $F(x) = 0$. For $x \geq 1$:

$$\begin{aligned} F(x) &= \int_1^x \frac{\theta}{t^{\theta+1}}\, dt \\ &= \left[-\frac{1}{t^\theta}\right]_1^x \\ &= -\frac{1}{x^\theta} + \frac{1}{1^\theta} \\ &= 1 - \frac{1}{x^\theta}. \end{aligned}$$

$$F(x) = \begin{cases} 0 & x < 1, \\ 1 - \dfrac{1}{x^\theta} & x \geq 1. \end{cases}$$

■

(ii) Verify that $F(x) \to 1$ as $x \to \infty$.

Solution. Since $\lim_{x\to\infty} \frac{1}{x^\theta} = 0$ (for $\theta > 0$),

$$\lim_{x\to\infty} F(x) = 1 - 0 = 1.$$

Therefore, $F(x)$ satisfies the properties of a CDF. ■

4.4 Gamma Function

1. **Notation**: $\Gamma(\alpha)$ for $\alpha > 0$
2. **Definition**:

$$\Gamma(\alpha) = \int_0^\infty y^{\alpha-1} e^{-y} dy$$

4.4.1 Properties of the Gamma Function

(i) $\Gamma(1) = 1$

Proof.

$$\Gamma(1) = \int_0^\infty y^{1-1} e^{-y} dy = \int_0^\infty e^{-y} dy = -e^{-y}]_0^\infty = (-0) - (-1) = 1$$

(ii) $\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$

Proof.

$$\begin{aligned} \Gamma(\alpha - 1) &= \int_0^\infty y^{\alpha-2} e^{-y} dy \\ &= \frac{1}{\alpha - 1} \int_0^\infty e^{-y} dy^{\alpha-1} \\ &= \frac{1}{\alpha - 1} \left\{ e^{-y} y^{\alpha-1} |_0^\infty + \int_0^\infty y^{\alpha-1} \underbrace{de^{-y}}_{=e^{-y}dy} \right\} \end{aligned}$$

$$= \frac{1}{\alpha - 1} \int_0^\infty y^{\alpha-1} e^{-y} dy$$
$$= \frac{1}{\alpha - 1} \Gamma(\alpha)$$

(iii) For $n \in \mathbb{Z}^+$ we have that $\Gamma(n) = (n-1)!$

Proof.

$$\begin{aligned}\Gamma(n) &= (n-1)\Gamma(n-1) \\ &= (n-1)(n-2)\Gamma(n-2) \\ &= (n-1) \cdot 2 \cdot 1 \cdot \Gamma(1) = (n-1)!\end{aligned}$$

4.5 Some Continuous Random Variables

4.5.1 Gamma Distribution

1. **Notation**: $X \sim \Gamma(\alpha, \beta)$ and $\alpha > 0, \beta > 0$
2. **Support**: $[0, +\infty)$
3. **Probability Density Function**:

$$f(x) = \frac{x^{\alpha-1} \exp(-x/\beta)}{\Gamma(\alpha)\beta^\alpha}, \quad x > 0$$

- $f(x) \geq 0$
- Verify that $\int_{-\infty}^{\infty} f(x)dx = 1$

Proof.

$$\begin{aligned}\int_{-\infty}^{+\infty} f(x)\,dx &= \int_{-\infty}^{+\infty} \frac{x^{\alpha-1}e^{-x/\beta}}{\Gamma(\alpha)\,\beta^\alpha}\,dx \\ &= \int_0^\infty \frac{x^{\alpha-1}e^{-x/\beta}}{\Gamma(\alpha)\,\beta^\alpha}\,dx \quad \text{since } f(x) = 0 \text{ for } x \leq 0 \\ &= \frac{1}{\Gamma(\alpha)\,\beta^\alpha} \int_0^\infty x^{\alpha-1}e^{-x/\beta}\,dx \\ \text{Let } y = \tfrac{x}{\beta} \quad &\Rightarrow \quad x = \beta y, \;\; dx = \beta dy \\ &\Rightarrow \frac{1}{\Gamma(\alpha)\,\beta^\alpha} \int_0^\infty (\beta y)^{\alpha-1}e^{-y}\,(\beta dy) \\ &= \frac{1}{\Gamma(\alpha)\,\beta^\alpha} \int_0^\infty \beta^{\alpha-1}y^{\alpha-1}e^{-y} \cdot \beta\,dy \\ &= \frac{1}{\Gamma(\alpha)\,\beta^\alpha} \cdot \beta^\alpha \int_0^\infty y^{\alpha-1}e^{-y}\,dy \\ &= \frac{1}{\Gamma(\alpha)} \int_0^\infty y^{\alpha-1}e^{-y}\,dy \\ &= \frac{1}{\Gamma(\alpha)} \cdot \Gamma(\alpha) \\ &= 1.\end{aligned}$$

4.5.2 Weibull Distribution

1. **Notation**: $X \sim \text{Weibull}(\theta, \beta)$ for $\theta > 0, \beta > 0$
2. **Support**: $\mathbb{R}$
3. **Probability Density Function**

$$f(x) = \frac{\beta}{\theta^\beta} x^{\beta-1} \exp\left\{-\left(\frac{x}{\theta}\right)^\beta\right\}, \quad x > 0$$

- $f(x) \geq 0$
- Verify that $\int_{-\infty}^{\infty} f(x)dx = 1$

Proof.

$$\begin{aligned}
\int_{-\infty}^{+\infty} f(x)\,dx &= \int_{-\infty}^{+\infty} \frac{\beta}{\theta^\beta} x^{\beta-1} \exp\Big\{-\left(\tfrac{x}{\theta}\right)^\beta\Big\}\,dx \\
&= \int_0^{\infty} \frac{\beta}{\theta^\beta} x^{\beta-1} \exp\Big\{-\left(\tfrac{x}{\theta}\right)^\beta\Big\}\,dx \quad \text{since } f(x) = 0 \text{ for } x \leq 0
\end{aligned}$$

$$\text{Let } y = \left(\tfrac{x}{\theta}\right)^\beta \;\Rightarrow\; x = \theta y^{1/\beta},\;\; dx = \frac{\theta}{\beta} y^{1/\beta-1} dy$$

$$\begin{aligned}
&\Rightarrow \int_0^{\infty} \frac{\beta}{\theta^\beta} \left(\theta y^{1/\beta}\right)^{\beta-1} e^{-y} \cdot \frac{\theta}{\beta} y^{1/\beta-1}\,dy \\
&= \int_0^{\infty} \frac{\beta}{\theta^\beta} \cdot \theta^{\beta-1} y^{(\beta-1)/\beta} e^{-y} \cdot \frac{\theta}{\beta} y^{1/\beta-1}\,dy \\
&= \int_0^{\infty} \frac{\beta}{\theta^\beta} \cdot \theta^{\beta} \cdot \frac{1}{\beta} \cdot y^{(\beta-1)/\beta + (1/\beta-1)} e^{-y}\,dy \\
&= \int_0^{\infty} y^0 e^{-y}\,dy \\
&= \int_0^{\infty} e^{-y}\,dy \\
&= \left[-e^{-y}\right]_0^{\infty} \\
&= (0 - (-1)) \\
&= 1.
\end{aligned}$$

4.5.3 Normal Distribution

1. **Notation**: $X \sim N(\mu, \sigma^2)$ and $\mu \in \mathbb{R}$ with $\sigma^2 > 0$
2. **Support**: $\mathbb{R}$
3. **Probability Density Function**

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}, \quad -\infty < x < \infty$$

- $f(x) \geq 0$
- Verify that $\int_{-\infty}^{+\infty} f(x)dx = 1$

Proof.

$$\int_{-\infty}^{+\infty} f(x)\,dx = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} dx$$

$$\text{Let } y = \frac{x-\mu}{\sigma} \quad \Rightarrow \quad x = \mu + \sigma y, \;\; dx = \sigma dy \Rightarrow \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(\sigma y)^2}{2\sigma^2}\right\} \cdot \sigma dy$$

$$= \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{y^2}{2}\right\} \cdot \sigma dy$$

$$= \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{y^2}{2}\right\} dy$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-y^2/2} dy$$

$$\text{Recall: } \int_{-\infty}^{+\infty} e^{-y^2/2} dy = \sqrt{2\pi} \Rightarrow \frac{1}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = 1.$$

General Case:

$$\int_{-\infty}^{+\infty} f(x)\,dx = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} dx$$

$$\text{Let } y = \frac{x-\mu}{\sigma} \quad \Rightarrow \quad x = \mu + \sigma y, \;\; dx = \sigma dy$$

$$\Rightarrow \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(\mu+\sigma y-\mu)^2}{2\sigma^2}\right\} \cdot \sigma dy$$

$$= \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(\sigma y)^2}{2\sigma^2}\right\} \cdot \sigma dy$$

$$= \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{y^2}{2}\right\} \cdot \sigma dy = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{y^2}{2}\right\} dy$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-y^2/2} dy = \frac{1}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = 1.$$

Remark 4.1. The **Normal Distribution** belongs to the location-scale family, where μ is the location parameter and σ is the scale parameter. Formally:

- If $X \sim N(\mu, \sigma^2)$, then the standardized variable
$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$
- Conversely, if $Z \sim N(0, 1)$, then the affine transformation
$$X = \mu + \sigma Z \sim N(\mu, \sigma^2).$$

5 Expectation

Definition 5.1. We consider the discrete and continuous cases for this.

(i) Take a **Discrete** random variable X with support A and pmf $f(x)$. Then, the expectation of X is defined as,
$$\mathbb{E}[X] = \sum_{x \in A} x \cdot f(x) \quad \text{if} \quad \sum_{x \in A} |x| f(x) < \infty$$

(ii) Take a **Continuous** random variable X with pdf $f(x)$
$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x \cdot f(x) \quad \text{if} \quad \int_{-\infty}^{+\infty} |x| \cdot f(x) dx < \infty$$

If,
$$\sum_{x \in A} |x| f(x) = \infty \quad \text{or} \quad \int_{-\infty}^{+\infty} |x| f(x) dx = \infty$$
then, we say that the **expectation** does not exist (DNE).

Example 5.1. Let $X \sim \text{Ber}(p)$ then $\mathbb{E}[X] = p \cdot 1 + (1 - p) \cdot 0 = p$.

Example 5.2. Let $X \sim \text{BIN}(n, p)$. Then, by definition,
$$\begin{aligned}
\mathbb{E}[X] &= \sum_{i=0}^{n} i \cdot \binom{n}{i} p^i (1-p)^{n-i} \\
&= \sum_{i=1}^{n} i \cdot \binom{n}{i} p^i (1-p)^{n-i} \quad \text{(since the } i = 0 \text{ term vanishes)} \\
&= \sum_{i=1}^{n} n \cdot \binom{n-1}{i-1} p^i (1-p)^{n-i} \quad \text{(using } i\binom{n}{i} = n\binom{n-1}{i-1}\text{)} \\
&= np \sum_{i=1}^{n} \binom{n-1}{i-1} p^{i-1} (1-p)^{(n-1)-(i-1)} \\
&= np \sum_{j=0}^{n-1} \binom{n-1}{j} p^j (1-p)^{(n-1)-j} \quad \text{(substitute } j = i-1\text{)} \\
&= np \cdot (p + (1-p))^{n-1} \quad \text{(by the Binomial Theorem)} \\
&= np \cdot 1^{n-1} = np.
\end{aligned}$$

Easier Method:

$$X \sim \text{BIN}(n,p) \quad \text{Note that } X = \sum_{i=1}^{n} X_i, \quad X_i \overset{\text{iid}}{\sim} \text{Bernoulli}(p).$$

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathbb{E}[X_i] \quad \text{(by linearity of expectation)}$$

$$= \sum_{i=1}^{n} p = np.$$

Example 5.3. X is a random variable with pdf $f(x) = \frac{\theta}{x^{\theta+1}}$ for $x \geq 1, \theta > 0$.

Proof. We check that $\int_{-\infty}^{\infty} f(x)\,dx = 1$.

$$\begin{aligned}
\int_{-\infty}^{\infty} f(x)\,dx &= \int_{-\infty}^{\infty} \frac{\theta}{x^{\theta+1}}\,dx \\
&= \int_{1}^{\infty} \frac{\theta}{x^{\theta+1}}\,dx \quad \text{(since } f(x) = 0 \text{ for } x < 1) \\
&= \theta \int_{1}^{\infty} x^{-(\theta+1)}\,dx \\
&= \theta \left[\frac{x^{-\theta}}{-\theta}\right]_{1}^{\infty} = -\left[x^{-\theta}\right]_{1}^{\infty} \\
&= -\left(\lim_{x\to\infty} x^{-\theta} - 1^{-\theta}\right) = -(0-1) = 1.
\end{aligned}$$

Observe that $\mathbb{E}(X)$ only exists when $\theta > 1$ so $\mathbb{E}(X) = \frac{\theta}{\theta-1}$. We compute

$$\begin{aligned}
\mathbb{E}[X] &= \int_{-\infty}^{\infty} x f(x)\,dx \\
&= \int_{1}^{\infty} x \cdot \frac{\theta}{x^{\theta+1}}\,dx \quad \text{(since } f(x) = 0 \text{ for } x < 1) \\
&= \theta \int_{1}^{\infty} x^{-\theta}\,dx \\
&= \theta \left[\frac{x^{-\theta+1}}{-\theta+1}\right]_{1}^{\infty} \quad \text{(valid when } -\theta+1 < 0, \text{ i.e. } \theta > 1) \\
&= \frac{\theta}{1-\theta}\left(\lim_{x\to\infty} x^{1-\theta} - 1^{1-\theta}\right) = \frac{\theta}{1-\theta}(0-1) \quad (\text{since } \theta > 1 \implies 1-\theta < 0) \\
&= \frac{\theta}{\theta-1}.
\end{aligned}$$

Thus, $\mathbb{E}[X]$ exists only when $\theta > 1$, and in that case

$$\mathbb{E}[X] = \frac{\theta}{\theta-1}.$$

Example 5.4. Take X with probability mass function given by $f(x) = \frac{1}{x(x+1)}$ and $A = \{1, 2, \dots\}$.

$$\begin{aligned}\sum_{i=1}^{\infty} f(i) &= \sum_{i=1}^{\infty} \frac{1}{i(i+1)} \\ &= \sum_{i=1}^{\infty} \left(\frac{1}{i} - \frac{1}{i+1}\right) \\ &= 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \cdots = 1\end{aligned}$$

However, notice that,

$$\begin{aligned}\mathbb{E}[X] &= \sum_{i=1}^{\infty} i f(i) \\ &= \sum_{i=1}^{\infty} \frac{1}{i+1} \\ &= \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots = \infty\end{aligned}$$

since the above is just a harmonic series. Then $\mathbb{E}[X]$ does not exist.

Example 5.5. Take X with pdf $f(x) = \frac{1}{\pi} \cdot \frac{1}{1+x^2}$ for $x \in \mathbb{R}$ which is just the Cauchy-Distribution. Let's calculate the expectation without checking the condition.

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} \frac{1}{\pi} \cdot \frac{x}{1+x^2} dx = 0$$

Proof.

$$\begin{aligned}\mathbb{E}[X] &= \int_{-\infty}^{+\infty} x \cdot f(x)\, dx \\ &= \int_{-\infty}^{+\infty} \frac{1}{\pi} \cdot \frac{x}{1+x^2}\, dx \\ &= \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{x}{1+x^2}\, dx = \frac{1}{\pi} \lim_{A\to\infty} \int_{-A}^{A} \frac{x}{1+x^2}\, dx \\ &\text{Let } u = 1 + x^2 \quad \Rightarrow \quad du = 2x\, dx, \quad x\, dx = \tfrac{1}{2} du \\ &\Rightarrow \frac{1}{\pi} \lim_{A\to\infty} \int_{1+A^2}^{1+(-A)^2} \frac{1}{2} \cdot \frac{1}{u}\, du \\ &= \frac{1}{2\pi} \lim_{A\to\infty} \Big[\ln(1+A^2) - \ln(1+(-A)^2)\Big] \\ &= \frac{1}{2\pi} \lim_{A\to\infty} \big[\ln(1+A^2) - \ln(1+A^2)\big] \\ &= \frac{1}{2\pi} \lim_{A\to\infty} 0 = 0.\end{aligned}$$

But since we know that $\int_{-\infty}^{\infty} \left|\frac{x}{1+x^2}\right| dx = \infty$, then the expectation is therefore undefined.

5.1 Expectation as a Function of a Random Variable

Theorem 5.1. ***Expectation as a Function of an R.V.***
If X is a random variable, with pmf/pdf $f(x)$ and support A, then for any function $g(\cdot)$,

$$\mathbb{E}[g(X)] = \begin{cases} \sum_{x\in A} g(x)f(x) & \text{if } \sum_{x\in A} |g(x)|f(x) < \infty \quad \textit{discrete} \\ \int_{-\infty}^{\infty} g(x)f(x)dx & \text{if } \int_{-\infty}^{\infty} |g(x)|f(x)dx < \infty \quad \textit{continuous} \end{cases}$$

5.2 Linearity of Expectation

Theorem 5.2. ***Linearity of Expectation***
We can exchange the order of $\mathbb{E}[\cdot]$ and $\sum$ under Linearity,

$$\mathbb{E}[ag(X) + bh(X)] = a\mathbb{E}[g(X)] + b\mathbb{E}[h(X)]$$

Theorem 5.3. ***Variance***

$$\begin{aligned} \textit{Var}(X) &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2 - 2X\mathbb{E}[X] + \{\mathbb{E}[X]\}^2] \\ &= \mathbb{E}[X^2] - \{\mathbb{E}[X]\}^2 \end{aligned}$$

Remark 5.1. Recall from STAT 231 that we often denote by $X_1, \ldots, X_n$ an i.i.d. sample from the distribution of X. Note, however, that the statistic

$$\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

is not the same as $\mathrm{Var}(X)$, which is a fixed constant determined by the distribution of X.

5.3 Moments

Theorem 5.4. ***Moments****:*

1. ***Default Definition****: k^{th} moment of X is $\mathbb{E}[X^k]$*
2. *The k**th central moment** of X is denoted as $\mathbb{E}[(X-\mu)^k]$ where $\mu = \mathbb{E}[X]$*
3. *The k**th factorial moment** is defined as,*

$$\mathbb{E}[(X)_k] = \mathbb{E}[X(X-1)(X-2)\cdots(X-k+1)]$$

Example 5.6. Let $X \sim \text{POI}(\theta)$ with pdf $f(x) = \frac{e^{-\theta}\theta^x}{x!}$ for $x \in \{0, 1, 2, \ldots\}$,

$$\begin{aligned} \mathbb{E}[(X)_k] &= \mathbb{E}[X(X-1)\cdots(X-k+1)] \\ &= \sum_{x=0}^{\infty} x(x-1)\cdots(x-k+1)\cdot e^{-\theta}\frac{\theta^x}{x!} \\ &= \sum_{x=k}^{\infty} \frac{1}{(x-k)!}e^{-\theta}\theta^x \\ &= \sum_{y=0}^{\infty} \frac{1}{y!}e^{-\theta}\theta^{x+k} \\ &= \theta^k \sum_{y=0}^{\infty} \frac{1}{y!}e^{-\theta}\theta^y = \theta^k \end{aligned}$$

Remark 5.2. $k = 1, 2, 3, \ldots, n$ **moments**

$$\begin{aligned} k = 1 &: \mathbb{E}[(X)_1] = \mathbb{E}[X] = \theta \\ k = 2 &: \mathbb{E}[(X)_2] = \mathbb{E}[X(X-1)] = \mathbb{E}[X^2] - \mathbb{E}[X] = \theta^2 \\ &\implies \mathbb{E}[X^2] = \theta^2 + \theta \\ &\implies \text{Var}(X) = \theta \end{aligned}$$

For Poisson random variables, we see that $\mathbb{E}[X] = \text{Var}(X) = \theta$.

Example 5.7. Let's say we want to find the Gamma Distribution of the kth moment, where,

$$\mathbb{E}[X^k] = \int_0^\infty x^k \frac{x^{\alpha-1}\exp(-x/\beta)}{\Gamma(\alpha)\beta^\alpha} dx = \int_0^\infty \frac{x^{\alpha+k-1}\exp\left(-\frac{x}{\beta}\right)}{\Gamma(\alpha)\beta^\alpha} dx$$

Method 1. Change of variable from $y = \frac{x}{\beta}$ to $x = \beta y$ and $dx = \beta dy$

$$\mathbb{E}[X^k] = \frac{\beta^k}{\Gamma(\alpha)} \underbrace{\int_0^\infty y^{\alpha+k-1}e^{-y}\, dy}_{\Gamma(\alpha+k)} = \beta^k \frac{\Gamma(\alpha+k)}{\Gamma(\alpha)}$$

Method 2. Let $\alpha' = \alpha + k$,

$$\begin{aligned} \mathbb{E}[X^k] &= \int_0^\infty \frac{x^{\alpha'-1}\exp\left(-\frac{x}{\beta}\right)}{\Gamma(\alpha)\beta^\alpha} dx \\ &= \int_0^\infty \frac{x^{\alpha'-1}\exp\left(-\frac{x}{\beta}\right)}{\Gamma(\alpha')\beta^{\alpha'}} \cdot \frac{\Gamma(\alpha')\beta^{\alpha'}}{\Gamma(\alpha)\beta^\alpha} dx \\ &= \int_0^\infty \underbrace{\frac{x^{\alpha-1}\exp\left(-\frac{x}{\beta}\right)}{\Gamma(\alpha')\beta^{\alpha'}}}_{\text{pdf of } \Gamma(\alpha',\beta)} dx \cdot \frac{\Gamma(\alpha')\beta^{\alpha'}}{\Gamma(\alpha)\beta^\alpha} \\ &= \beta^k \cdot \frac{\Gamma(\alpha+k)}{\Gamma(\alpha)} \end{aligned}$$

Let $k = 1$, then,

$$\mathbb{E}[X] = \beta \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)} = \alpha\beta,$$

since we know that $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$ for $\alpha \geq 1$. Similarly, for $k = 2$, then,

$$\mathbb{E}[X^2] = \beta^2 \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)} = \alpha(\alpha+1)\beta^2$$

by the same property of $\Gamma(\cdot)$ utilized above, where we apply the property twice,

$$\Gamma(\alpha+2) = (2+1)\Gamma(\alpha+1) = \alpha(\alpha+1)\Gamma(\alpha)$$

Thus, we derive the variance as,

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \alpha(\alpha+1)\beta^2 - \alpha^2\beta^2 = \alpha\beta^2$$

6 Moment Generating Functions

Definition 6.1. Suppose that X is a random variable, and its moment-generating function (mgf) is defined as,

$$M(t) = \mathbb{E}[e^{tX}]$$

if $M(t)$ exists on $t \in (-h, h)$ for some $h > 0$.

Remark 6.1. h can be arbitrarily small, but $M(t)$ must exist at $t = 0$. Note that we must state the range of t when determining the mgf of a given random variable.

Example 6.1. Find the MGF of $X \sim \text{Gamma}(\alpha, \beta)$,

$$M(t) = \mathbb{E}[e^{tX}] = \int_0^\infty e^{tx} \frac{x^{\alpha-1} e^{-x/\beta}}{\Gamma(\alpha)\beta^\alpha} dx = \int_0^\infty \frac{x^{\alpha-1} \exp\left\{-x\left(\frac{1}{\beta} - t\right)\right\}}{\Gamma(\alpha)\beta^\alpha} dx.$$

Solution. We note that $\frac{1}{\beta} - t > 0$, otherwise the MGF does not exist.

Method 1. Let $y = x\left(\frac{1}{\beta} - t\right)$, so that

$$x = \frac{y}{\frac{1}{\beta} - t}, \qquad dx = \frac{1}{\frac{1}{\beta} - t}\, dy.$$

$$M(t) = \int_0^\infty \frac{\left(\frac{1}{\beta} - t\right)^{-\alpha}}{\Gamma(\alpha)\beta^\alpha} y^{\alpha-1} e^{-y}\, dy = \frac{\left(\frac{1}{\beta} - t\right)^{-\alpha}}{\Gamma(\alpha)\beta^\alpha} \cdot \Gamma(\alpha) = (1 - \beta t)^{-\alpha}.$$

Method 2. Define $\beta' = \left(\frac{1}{\beta} - t\right)^{-1}$,

$$\begin{aligned} M(t) &= \int_0^\infty \frac{x^{\alpha-1} \exp\left(-\frac{x}{\beta'}\right)}{\Gamma(\alpha)\beta^\alpha} dx \\ &= \frac{\Gamma(\alpha)(\beta')^\alpha}{\Gamma(\alpha)\beta^\alpha} \int_0^\infty \frac{x^{\alpha-1} \exp(-x/\beta')}{\Gamma(\alpha)(\beta')^\alpha} dx \\ &= \frac{\Gamma(\alpha)(\beta')^\alpha}{\Gamma(\alpha)\beta^\alpha} \\ &= (1 - \beta t)^{-\alpha}. \end{aligned}$$

Note that, $M(t)$ eists on $(-\infty, 1/\beta)$ so $M(t)$ exists on the interval $(-h, h)$ for any $h \le 1/\beta$ for example we can let $h = \frac{1}{2\beta}$ as long it is less than $1/\beta$. ■

Example 6.2. Find the MGF of the Poisson distribution.

Let $X \sim \text{Poi}(\theta)$ so that we have pmf given by,

$$f(x) = e^{-\theta} \frac{\theta^x}{x!}, \quad x \in \{0, 1, 2, \dots\}$$

Solution.

$$\begin{aligned} M(t) = \mathbb{E}[e^{tX}] &= \sum_{k=0}^\infty e^{-\theta} \frac{\theta^x}{x!} dx \\ &= \sum_{k=0}^\infty e^{-\theta} \frac{(\theta e^t)^x}{x!} = e^{-\theta} \sum_{x=0}^\infty \frac{(\theta')^x}{x!} \quad \text{Let } \theta' = \theta e^t \\ &= e^{\theta' - \theta} \sum_{k=0}^\infty \underbrace{e^{-\theta'} \cdot \frac{(\theta')^x}{x!}}_{\text{pmf of Poi}(\theta')} = e^{\theta' - \theta} = \exp\left\{\theta(e^t - 1)\right\} \end{aligned}$$

■

6.1 Properties of the MGF

Let's consider the MGF after a location-scale transformation. Suppose that X has MGF $M_X(t)$, let's find the mgf of $Y = aX + b$ where a, b are constants.

$$M_Y(t) = \mathbb{E}[e^{tY}] = \mathbb{E}[e^{atX+bt}] = e^{bt}\mathbb{E}[e^{atX}] = e^{bt}M_X(at)$$

Example 6.3. $Z \sim N(0,1)$, Find the MGF,

$$\begin{aligned} M_Z(t) = \mathbb{E}[e^{tZ}] &= \int_{-\infty}^{\infty} e^{tz}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}z^2}dz \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{1}{2}(z^2-2tz)\right\}dz \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{1}{2}(z-t)^2+\frac{1}{2}t^2\right\}dz \\ &= \int_{-\infty}^{\infty} \underbrace{\frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{1}{2}(z-t)^2\right\}dz}_{\text{pdf of } N(t,1)} \cdot e^{\frac{1}{2}t^2} = e^{\frac{1}{2}t^2} \end{aligned}$$

Example 6.4. Let $X \sim N(\mu, \sigma^2)$, find the MGF of $X = \sigma Z + \mu$,

$$M_X(t) = e^{\mu t} \quad M_Z(\sigma t) = e^{\mu t} \cdot e^{\frac{1}{2}\sigma^2 t^2} = e^{\frac{1}{2}\sigma^2 t^2 + \mu t}$$

6.1.1 MGF to Moment

Find moments using mgf. Suppose that X is a random variable with mgf given by $M_X(t)$ then,

$$\mathbb{E}[X^k] = M_X^{(k)}(0)$$

is the kth derivative of $M_X(t)$ at $t = 0$.

Example 6.5. Suppose that we have a $\Gamma(\alpha, \beta)$ distribution with the known,

$$\mathbb{E}[X] = \alpha\beta, \quad \mathbb{E}[X^2] = \alpha(\alpha+1)\beta^2, \quad M_X(t) = (1-\beta t)^{-\alpha}$$

Then, using the mgf and calculus we can derive,

$$\begin{aligned} &M_X'(t) = \alpha\beta(1-\beta t)^{-\alpha-1} \implies M_X'(0) = \alpha\beta = \mathbb{E}[X] \\ &M_X''(t) = \alpha(\alpha+1)\beta^2(1-\beta t)^{-\alpha-2} \implies M_X''(0) = \alpha(\alpha+1)\beta^2 = \mathbb{E}[X^2] \end{aligned}$$

Example 6.6. Suppose that we have a Poisson(θ) distribution with the known,

$$\mathbb{E}[X] = \theta, \quad \mathbb{E}[X^2] = \theta^2 + \theta, \quad M_X(t) = \exp\left\{\theta(e^t - 1)\right\}$$

Then, using the mgf and calculus we can derive,

$$\begin{aligned} M_X'(t) &= \theta e^t \cdot \exp\left\{\theta(e^t-1)\right\} \implies M_X'(0) = \theta \\ M_X''(t) &= \theta e^t \cdot \exp\left\{\theta(e^t-1)\right\} + \theta^2 e^{2t}\exp\left\{\theta(e^t-1)\right\} \\ \implies M_X''(0) &= \theta + \theta^2 = \mathbb{E}[X^2] \end{aligned}$$

6.1.2 MGF vs. Distribution

If the mgf exists, X and Y have the same distribution, equivalently, X, Y have the same mgf. MGF and Distribution are one-to-one

Example 6.7. X has $M_X(t) = e^{\frac{1}{2}t^2}$ so $X \sim N(0,1)$ by the uniqueness of the mgf.

1. Find the mgf of $Y = 2X - 1$

$$M_Y(t) = e^{-t}M_X(2t) = e^{-t}e^{2t^2} = e^{2t^2-t}$$

2. Find $\mathbb{E}(Y)$ and $\text{Var}(Y)$.

$$\begin{aligned}
\mathbb{E}(Y) &= \mathbb{E}(2X-1) = 2\mathbb{E}(X) - 1 = -1 \\
\text{Var}(Y) &= 4\text{Var}(X) = 4 \\
M_Y'(t) &= (4t-1)e^{2t^2-t} \implies \mathbb{E}(Y) = M_Y'(0) = -1 \\
M_Y''(t) &= 4e^{2t^2-t} + (4t-1)^2e^{2t^2-t} \implies \mathbb{E}(Y^2) = M_Y''(0) = 5
\end{aligned}$$

3. Find the distribution of Y. Evidently, the mgf of $N(\mu, \sigma^2)$ is $e^{\mu t + \frac{1}{2}\sigma^2 t^2}$

7 Joint Distributions

In this lecture, we wish to characterize the behaviours of two (or multiple) random variables simultaneously.

7.1 The Joint CDF

Definition 7.1. Given two random variables, X, Y, the joint cdf of the two random variables, can be written as,

$$F(x,y) = P(X \le x, Y \le y) = P(\{X \le x\} \cap \{Y \le y\})$$

Suppose that we have $X, \ldots, X_m$ random variables, so that their joint cdf is written as,

$$F(x_1, \ldots, x_m) = \mathbb{P}(X_1 \le x_1, \ldots, X_m \le x_m) = P(\{X_1 \le x_1\} \cap \cdots \cap \{X_m \le x_m\})$$

Theorem 7.1. ***Properties of the Joint CDF****:*

1. *Fix x, $F(x,y)$ is a non-decreasing function of y,*
   ***or** Fix y, so that $F(x,y)$ is a non-decreasing function of x.*
2. ***Limit of** $F(\cdot,\cdot)$*

$$\underbrace{\lim_{x\to\infty}}_{\{X\le x\}\to\emptyset} F(x,y) = \underbrace{\lim_{y\to-\infty}}_{\{Y\le y\}\to\emptyset} F(x,y) = y$$

3. ***Joint Limit of** $F(\cdot,\cdot)$*

$$\underbrace{\lim_{x\to\infty, y\to\infty}}_{\{X\le x\},\{Y\le y\}\to S \quad \textit{(sample space)}} F(x,y) = 1$$

4. ***Relationship between** $F(x,y)$ **and** $F_X(x), F_Y(y)$*

$$\begin{aligned}
F_X(x) &= P(X \le x) \\
&= P(X \le x, Y < \infty) = \lim_{y\to\infty} F(x,y) \\
F_Y(y) &= P(Y \le y) = \lim_{x\to\infty} F(x,y)
\end{aligned}$$

Note that,

$$F(x,y) \implies F_X(x), F_Y(y) \quad \textit{but} \quad F_X(x), F_Y(y) \impliedby F(x,y) \textit{ does NOT hold!}$$

7.2 Joint Discrete Random Variables

Definition 7.2. X, Y are both discrete random variables,
then the **Joint PMF** of the two random variables can be written as,

$$f(x,y) = P(X = x, Y = y)$$

with **support**,

$$A = \{(x,y) \mid f(x,y) > 0\}$$

Theorem 7.2. ***Properties of the PMF of Joint Random Variables***

1. $f(x,y) \geq 0$
2. ***Total Sum of PMF***
$$\sum_{(x,y)\in A} f(x,y) = 1$$
3. ***Mass Property***
$$P\{(X,Y) \in B\} = \sum_{(x,y)\in B} f(x,y)$$

Example 7.1.

$$P(X \leq Y) = \sum_{(x,y)\in B} f(x,y), \quad B = \{(x,y) \mid x \leq y\}$$

Definition 7.3. Marginal PMF: Let $f(x,y)$ be a joint pmf,

$$\text{Marginal PMF of } X: \quad f_X(x) = P(X = x) = \sum_y P(X = x, Y = y) = \sum_y f(x,y)$$

$$\text{Marginal PMF of } Y: \quad f_Y(y) = P(Y = y) = \sum_x f(x,y)$$

Example 7.2. Let X, Y be discrete random variables with joint pmf given by,

$$f(x,y) = kq^2p^{x+y}, \quad x,y \in \{0,1,\dots\}, \quad p \in (0,1), \quad q = 1-p$$

1. Find the value of k
2. Find the marginal pmfs of X, Y
3. Find $\Pr(X \leq Y)$

Solution.

1. Find the value of k

$$\sum_{(x,y)\in A} f(x,y) = 1$$

$$\implies \sum_{x=0}^{\infty}\sum_{y=0}^{\infty} kq^2p^{x+y} = kq^2\sum_{x=0}^{\infty}\sum_{y=0}^{\infty} p^{x+y} = kq^2 \underbrace{\left(\sum_{x=0}^{\infty} p^x\right)\left(\sum_{y=0}^{\infty} p^y\right)}_{p^0+p^1+\cdots=\frac{1}{1-p}}$$

$$= kq^2 \cdot \left(\frac{1}{1-p}\right)^2 = k \implies k = 1$$

2. Find the marginal pmfs of X, Y

$$f_X(x) = \sum_{y=0}^{\infty} f(x,y) = \sum_{y=0}^{\infty} q^2 p^{x+y} = q^2 p^x \sum_{y=0}^{\infty} p^y = \frac{q^2 p^x}{1-p} = qp^x$$
$$f_Y(y) = qp^y$$

So, $X, Y \sim \text{Geo}(1-p) = \text{Geo}(q)$.

3. Find $\Pr(X \le Y)$.

$$B = \{(x,y) \mid x \le y\}$$
$$P(X \le Y) = \sum_{(x,y)\in B} f(x,y) = \sum_{x=0}^{\infty}\sum_{y=x}^{\infty} q^2 p^{x+y}$$
$$= \sum_{x=0}^{\infty} q^2 p^x \cdot \underbrace{\left(\sum_{y=x}^{\infty} p^y\right)}_{p^x + p^{x+1} + \cdots = p^x(p^0+p^1+\dots) = \frac{p^x}{1-p}}$$
$$= \sum_{x=0}^{\infty} q^2 p^x \cdot \frac{p^x}{1-p} = \underbrace{q \cdot \sum_{x=0}^{\infty} p^{2x}}_{p^0+p^2+\cdots=\frac{1}{1-p^2}} = \frac{q}{1-p^2} = \frac{1}{1+p}$$

■

7.3 Joint Continuous Random Variables

Definition 7.4. If X, Y are continuous random variables, and if their joint cdf can be written as,

$$F(x,y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f(s,t)dtds \quad \text{for some } f(\cdot,\cdot)$$

then, $f(x,y)$ is the joint pdf.

$$f(x,y) = \begin{cases} \frac{\partial^2 F(x,y)}{\partial x \partial y}, & \text{if derivative exists} \\ 0, & \text{otherwise} \end{cases}$$

with support $A = \{(x,y) \mid f(x,y) > 0\}$.

1. $$f(x,y) \ge 0$$

2. **Normalization Condition**
$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)dxdy = 1$$

3. **Partition Condition**
$$P\{(X,Y) \in B\} = \int\int_{(X,Y)\in B} f(x,y)dxdy$$

8 Joint CDF, PMF/PDFs

Example 8.1. Suppose that we have the joint pdf for (X, Y) which is defined as,

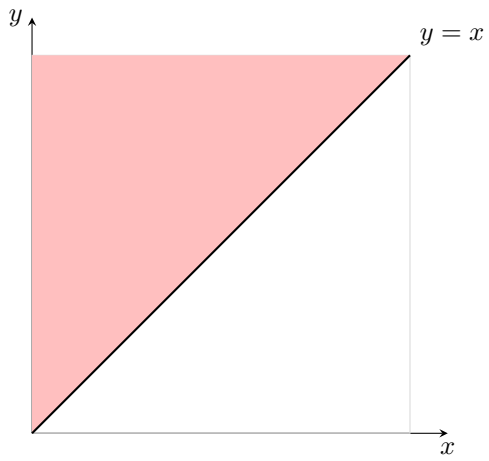
$$f(x,y) = \begin{cases} k\exp(-x-y), & 0 < x < y < \infty, \\ 0, & \text{otherwise} \end{cases}$$

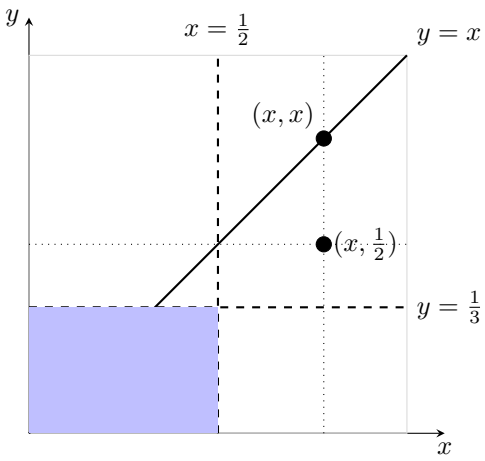
Find k**,**

$$\begin{aligned} 1 &= \int\int f(x,y)dxdy = \int_0^\infty \int_x^\infty ke^{-x-y}dydx \\ &= \int_0^\infty ke^{-2x}dx = \frac{k}{2} \implies k = 2 \end{aligned}$$

Find Pr$(X \le 1/3, Y \le 1/2)$**,**

$$\begin{aligned} P(X \le 1/3, Y \le 1/2) &= \int_0^{\frac{1}{3}} \int_x^{\frac{1}{2}} 2e^{-x-y}dydx \\ &= \int_0^{\frac{1}{3}} 2e^{-x}(e^{-x} - e^{-1/2})dx \\ &= \int_0^{\frac{1}{3}} (2e^{-2x} - 2e^{-x-1/2})dx \\ &= (-e^{-2x} + 2e^{-x-1/2}) \,|_0^{\frac{1}{3}} \\ &= 1 - e^{-2/3} + 2e^{-5/6} - 2e^{-1/2} \end{aligned}$$



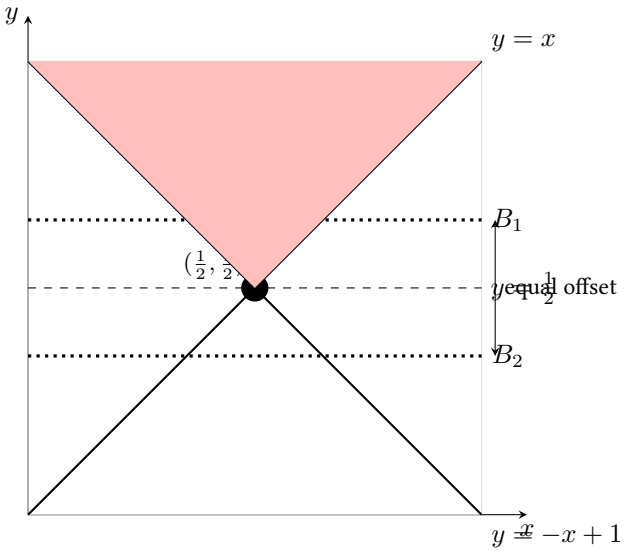


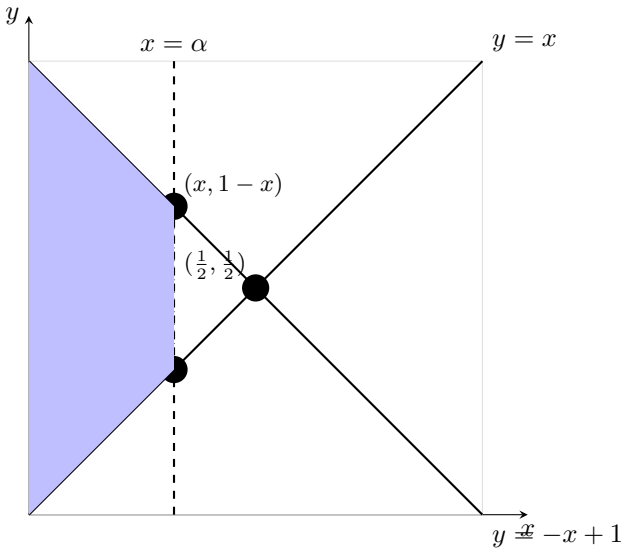
Compute Pr$(X+Y\geq 1)$**,**

$$\begin{aligned}
P(X+Y\geq 1) &= \int\int_{B_1} f(x,y)dxdy + \int\int_{B_2} f(x,y)dxdy \\
&= \int_0^{\frac{1}{2}}\int_{1-x}^{\infty} 2e^{-x-y}dydx + \int_{\frac{1}{2}}^{\infty}\int_{x}^{\infty} 2e^{-x-y}dydx \\
&= \int_0^{\frac{1}{2}} 2e^{-x}\cdot e^{x-1}dx + \int_{\frac{1}{2}}^{\infty} 2e^{-2x}dx \\
&= e^{-1} - e^{-2x}\,|_{\frac{1}{2}}^{\infty} = 2e^{-1}
\end{aligned}$$

OR

$$\begin{aligned}
P(X+Y<1) &= \int_0^{\frac{1}{2}}\int_{x}^{1-x} 2e^{-x-y}dydx \\
&= \int_0^{1} 2e^{-x}(e^{-x}-e^{x-1})dx \\
&= \int_0^{1} (2e^{-2x}-2e^{-1})dx = 1-2e^{-1} \\
\implies P(X+Y\geq 1) &= 1-P(X+Y<1) = 2e^{-1}
\end{aligned}$$





8.1 Independence

Definition 8.1. For any two random variables X, Y, we can call them independent if,

$$\Pr(X \in A, Y \in B) = \Pr(X \in A)\Pr(Y \in B)$$

for any two sets A and B which are two sets of real numbers.

Remark 8.1. We can express this equality as $\{X \in A\} \cup \{Y \in B\}$ which are two disjoint events.

$$\{X \in A, Y \in B\} = \{X \in A\} \cap \{Y \in B\}$$

Notation: $X \perp Y$.

Theorem 8.1. *Random variables X and Y are said to be independent if and only if, one of the following three hold (equivalent),*

(i) *X and Y are independent.*

(ii) $f(x, y) = f_X(x) f_Y(y)$

(iii) $F(x, y) = F_X(x) F_Y(y)$

Remark 8.2. We can say that,

$$(i) \overset{\text{iff}}{\iff} (ii) \overset{\text{iff}}{\iff} (iii)$$

where $f(x, y), f_X(x), f_Y(y)$ denotes the pdf/pmf for discrete/continuous distributions, and where $F(x, y), F_X(x), F_Y(y)$ denotes the cdf for both discrete and continuous distributions.

Theorem 8.2 (Functions of Independent Random Variables)**.** *If X, Y are independent, then $g(X), h(Y)$ are independent as well, where $g(\cdot), h(\cdot)$ are functions.*

Proof. Let $A, B \subseteq \mathbb{R}$ be Borel sets. Define the preimages,

$$\begin{aligned} g^{-1}(A) &= \{x \in \mathbb{R} : g(x) \in A\}, \\ h^{-1}(B) &= \{y \in \mathbb{R} : h(y) \in B\}, \\ \{g(X) \in A\} &= \{X \in g^{-1}(A)\}, \\ \{h(Y) \in B\} &= \{Y \in h^{-1}(B)\}. \end{aligned}$$

$$\begin{aligned} P(g(X) \in A, h(Y) \in B) &= P(X \in g^{-1}(A),\, Y \in h^{-1}(B)) \\ &\overset{(1)}{=} P(X \in g^{-1}(A))\, P(Y \in h^{-1}(B)) \\ &= P(g(X) \in A)\, P(h(Y) \in B), \end{aligned}$$

where step (1) holds because X and Y are independent, hence

$$P(X \in E, Y \in F) = P(X \in E) P(Y \in F) \quad \text{for all measurable } E, F.$$

This shows $g(X)$ and $h(Y)$ are independent.

Example 8.2. Consider the joint discrete random variable example,

$$f(x,y) = q^2 p^{x+y}, \quad q = 1-p, \quad x,y = 0,1,2,\dots$$

Are X, Y independent?

$$f_X(x) = qp^x, \quad f_Y(y) = qp^y$$
$$f(x,y) = (qp^x)(qp^y) = f_X(x)f_Y(y) \implies X \perp Y$$

Example 8.3. Consider the previous example,

$$f(x,y) = \begin{cases} x+y, & x,y \in [0,1], \\ 0, & \text{otherwise} \end{cases}$$

Are X, Y independent?

1. Find the marginal PDFs

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)dy = \begin{cases} x+\frac{1}{2}, & \text{if } x \in [0,1], \\ 0, & \text{otherwise} \end{cases}$$
$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)dx = \begin{cases} y+\frac{1}{2}, & \text{if } y \in [0,1] \\ 0, & \text{otherwise} \end{cases}$$

2. Determine if the pmf is the product of the marginal pdfs

$$f(x,y) \neq f_X(x)f_Y(y) \implies X \not\perp Y$$

   Therefore, we conclude that X, Y are **dependent** random variables.

Theorem 8.3 (Factorization Theorem for Independence)**.** *Suppose that we have two random variables, X and Y. The support of X is denoted as A_X and the support of Y is denoted as A_Y, and the support of (X,Y) is denoted as A. Then the following two conditions together,*

(i) $f(x,y) = cg(x)h(y)$ *where c is a constant*

(ii) $A = A_X \times A_Y$ *(the Cartesian product)*

are the necessary and sufficient conditions for X and Y to be independent.

Remark 8.3. **Conditions for Factorization Theorem of Independence**

1. c is a constant in (i) that does not depend on x and y and $g(\cdot), h(\cdot)$ are functions.
2. A_X is the marginal support of X and A_Y is the marginal support of Y so the marginal support of the joint (X, Y) can be expressed as an explicit cartesian product.

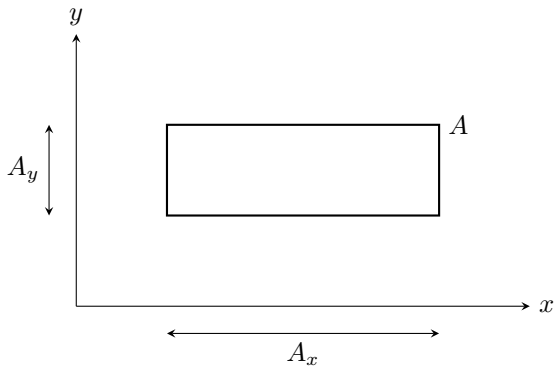


Figure 10: $A_X \times A_Y = A$ Support of (X, Y) Cartesian Product

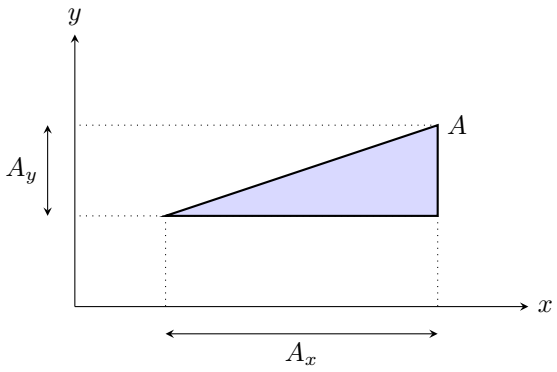


Figure 11: Triangular support: $A = \frac{1}{2} A_x A_y \neq A_x A_y$

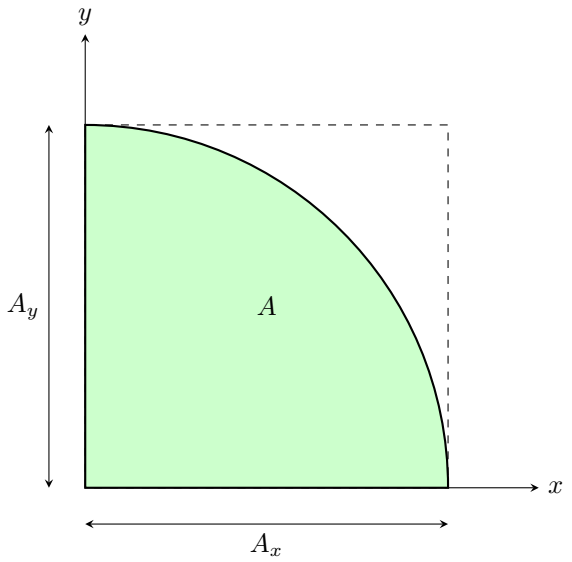


Figure 12: Quarter-circle support: $A = \frac{\pi}{4} A_x^2 \neq A_x A_y$

Example 8.4. Given the joint pmf,

$$f(x,y) = \frac{\theta^{x+y}e^{-2\theta}}{x!y!}$$

where $x, y = 0, 1, \dots$. Are X and Y independent? Find the marginal pmf of X and Y.

Solution. We verify the two conditions:

(i) Split the joint pmf into two explicit parts,

$$f(x,y) = e^{-2\theta}\frac{\theta^x}{x!} \cdot \frac{\theta^y}{y!}, \quad x, y = 0, 1, 2, \dots$$

(ii) Verify the support of $X, Y, (X, Y)$ is indeed a Cartesian product,

$$\begin{aligned} A &= \{(x,y) : x, y = 0, 1, \dots\} \\ &= \{x : x = 0, 1, \dots\} \times \{y : y = 0, 1, \dots\} \\ &= A_X \times A_Y \implies X \perp Y \end{aligned}$$

Also, note that the individual marginal pmfs for $f(x,y)$ are,

$$\begin{aligned} f_X(x) &= e^{-\theta}\frac{\theta^x}{x!}, \quad x = 0, 1, 2, \dots, \\ f_Y(y) &= e^{-\theta}\frac{\theta^y}{y!}, \quad y = 0, 1, 2, \dots. \end{aligned}$$

■

Example 8.5. Suppose X, Y have the joint pdf,

$$f(x,y) = \frac{3}{2}y(1-x^2), \quad x \in [-1,1], \quad y \in [0,1]$$

Are they independent? Find the marginal pdfs of X and Y.

Solution. We check the two conditions, then evaluate their marginal pdfs with 2 methods,

1. Check the joint pdf is separable
$$f(x,y) = \frac{3}{2} \cdot (1-x^2) \cdot y$$
2. Check $A = A_X \times A_Y$,
$$\begin{aligned} A &= \{(x,y) : x \in [-1,1], \quad y \in [0,1]\} \\ &= \{x : x \in [-1,1]\} \times \{y : y \in [0,1]\} \\ &= A_X \times A_Y \implies X \perp Y. \end{aligned}$$

Method 1.

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$$
$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$$

Method 2.

$$f(x,y) = f_X(x)f_Y(y)$$
$$f_Y(y) = c \cdot y \implies \int_{-\infty}^{\infty} f_Y(y)\,dy = \frac{c}{2} \implies c = 2$$
$$f_X(x) = \frac{f(x,y)}{f_Y(y)} = \frac{3}{4}(1-x)^2, \quad x \in [-1,1]$$

■

Example 8.6. The joint pdf of (X,Y) is,

$$f(x,y) = \frac{2}{\pi}, \quad x \in [0, \sqrt{1-y^2}], \quad y \in (-1,1).$$

Are X and Y independent? Find the marginal pdfs for X and Y.

Solution. We have the marginal pdfs of X and Y and show that,

$$A_X = [0,1], \quad A_Y = (-1,1) \implies A_X \times A_Y = \{(x,y) : x \in [0,1], \quad y \in (-1,1)\}$$

However, notice that,

$$A = \{(x,y) : x^2 + y^2 \le 1, \quad x \in [0,1], \quad y \in (-1,1)\} \implies A \neq A_X \times A_Y$$

Therefore, X and Y are not independent, and are dependent.

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f(x,y)dy \\ &= \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{2}{\pi}dy = \frac{4}{\pi}\sqrt{1-x^2} \quad x \in [0,1] \\ f_Y(y) &= \int_{-\infty}^{\infty} f(x,y)dx \\ &= \int_{0}^{\sqrt{1-y^2}} \frac{2}{\pi}dx = \frac{2}{\pi}\sqrt{1-y^2} \quad y \in (-1,1) \end{aligned}$$

■

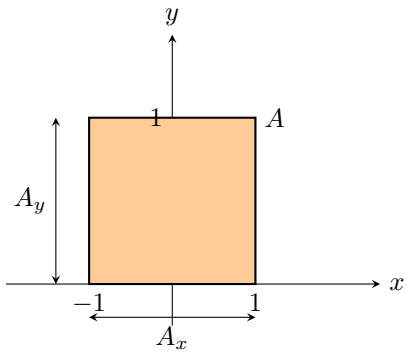


Figure 13: Rectangular support centered at the origin: $A = A_x A_y$

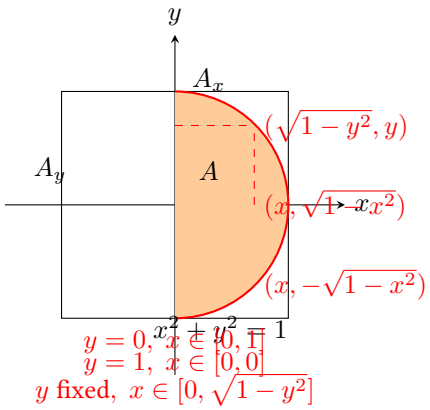


Figure 14: Shaded region with circular boundary and labeled coordinates.

9 Joint Expectation, Variance and Conditionals

9.1 Joint Expectation

Definition 9.1. We extend the definition of expectation from univariate to bivariate (multivariate) cases. Suppose that $h(x,y)$ is a real function, then we have that,

$$\mathbb{E}[h(X,Y)] = \begin{cases} \sum_x \sum_y h(x,y)f(x,y), & \text{discrete} \\ \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(x,y)f(x,y)dxdy, & \text{continuous} \end{cases}$$

If the expectation exists, then we have that,

$$\begin{cases} \sum_x \sum_y |h(x,y)|f(x,y) < \infty, & \text{discrete} \\ \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} |h(x,y)|f(x,y)dxdy, & \text{continuous} \end{cases}$$

Example 9.1. **Example Functional Expectations**
Take, for example, if we have that $h(x,y) = xy$, then the expectation could be defined as,

$$\mathbb{E}[XY] = \begin{cases} \sum_x \sum_y xyf(x,y), & \text{discrete} \\ \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xyf(x,y)dxdy, & \text{continuous} \end{cases}$$

Or, we could take that $h(x,y) = x$, in which have that,

$$\mathbb{E}[X] = \begin{cases} \sum_x \sum_y xf(x,y), & \text{discrete} \\ \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xf(x,y)dxdy, & \text{continuous} \end{cases}$$

So, in the discrete and continuous, respectively, we have that,

$$\implies \begin{cases} f_X(x) = \sum_y f(x,y), \\ \mathbb{E}[X] = \sum_x xf(x,y)f_X(x) \end{cases} \qquad \text{or} \quad \begin{cases} f_X(x) = \int_{-\infty}^{\infty} f(x,y)dy, \\ \mathbb{E}[X] = \int_{-\infty}^{\infty} xf_X(x)dx \end{cases}$$

9.2 Properties of Joint Expectation

1. Change the Order of Summation and Expectation

$$\mathbb{E}\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} \mathbb{E}[a_i X_i] = \sum_{i=1}^{n} a_i \mathbb{E}[X_i]$$

   Similarly, we can write that,

$$\mathbb{E}[ag(X,Y) + bh(X,Y)] = a\mathbb{E}[g(X,Y)] + b\mathbb{E}[h(X,Y)]$$

2. $\mathbb{E}[XY] =\neq \mathbb{E}[X]\mathbb{E}[Y]$, in general. However, if we know that X and Y are independent random variables, then we can write that $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

   Proof.

$$\begin{aligned} \mathbb{E}[XY] &= \int\int_{(x,y)\in A} xyf(x,y)dxdy \\ &= \int\int_{(x,y)\in A} xyf_X(x)f_Y(y)dxdy, \qquad f(x,y) = f_X(x)f_Y(y) \\ &= \int_{y\in A_Y}\int x \in A_X xyf_X(x)f_Y(y)dxdy, \qquad A = A_X \times A_Y \\ &= \left(\int_{x\in A_X} xf_X(x)dx\right)\cdot\left(\int_{y\in A_Y} yf_Y(y)dy\right) \\ &= \mathbb{E}[X]\cdot\mathbb{E}[Y] \end{aligned}$$

Remark 9.1. For the same reason, if $X \perp Y$, then $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$,
so, we can write that,

$$X \perp Y \implies g(X) \perp h(Y) \implies \mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$$

Remark 9.2. If $X_1, \ldots, X_n$ are independently distributed random variables,

$$\mathbb{E}\left[\prod_{i=1}^{n} X_i\right] = \prod_{i=1}^{n} \mathbb{E}[X_i]$$

Definition 9.2 (Covariance)**.** $\text{Cov}(X, Y)$ is the covariance between X, Y, such that,

- If $\text{Cov}(X, Y) > 0$, when X is above its mean, then Y tends to be above its mean as well (they move together)
- If $\text{Cov}(X, Y) < 0$, when X is above its mean, then Y tends to be below its mean (they move in opposite directions)
- If $\text{Cov}(X, y) \approx 0$, there is no linear relationship.

The covariance between two random variables X and Y can be written as,

$$\begin{aligned}\text{Cov}(X, Y) &= \mathbb{E}\left[\{X - \mathbb{E}[X]\}\{Y - \mathbb{E}[Y]\}\right] \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]\end{aligned}$$

If $X \perp Y$, then $\text{Cov}(X, Y) = 0$. The forward implication holds: independence implies zero covariance. However, the converse is false: zero covariance does not imply independence, since covariance only captures linear dependence. Variables can be dependent in non-linear ways while still having zero covariance.

Remark 9.3.

$$\text{Cov}(X, X) = \mathbb{E}[\{X - \mathbb{E}[X]\}^2] = \text{Var}(X)$$

Theorem 9.1. ***Variance Formulas***

(i) Linear Combinations of Random Variables

$$\underbrace{Var(aX + bY)}_{aX+bY} = \underbrace{a^2 Var(X)}_{a^2X^2} + \underbrace{b^2 Var(Y)}_{b^2Y^2} + \underbrace{2abCov(X, Y)}_{2abXY}$$

(ii) Sum of Linear Combination of Random Variables

$$\begin{aligned}\underbrace{Var\left(\sum_{i=1}^{n} a_i X_i\right)}_{\left(\sum_{i=1}^{n} a_i X_i\right)^2} &= \underbrace{\sum_{i=1}^{n} a_i^2 Var(X_i)}_{\sum_{i=1}^{n} a_i^2 X_i^2} + \underbrace{\sum_{i \neq j} (a_i a_j) Cov(X_i, X_j)}_{\sum_{i \neq j} a_i a_j X_i X_j} \\ &= \sum_{i=1}^{n} a_i^2 Var(X_i) + 2 \sum_{1 \leq i \leq j \leq n} a_i a_j Cov(X_i, X_j)\end{aligned}$$

(iii) If $X_1, \ldots, X_n$ are independent random variables, and $Cov(X_i, X_j) = 0$ for $i \neq j$, then,

$$Var\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 Var(X_i)$$

Example 9.2. Suppose that we have joint pmf for (X, Y), given by,

$$f(x, y) = \frac{\theta^{x+y} e^{-2\theta}}{x! y!}, \quad x, y = 0, 1, 2, \ldots$$

Find $\text{Var}(2X + 3Y)$.

Solution.

$$\begin{aligned} f(x,y) = e^{-\theta}\frac{\theta^x}{x!} \cdot e^{-\theta}\frac{\theta^y}{y!} = f_X(x) \cdot f_Y(y) \implies X \perp Y \\ \text{Var}(2X+3Y) &= 4\text{Var}(X) + 9\text{Var}(Y) \\ &= 4\theta + 9\theta, \qquad \text{Var}(X) = \text{Var}(Y) = \theta \\ &= 13\theta \end{aligned}$$

■

Example 9.3. Suppose that we have random variables (X, Y) with joint pdf,

$$f(x,y) = \begin{cases} x+y, & 0 \le x \le 1, \quad 0 \le y \le 1 \\ 0, & \text{otherwise.} \end{cases}$$

Find $\text{Var}(X+Y)$.

Solution. Notice that X and Y are independent because $f(x,y) \neq f_X(x) \cdot f_Y(y)$, so,

$$\begin{aligned} \text{Var}(X+Y) &= \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X,Y) \\ \text{Var}(X) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \qquad \text{Var}(Y) = \mathbb{E}[Y^2] - \mathbb{E}[Y]^2 \\ \text{Cov}(X,Y) &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] \end{aligned}$$

So we need $\mathbb{E}[X] = \mathbb{E}[Y]$ and $\mathbb{E}[X^2] = \mathbb{E}[Y^2]$ as well as $\mathbb{E}[XY]$,

$$\begin{aligned} \mathbb{E}[X] = \mathbb{E}[Y] &= \int_0^1 x f_X(x)dx = \int_0^1 x\left(x + \frac{1}{2}\right)dx = \frac{7}{12} \\ \mathbb{E}[X^2] = \mathbb{E}[Y^2] &= \int_0^1 x^2 f_X(x)dx = \int_0^1 x\left(x + \frac{1}{2}\right)dx = \frac{5}{12} \\ \mathbb{E}[XY] &= \int_0^1\int_0^1 xy(x+y)dxdy = \int_0^1\int_0^1 (x^2y + xy^2)dxdy \\ &= \int_0^1 \left(\frac{1}{3}y + \frac{1}{2}y^2\right)dy = \frac{1}{3} \\ \text{Cov}(X,Y) &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \frac{1}{3} - \left(\frac{7}{12}\right)^2 = \frac{-1}{144} \\ \text{Var}(X) &= \frac{5}{12} - \left(\frac{7}{12}\right)^2 = \frac{11}{144} \\ \implies \text{Var}(X+Y) &= 2\text{Var}(X) + 2\text{Cov}(X,Y) = \frac{5}{36} \end{aligned}$$

■

9.3 Correlation Coefficient

Definition 9.3. The correlation coefficient of X and Y is defined as,

$$\rho(X,Y) = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}}$$

If X and Y are independent random variables, then $\rho(X,Y) = 0$ because $\text{Cov}(X,Y) = 0$.

Theorem 9.2. ***Properties of the Correlation Coefficient***

1. $\rho(X,Y) \in [-1,1]$ *(i.e., bounded by* -1 *and* 1*)*

Proof.

$$\begin{aligned}
\rho(X,Y) &= \frac{\operatorname{Cov}(X,Y)}{\sqrt{\operatorname{Var}(X)}\sqrt{\operatorname{Var}(Y)}} \\
&= \frac{\mathbb{E}\big[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])\big]}{\sqrt{\mathbb{E}\big[(X-\mathbb{E}[X])^2\big]}\sqrt{\mathbb{E}\big[(Y-\mathbb{E}[Y])^2\big]}} \\
&\equiv \frac{\langle X-\mathbb{E}[X],\, Y-\mathbb{E}[Y]\rangle}{\|X-\mathbb{E}[X]\|\,\|Y-\mathbb{E}[Y]\|} \\
&\le \frac{\|X-\mathbb{E}[X]\|\,\|Y-\mathbb{E}[Y]\|}{\|X-\mathbb{E}[X]\|\,\|Y-\mathbb{E}[Y]\|} \quad \text{(by Cauchy–Schwarz)} \\
&= 1, \\
\rho(X,Y) &\ge -\frac{\|X-\mathbb{E}[X]\|\,\|Y-\mathbb{E}[Y]\|}{\|X-\mathbb{E}[X]\|\,\|Y-\mathbb{E}[Y]\|} \quad \text{(since absolute value)} \\
&= -1, \\
\therefore \quad -1 \;&\le\; \rho(X,Y) \;\le\; 1.
\end{aligned}$$

2. *Let* $Y = aX + b$ *with* $a \neq 0$*, then,*

$$\begin{aligned}
\mathit{Cov}(X,Y) &= \mathit{Cov}(X, aX+b) = \mathit{Cov}(X, aX) \\
&= a\mathit{Cov}(X,X) = a\mathit{Var}(X) \\
\implies \rho(X,Y) &= \frac{a\mathit{Var}(X)}{\sqrt{a^2\mathit{Var}(X)\mathit{Var}(X)}} \\
&= \frac{a}{|a|} = \begin{cases} 1, & a > 0 \\ -1, & a < 0 \end{cases}
\end{aligned}$$

Proof.

$$\begin{aligned}
\operatorname{Cov}(X,Y) &= \operatorname{Cov}(X, aX+b) \\
&= \operatorname{Cov}(X,aX) + \operatorname{Cov}(X,b) \\
&= a\operatorname{Cov}(X,X) + 0 \quad \text{(since } b \text{ is constant)} \\
&= a\operatorname{Var}(X), \\
\operatorname{Var}(Y) &= \operatorname{Var}(aX+b) = \operatorname{Var}(aX) = a^2\operatorname{Var}(X), \\
\rho(X,Y) &= \frac{\operatorname{Cov}(X,Y)}{\sqrt{\operatorname{Var}(X)}\sqrt{\operatorname{Var}(Y)}} = \frac{a\operatorname{Var}(X)}{\sqrt{\operatorname{Var}(X)}\sqrt{a^2\operatorname{Var}(X)}} = \frac{a\operatorname{Var}(X)}{|a|\operatorname{Var}(X)} \\
&= \frac{a}{|a|} = \begin{cases} 1, & a > 0, \\ -1, & a < 0, \end{cases}
\end{aligned}$$

Remark 9.4. $\rho(X,Y)$ is a measure of the linear relationship between X and Y.

Example 9.4. Suppose that we have the joint pdf for X and Y given as,

$$f(x,y) = \begin{cases} x+y, & x \in [0,1], \quad y \in [0,1] \\ 0, & \text{otherwise} \end{cases}$$

Solution.

$$\rho(X,Y) = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{-1/144}{11/144} = \frac{-1}{11}$$

Therefore, we see that X and Y are negatively correlated. ■

9.4 Conditional Distributions

Definition 9.4. Suppose that X and Y are joint random variables with joint pdf/pmf $f(x,y)$. Then the conditional pdf/pmf of X given $Y = y$ and Y given $X = x$ is written as,

$$f(x|y) = \frac{f(x,y)}{f_Y(y)} \qquad f(y|x) = \frac{f(x,y)}{f_X(x)}$$

given that $f_Y(y) > 0$ and $f_X(x) > 0$, respectively.

Remark 9.5. The conditional pdf/pmf are still valid pdf/pmf's, i.e.,

$$f(x \mid y) \geq 0 \implies \begin{cases} \sum_x f(x \mid y) = 1, & \text{discrete,} \\ \int_{-\infty}^{\infty} f(x \mid y) = 1, & \text{continuous} \end{cases}$$

10 Conditional Distributions and their Expectation

Theorem 10.1.

(i) If X and Y are independent, it is equivalent to say that $f(x \mid y) = f_X(x)$ which is also equivalent to saying that $f(y \mid x) = f_Y(y)$.

(ii) We can use the conditional distribution to find the joint and marginal pdf/pmf's

$$f(x \mid y) = \frac{f(x,y)}{f_Y(y)} \implies f(x,y) = f_Y(y) f(x \mid y)$$

Example 10.1. Suppose that $Y \sim \text{POI}(\theta)$ and that $(X \mid Y = y) \sim \text{BIN}(y, p)$.
Find the marginal distribution of X.

Solution.

1. **Step 1: Find the Joint PDF**

$$\begin{aligned} f(x,y) &= f_Y(y) f(x \mid y) \\ &= e^{-\theta} \frac{\theta^y}{y!} \cdot \binom{y}{x} p^x (1-p)^{y-x}, \qquad 0 \le x \le y \in \mathbb{Z}^+ \end{aligned}$$

2. **Step 2: Joint to Marginal**

$$\begin{aligned} f_X(x) &= \sum_{y \in A_Y} f(x,y) \\ &= \sum_{y=x}^{\infty} \binom{y}{x} p^x (1-p)^{y-x} e^{-\theta} \frac{\theta^y}{y!} \\ &= \sum_{y=x}^{\infty} \frac{y!}{x!(y-x)!} p^x (1-p)^{y-x} \cdot e^{-\theta} \frac{\theta^y}{y!} \\ &= e^{-\theta} \frac{\theta^x}{x!} \sum_{y=x}^{\infty} \frac{1}{(y-x)!} p^x [\theta(1-p)]^{y-x} \\ &= e^{-\theta} \frac{(\theta p)^x}{x!} \cdot e^{\theta(1-p)} \sum_{y=x}^{\infty} e^{-\theta(1-p)} \frac{[\theta(1-p)]^{y-x}}{(y-x)!} \\ &= e^{-\theta p} \cdot \frac{(\theta p)^x}{x!} \implies X \sim \text{POI}(\theta p) \end{aligned}$$

■

Example 10.2. Suppose that Y has the pdf,

$$f_Y(y) = \frac{y^{\alpha-1} \exp(-y)}{\Gamma(\alpha)}, \quad y > 0 \qquad \text{and} \qquad f(x \mid y) = y e^{-xy}, \quad x > 0, y > 0$$

So, $Y \sim \text{Gamma}(\alpha, 1)$ and $X \mid (Y = y) \sim \text{EXP}(1/Y)$. Find the marginal pdf of X.

Solution. Solve in Steps

1. **Step 1:** Find the joint pdf,
$$f(x,y) = f_Y(y)f(x \mid y) = \frac{1}{\Gamma(\alpha)}y^{\alpha}e^{-(x+1)y}, \quad x > 0, \quad y > 0$$

2. **Step 2:** Joint to marginal,
$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)dy = \int_0^{\infty} \frac{1}{\Gamma(\alpha)}y^{\alpha}e^{-(x+1)y}dy$$

3. **Method 1:** Make the substitution $t = (x+1)y$. Then
$$y = \frac{t}{x+1}, \qquad dy = \frac{1}{x+1}\,dt.$$
When $y = 0$, we have $t = 0$, and when $y \to \infty$, we have $t \to \infty$.
Thus the limits remain from 0 to ∞.
$$\begin{aligned} f_X(x) &= \int_0^{\infty} \frac{1}{\Gamma(\alpha)}y^{\alpha-1}e^{-(x+1)y}\,dy \\ &= \int_0^{\infty} \frac{1}{\Gamma(\alpha)}\left(\frac{t}{x+1}\right)^{\alpha-1}e^{-t}\cdot\frac{1}{x+1}\,dt \\ &= \frac{1}{\Gamma(\alpha)}\cdot\frac{1}{(x+1)^{\alpha-1}}\cdot\frac{1}{x+1}\int_0^{\infty} t^{\alpha-1}e^{-t}\,dt \\ &= \frac{1}{\Gamma(\alpha)(x+1)^{\alpha}}\cdot\frac{1}{x+1}\int_0^{\infty} t^{\alpha-1}e^{-t}\,dt \\ &= \frac{1}{\Gamma(\alpha)(x+1)^{\alpha+1}}\int_0^{\infty} t^{\alpha-1}e^{-t}\,dt \\ &= \frac{1}{\Gamma(\alpha)(x+1)^{\alpha+1}}\cdot\Gamma(\alpha) = \frac{1}{(x+1)^{\alpha+1}}. \end{aligned}$$
recognizing that $\int_0^{\infty} t^{\alpha-1}e^{-t}\,dt = \Gamma(\alpha)$, which leads to the simplification above.

4. **Method 2:** Notice that integrands are usually constructed via const $\times$ pdf, so, we could try and find the $\Gamma(\cdot)$ pdf for Gamma$(?,?)$. Observe from the equation,
$$\int_0^{\infty} \frac{1}{\Gamma(\alpha)}y^{\alpha}e^{-(x+1)y}dy$$
is remotely equivalent to a Gamma$(\alpha+1, \frac{1}{x+1})$ distributed pdf, so we attempt to modify our equations in a way that leads to integrating over a pdf that equates to 1.
$$\begin{aligned} &= \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)(x+1)^{\alpha+1}} \times \int_0^{\infty} \frac{(x+1)^{\alpha+1}}{\Gamma(\alpha+1)}y^{\alpha}e^{-(x+1)y}\,dy \\ &= \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)(x+1)^{\alpha+1}}\cdot\frac{(x+1)^{\alpha+1}}{\Gamma(\alpha+1)}\int_0^{\infty} y^{\alpha}e^{-(x+1)y}\,dy \\ &= \frac{1}{\Gamma(\alpha)}\int_0^{\infty} y^{\alpha}e^{-(x+1)y}\,dy \\ &= \frac{1}{\Gamma(\alpha)}\cdot\frac{\Gamma(\alpha+1)}{(x+1)^{\alpha+1}} = \frac{\alpha\,\Gamma(\alpha)}{\Gamma(\alpha)(x+1)^{\alpha+1}} = \frac{\alpha}{(x+1)^{\alpha+1}}. \end{aligned}$$

■

10.1 Conditional Expectation

The conditional pmf/pdf is also a valid pmf/pdf. Thus, we can define the conditional expectation based on the conditional pmf/pdf.

Definition 10.1. The conditional expectation of $g(Y)$ given $X = x$ is defined as,

$$\mathbb{E}[g(Y) \mid X = x] = \begin{cases} \sum_{y \in A_Y} g(y) f(y \mid x), & Y \text{ is discrete} \\ \int_{y \in A_Y} g(y) f(y \mid x) dy, & Y \text{ is continuous} \end{cases}$$

Example 10.3. Suppose that we have functions $g(y)$ and $\mathbb{E}[Y \mid X = x]$, then,

1. Conditional Variance,

$$\begin{aligned} \text{Var}(Y \mid X = x) &= \mathbb{E}[Y^2 \mid X = x] - (\mathbb{E}[Y \mid X = x])^2 \\ \text{Var}(Y) &= \mathbb{E}[Y^2] - \mathbb{E}[Y]^2 \end{aligned}$$

2. Conditional MGF,

$$\mathbb{E}[e^{tY} \mid X = x]$$

Theorem 10.2. ***Properties of Conditional Expectation***

1. *If X and Y are independent random variables,*

$$\begin{aligned} &\mathbb{E}[g(Y) \mid X = x] = \mathbb{E}[g(Y)], \quad f(y \mid x) = f_Y(y) \\ &\mathbb{E}[h(X) \mid Y = y] = \mathbb{E}[h(X)] \end{aligned}$$

2. *Suppose that $h(X, Y) \in \mathbb{R}$, then,*

$$\mathbb{E}[h(X, Y) \mid X = x] = \mathbb{E}[h(x, Y) \mid X = x] \quad \textit{since } x \textit{ is constant}$$

3. *Conditional Expectation has all the properties of expectation, i.e., if needed, you can change the order of summation and expectation if needed,*

$$\mathbb{E}[ag(Y) + bh(Y) \mid X = x] = a\mathbb{E}[g(Y) \mid X = x] + b\mathbb{E}[h(Y) \mid X = x]$$

Example 10.4. Suppose that we have random variables X and Y, then,

$$\begin{aligned} \mathbb{E}[X + Y \mid X = x] &= \mathbb{E}[x + Y \mid X = x] = x + \mathbb{E}[Y \mid X = x] \\ \mathbb{E}[XY \mid X = x] &= \mathbb{E}[xY \mid X = x] = x\mathbb{E}[Y \mid X = x] \\ \mathbb{E}[e^{XY} \mid X = x] &= \mathbb{E}[e^{xY} \mid X = x] = \int e^{xy} f(y \mid x) dy \end{aligned}$$

Example 10.5. Suppose that we have the joint pdf for X and Y,

$$f(x,y) = \begin{cases} 8xy, & 0 < y < x < 1 \\ 0, & \text{otherwise.} \end{cases}$$

Find $\mathbb{E}[X \mid Y = y]$ and $\text{Var}(X \mid Y = y)$.

Solution.

$$\begin{aligned} f_Y(y) &= \int_y^1 f(x,y)dx = \int_0^1 8xydx = 4y(1-y^2), \quad y \in (0,1) \\ f(x \mid y) &= \frac{f(x,y)}{f_Y(y)} = \frac{2x}{1-y^2}, \quad x \in (y,1) \\ \mathbb{E}[X \mid Y = y] &= \int_{x \in A_X} x^2 f(x \mid y)dx = \int_y^1 \frac{2x^3}{1-y^2}dx = \frac{1-y^4}{2(1-y^2)} \\ \mathbb{E}[X^2 \mid Y = y] &= \int_{x \in A_X} x^2 f(x \mid y)dx = \int_y^1 \frac{2x^3}{1-y^2}dx = \frac{1-y^4}{2(1-y^2)} \\ \text{Var}(X \mid Y = y) &= \mathbb{E}[X^2 \mid Y = y] - (\mathbb{E}[X \mid Y = y])^2 \\ &= \frac{1-y^4}{2(1-y^2)} - \frac{4}{9}\left(\frac{1-y^3}{1-y^2}\right)^2 \end{aligned}$$

■

Example 10.6. In questions where we know the distribution of our random variables, take $Y \sim \text{POI}(\theta)$ and $X \mid Y = y \sim \text{BIN}(y,p)$, we can find $\mathbb{E}[X \mid Y = y]$ and $\text{Var}(X \mid Y = y)$ directly using the definitions of the distribution's mean and variance.

$$\mathbb{E}[X \mid Y = y] = yp, \quad \text{Var}(X \mid Y = y) = yp(1-p)$$

10.2 Double Expectation Theorem

Theorem 10.3. ***Double Expectation Theorem*** *Suppose that we have random variables Y and X. Then, we can define $\mathbb{E}[Y \mid X = x]$ and $\mathbb{E}[Y \mid X]$.*

1. *$\mathbb{E}[Y \mid X = x]$ is a function of x and x only (lower case constant), where we can denote this value as $h(x) = \mathbb{E}[Y \mid X = x]$.*
2. *$\mathbb{E}[Y \mid X]$ where we replace x in $h(x)$ with X, then $\mathbb{E}[Y \mid X] = h(X)$ which is a random variable itself*

Remark 10.1. **Remark on $\mathbb{E}[Y \mid X]$**

1. $\mathbb{E}[Y \mid X]$ is a function of the random variable X and is a random variable.
2. $\mathbb{E}[Y \mid X]$ is a function of X and only X.

Theorem 10.4. *Suppose that Y is a random variable and $g(\cdot)$ is any real function. Then,*

$$\mathbb{E}[g(Y)] = \mathbb{E}[\mathbb{E}[g(Y) \mid X]]$$

Proof.

$$\begin{aligned}
\mathbb{E}[g(Y)] &= \int g(y) f_Y(y)\, dy \qquad \left(f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\, dx\right) \\
&= \iint g(y) f(x,y)\, dx\, dy \qquad \left(f(x,y) = f(y|x) f(x)\right) \\
&= \iint g(y) f(y|x) f(x)\, dx\, dy \\
&= \iint g(y) f(y|x) f(x)\, dy\, dx \\
&= \int f(x) \left[\int g(y) f(y|x)\, dy \right] dx \\
&= \int f(x)\, \mathbb{E}[g(Y) \mid X = x]\, dx \\
&= \int f(x) h(x)\, dx \quad \left(\text{Let } h(x) = \mathbb{E}[g(Y) \mid X = x]\right) \\
&= \mathbb{E}[h(X)] = \mathbb{E}\Big[\mathbb{E}[g(Y) \mid X]\Big].
\end{aligned}$$

Example 10.7. Suppose $Y \sim \text{Poisson}(\theta)$ and, conditional on $Y = y$, we have

$$X \mid Y = y \sim \text{Binomial}(y, p).$$

We want to compute $\mathbb{E}[X]$.

$$\mathbb{E}[X \mid Y = y] = yp \quad \Rightarrow \quad \mathbb{E}[X \mid Y] = Yp.$$

By the Double Expectation Theorem,

$$\begin{aligned}
\mathbb{E}[X] &= \mathbb{E}[\,\mathbb{E}[X \mid Y]\,] \\
&= \mathbb{E}[Yp] \\
&= p\,\mathbb{E}[Y] \\
&= \theta p.
\end{aligned}$$

11 (Cont.) Conditional Expectation

Theorem 11.1 (Variance Decomposition Formula (Law of Total Variance))**.**

$$\mathit{Var}(Y) = \mathbb{E}[\mathit{Var}(Y \mid X)] + \mathit{Var}(\mathbb{E}[Y \mid X])$$

Proof. Recall that we have that $\mathrm{Var}(Y \mid X) = \mathbb{E}[Y^2 \mid X] + (\mathbb{E}[Y \mid X])^2$

$$(1)\ E[\mathrm{Var}(Y|X)] = E\left[E[Y^2|X]\right] - E\left[(E[Y|X])^2\right]$$
$$(2)\ \mathrm{Var}\left(E[Y|X]\right) = E\left[(E[Y|X])^2\right] - (E[Y])^2$$
$$(1)+(2)\ \Rightarrow\ E[\mathrm{Var}(Y|X)] + \mathrm{Var}\left(E[Y|X]\right) = E[Y^2] - (E[Y])^2 = \mathrm{Var}(Y)$$

Interpretation: Y is the response variable, X is an explanatory variable
Aim: We wish to study the association between Y and X

(i) $\mathbb{E}[Y \mid X]$: mean of each group. $\mathbb{E}[Y \mid X = 1], \mathbb{E}[Y \mid X = 2], \mathbb{E}[Y \mid X = 3]$.

(ii) $\mathrm{Var}\{\mathbb{E}[Y \mid X]\}$: measures how much conditional means differ (*between-group variance*).

(iii) $\mathrm{Var}(Y \mid X)$: variance within each group. $\mathrm{Var}(Y \mid X = 1), \mathrm{Var}(Y \mid X = 2), \mathrm{Var}(Y \mid X = 3)$.

(iv) $\mathbb{E}[\mathrm{Var}(Y \mid X)]$: mean variance within each group (*within-group variance*).

1. $\mathrm{Var}(Y) \geq \mathbb{E}[\mathrm{Var}(Y \mid X)]$. (If $\mathbb{E}[\mathrm{Var}(Y \mid X)] = 0$, then X can predict Y perfectly.)
   Generally, using X, we can reduce the variance of Y.
2. When $X \perp\!\!\!\perp Y$, $\mathrm{Var}(\mathbb{E}[Y \mid X]) = \mathrm{Var}(\mathbb{E}[Y]) = 0$.

$$\Rightarrow \mathrm{Var}(Y) = \mathbb{E}[\mathrm{Var}(Y \mid X)]$$

If X is independent of Y, knowing X will not reduce variance (X *does not have any prediction power*).

Remark 11.1. "Total variance" = "within-group variance" + "between-group variance":

$$\mathrm{Var}(Y) = \mathbb{E}[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(\mathbb{E}[Y \mid X])\,.$$

Example 11.1. Y is a final grade, X is the grouping variable, for $X \in \{1, 2, 3\}$

Example 11.2. $Y \sim \mathrm{Poi}(\theta), \quad X \mid Y = y \sim \mathrm{Bin}(y, p)$. Find $\mathrm{Var}(X)$.

Solution. From previous lectures, $X \sim \mathrm{Poi}(\theta p)$, so $\mathrm{Var}(X) = \theta p$.

If we don't know the marginal of X, use the law of total variance:

$$\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(\mathbb{E}[X \mid Y])\,.$$

For $X \mid Y = y \sim \mathrm{Bin}(y, p)$,

$$\mathbb{E}[X \mid Y = y] = yp, \qquad \mathrm{Var}(X \mid Y = y) = yp(1-p).$$

Hence

$$\mathrm{Var}(X) = \mathbb{E}[Yp(1-p)] + \mathrm{Var}(Yp) = p(1-p)\,\mathbb{E}[Y] + p^2\,\mathrm{Var}(Y) = \theta p(1-p) + p^2\theta = p\theta.$$

■

Example 11.3. $X \sim \mathrm{Unif}(0, 1), \quad Y \mid X = x \sim \mathrm{Bin}(10, x)$. Find $\mathbb{E}(Y)$ and $\mathrm{Var}(Y)$.

Solution.

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]] = \mathbb{E}[10X] = 10 \cdot \tfrac{1}{2} = 5.$$

$$\begin{aligned}\mathrm{Var}(Y) &= \mathbb{E}[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(\mathbb{E}[Y \mid X]) \\ &= \mathbb{E}[10X(1-X)] + \mathrm{Var}(10X) \\ &= 10\,\mathbb{E}[X] - 10\,\mathbb{E}[X^2] + 100\ \mathrm{Var}(X) \\ &= 10\big(\tfrac{1}{2}\big) - 10\big(\tfrac{1}{3}\big) + 100\big(\tfrac{1}{3} - \tfrac{1}{4}\big) \\ &= 10.\end{aligned}$$

■

Remark 11.2. **Summary:**
If we know

1. the marginal distribution of X,
2. the conditional distribution of $Y \mid X = x$,

then:

$$\text{To find } \mathbb{E}[Y], \text{ use the double expectation theorem.}$$

$$\text{To find } \mathrm{Var}(Y), \text{ use the variance decomposition formula.}$$

Example 11.4. $Y \sim \mathrm{Poi}(\theta), \quad X \mid Y = y \sim \mathrm{Bin}(y, p)$. Find the mgf of X.

Solution.

$$M_X(t) = \mathbb{E}[e^{tX}] = \mathbb{E}\big[\mathbb{E}[e^{tX} \mid Y]\big].$$

$$\mathbb{E}[e^{tX} \mid Y = y] = \sum_{x=0}^{y} \binom{y}{x} p^x (1-p)^{y-x} e^{tx} = \sum_{x=0}^{y} \binom{y}{x} (pe^t)^x (1-p)^{y-x}.$$

Using $(a+b)^y = \sum_{x=0}^{y} \binom{y}{x} a^x b^{y-x}$, let $a = pe^t$ and $b = 1-p$, then

$$\mathbb{E}[e^{tX} \mid Y = y] = (pe^t + 1 - p)^y, \quad t \in \mathbb{R}.$$

Hence,

$$\begin{aligned}M_X(t) &= \mathbb{E}\big[(pe^t + 1 - p)^Y\big] \\ &= \sum_{y=0}^{\infty} e^{-\theta} \frac{\theta^y}{y!} (pe^t + 1 - p)^y \\ &= e^{-\theta} \sum_{y=0}^{\infty} \frac{\big[\theta(pe^t + 1 - p)\big]^y}{y!} \\ &= e^{-\theta} \cdot e^{\theta(pe^t + 1 - p)} \\ &= \exp\{\theta p(e^t - 1)\}.\end{aligned}$$

This again verifies that $X \sim \mathrm{Poi}(\theta p)$. ■

11.1 Joint Moment Generating Function

Definition 11.1. Bivariate and Multivariate Definitions

1. **Bivariate** X, Y are random variables, then the joint mgf is defined as,

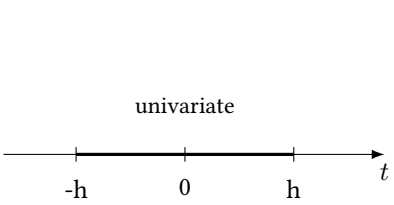
$$M(t_1, t_2) = \mathbb{E}[e^{t_1 X + t_2 Y}]$$

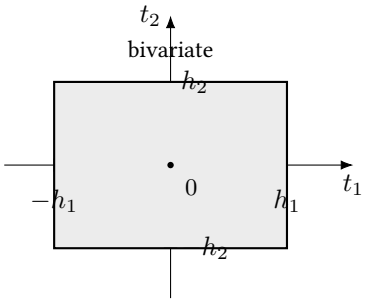
If $M(t_1, t_2)$ exists, for some $h_1 > 0$ and $h_2 > 0$, such that $|t_1| < h_1$ and $|t_2| < h_2$, then $M(0, 0) = 1$ by definition.

2. **Multivariate**: We have $X_1, \ldots, X_n$ random variables, thus we have,

$$M(t_1, \ldots, t_n) = \mathbb{E}\left[e^{\,t^T X}\right] = \mathbb{E}\left[e^{\sum_{i=1}^n t_i X_i}\right] = \mathbb{E}\left[\prod_{i=1}^n e^{t_i X_i}\right].$$

for $t^\top = [t_1, t_2, \ldots, t_n]$ and $X^\top = [X_1, X_2, \ldots, X_n]$. If M exists on $|t_i| < h_i$, then for some $h_i > 0$, we have that $i = 1, 2, \ldots, n$.





Theorem 11.2. ***Properties of the Joint MGF***

1. ***From joint mgf to marginal mgf.***

Suppose $M(t_1, t_2)$ is the joint mgf for (X, Y). Then

$$M_X(t_1) = M(t_1, 0), \quad |t_1| < h_1, \qquad M_Y(t_2) = M(0, t_2), \quad |t_2| < h_2.$$

2. ***Independence.***

$X \perp\!\!\!\perp Y \iff M(t_1, t_2) = \mathbb{E}[e^{t_1 X + t_2 Y}] = \mathbb{E}[e^{t_1 X}]\,\mathbb{E}[e^{t_2 Y}] = M_X(t_1)\, M_Y(t_2)$.

3. ***Uniqueness.***

The joint mgf and the joint distribution are one-to-one, provided the joint mgf exists.

Example 11.5. Let the joint density be $f(x,y) = e^{-y}$ on the support $0 < x < y$ (and 0 otherwise). Find the *joint* mgf $M(t_1,t_2) = \mathbb{E}[e^{t_1X+t_2Y}]$ and the *marginal* mgfs M_X, M_Y.

Solution.

$$\begin{aligned} M(t_1,t_2) &= \iint e^{t_1x+t_2y} f(x,y)\,dx\,dy = \int_0^\infty \int_0^y e^{t_1x+t_2y}\, e^{-y}\,dx\,dy \\ &= \int_0^\infty e^{(t_2-1)y}\left[\frac{e^{t_1x}}{t_1}\right]_{x=0}^{x=y} dy = \frac{1}{t_1}\int_0^\infty \left(e^{(t_1+t_2-1)y} - e^{(t_2-1)y}\right) dy \\ &\quad \text{(converges when } t_1+t_2<1 \text{ and } t_2<1\text{)} \\ &= \frac{1}{t_1}\left(\frac{1}{1-(t_1+t_2)} - \frac{1}{1-t_2}\right) = \frac{1}{(1-t_2)(1-t_1-t_2)}. \end{aligned}$$

$$M_X(t_1) = M(t_1,0) = \frac{1}{1-t_1} \qquad (t_1<1), \qquad M_Y(t_2) = M(0,t_2) = \frac{1}{(1-t_2)^2} \qquad (t_2<1).$$

So $X \sim \text{Exp}(1)$ and $Y \sim \text{Gamma}(2,1)$. ■

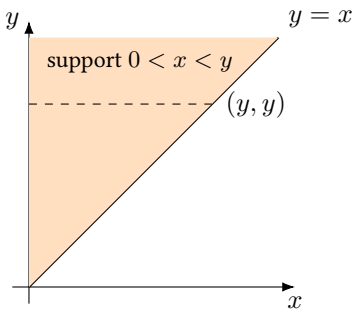


Figure 15: *
Support in the (x,y)–plane

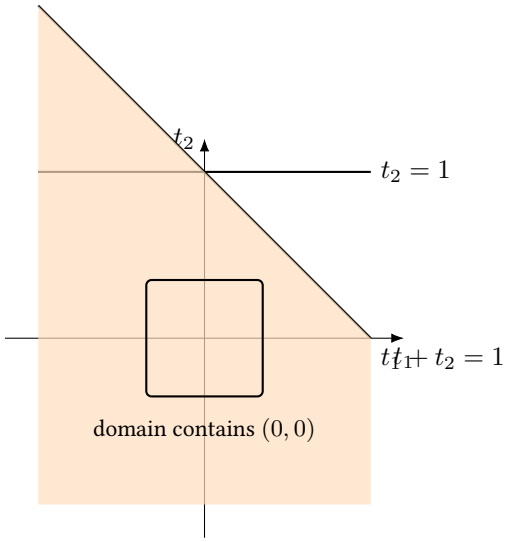


Figure 16: *
Domain for $M(t_1,t_2)$: $t_2<1,\ t_1+t_2<1$

Example 11.6. Suppose $X \sim \text{Poi}(\theta_1)$ and $Y \sim \text{Poi}(\theta_2)$, and X and Y are independent. Find the distribution of $X+Y$.

Solution.

$$\begin{aligned} M_{X+Y}(t) &= \mathbb{E}\left[e^{t(X+Y)}\right] \\ &= \mathbb{E}\left[e^{tX}\right]\cdot\mathbb{E}\left[e^{tY}\right] \quad \text{(independence)} \\ &= \exp\{\theta_1(e^t-1)\}\cdot\exp\{\theta_2(e^t-1)\} \\ &= \exp\{(\theta_1+\theta_2)(e^t-1)\} \implies X+Y \sim \text{Poi}(\theta_1+\theta_2). \end{aligned}$$

If $X_i \sim \text{Poi}(\theta_i)$ for $i = 1,\dots,n$, and X_i are independent of each other, then

$$\sum_{i=1}^n X_i \sim \text{Poi}\left(\sum_{i=1}^n \theta_i\right).$$

■

Theorem 11.3. ***Using MGF to Calculate Cross Moments***
For the bivariate joint mgf $M(t_1, t_2)$ of (X, Y):

$$\mathbb{E}[X^{k_1} Y^{k_2}] = \left. \frac{\partial^{k_1+k_2} M(t_1, t_2)}{\partial t_1^{k_1} \, \partial t_2^{k_2}} \right|_{t_1=0,\, t_2=0}.$$

$$\mathbb{E}[X] = \left. \frac{\partial M(t_1, t_2)}{\partial t_1} \right|_{t_1=0,\, t_2=0}, \qquad \mathbb{E}[XY] = \left. \frac{\partial^2 M(t_1, t_2)}{\partial t_1 \, \partial t_2} \right|_{t_1=0,\, t_2=0}.$$

More generally, for $X_1, \ldots, X_n$ with joint mgf $M(t_1, \ldots, t_n)$, let $k = k_1 + k_2 + \cdots + k_n$. Then

$$\mathbb{E}[X_1^{k_1} X_2^{k_2} \cdots X_n^{k_n}] = \left. \frac{\partial^k M(t_1, \ldots, t_n)}{\partial t_1^{k_1} \, \partial t_2^{k_2} \cdots \partial t_n^{k_n}} \right|_{t_1=0,\, t_2=0,\, \ldots,\, t_n=0}.$$

12 Multinomial Distribution

Example 12.1. If we have two boxes, both having many balls. The probability of choosing a ball from Box 1 is p. Suppose that we choose n times and get n balls, and denote the number of balls from box 1 as X. What is the distribution of X?

Solution. Say that we have a 2nd box, named Box 2, that has probability of being chosen with $1-p$. Then, let $X =$ the number of balls from Box 1. From this, we can see that $X \sim \text{BIN}(n, p)$. ■

Example 12.2. Now, let's consider the case where we have $k+1$ boxes, all of which have many balls. The probability of choosing a ball from box i is p_i. After n draws, let's denote the number of balls from box i as X_i. Let's introduce the following notation,

1. $p_i \in (0,1)$ such that $\sum_{i=1}^{k+1} p_i = 1$
2. $x_i \in \{0, 1, \ldots, n\}$ such that $\sum_{i=1}^{k+1} X_i = n$

Notice that we only need $p_1, p_2, \ldots, p_k$ since we have that $p_{k+1} = 1 - p_1 - \cdots - p_k$ and we also know that $X_1, X_2, \ldots, X_{k+1}$ are random variables representing the number of balls we pick out of box i, so we know that, $X_{k+1} = n - \cdots - x_k$.

Definition 12.1. Multinomial Definition:
Let's say we have random variables $X_1, \ldots, X_{k+1} \sim \text{Mult}(n, p_1, \ldots, p_{k+1})$, i.e., distributed by the Multinomial distribution, then we can write the pmf $f(x_1, \ldots, x_{k+1})$ as,

$$\begin{aligned} f(x_1, \ldots, x_{k+1}) &= P(X_1 = x_1, X_2 = x_2, \ldots, X_{k+1} = x_{k+1}) \\ &= \frac{n!}{x_1! x_2! \cdots x_{k+1}!} \cdot p_1^{x_1} p_2^{x_2} \cdots p_{k+1}^{x_{k+1}} \end{aligned}$$

with properties,

1. For $x_i \in \{0, 1, \ldots, n\}$ and $i = 1, 2, \ldots, k$ we have that $0 \leq \sum_{i=1}^{k} X_i \leq n$
   which implies that $0 \leq X_{k+1} \leq n$
2. For $p_i \in (0,1)$, we have that $0 \leq \sum_{i=1}^{k} p_i \leq 1$ which implies that $0 < p_{k+1} < 1$

Remark 12.1. **PMF Explanation**

- The number of combinations for the x_i balls that are from box i is represented by,

$$\frac{n!}{x_1! x_2! \cdots x_{k+1}!}$$

- The probability of each combination, is written as,

$$p_1^{x_1} p_2^{x_2} \cdots p_{k+1}^{x_{k+1}}$$

When $k = 1$, the multinomial distribution simplifies down to the **Binomial distribution**.

Theorem 12.1. ***Properties of the Multinomial***

1. *The MGF is defined as,*

$$M(t_1, t_2, \ldots, t_k) = \mathbb{E}[e^{t_1 X_2 + t_2 X_2 + \ldots t_k X_k}] = [(p_1 e^{t_1} + p_2 e^{t_2} + \ldots p_k e^{t_k} + p_{k+1})]^n$$

   Recall we can derive this using the same method of the Binomial mgf.

2. *Marginal Distribution*

   (a) *Find X_1 so we can use $M_{X_1}(t_1) = M(t_1, 0, \ldots, 0) = (p_1 e^{t_1} + 1 - p_1)^n$ which implies that $X_1 \sim BIN(n, p_1)$ as required.*

   (b) *Find (X_1, X_2) so $M_{X_1,X_2}(t_1, t_2) = M(t_1, t_2, 0, \ldots, 0) = (p_1 e^{t_1} + p_2 e^{t_2} + 1 - p_1 - p_2)^n$ which implies that $(X_1, X_2) \sim Multi(n, p_1, p_2)$ as required.*

3. *$T = X_i + X_j$ for $i \neq j$, let's take the example where $T = X_1 + X_2$,*

$$\begin{aligned} M_T(t) &= \mathbb{E}[e^{tT}] = \mathbb{E}[e^{tX_1 + tX_2}] \\ &= M(t_1, t_2, 0, \ldots, 0) \\ &= \{(p_1 + p_2)e^t + 1 - p_1 - p_2\}^n \end{aligned}$$

   so we observe that $T \sim BIN(n, p_1 + p_2)$ as required.

4. *Mean, Variance, and Covariance Suppose that we have $\mathbb{E}[X_i] = np_i$ since each X_i is distributed binomially with n and p_i. Then, the variance can be written as $Var(X_i) = np_i(1 - p_i)$ so,*

$$\begin{aligned} X_i + X_j \sim Bin(n, p_i + p_j) &\implies Var(X_i + X_j) = n(p_i + p_j) \cdot (1 - p_i - p_j) \\ Var(X_i + X_j) &= Var(X_i) + Var(X_j) + 2Cov(X_i, X_j) \\ Cov(X_i, X_j) &= \frac{1}{2}\{Var(X_i + X_j) - Var(X_i) - Var(X_j)\} \\ &= -np_i p_j > 0 \end{aligned}$$

   which is intuitive since as X_i goes up, for a fixed number of balls, X_j must go down.

5. *Conditional Distribution, for n draws, X_j balls from box j and $n - X_j$ balls from the other boxes, naturally, we want $X_i \mid X_j = x_j$ for $i \neq j$.*

$$X_i \mid X_j = x_j \quad (i \neq j)$$

$$f_{X_i|X_j}(x_i \mid x_j) = \frac{f_{X_i,X_j}(x_i, x_j)}{f_{X_j}(x_j)} = \frac{\dfrac{n!}{x_i!\, x_j!\, (n - x_i - x_j)!}\, p_i^{x_i} p_j^{x_j} (1 - p_i - p_j)^{n - x_i - x_j}}{\dfrac{n!}{x_j!\, (n - x_j)!}\, p_j^{x_j} (1 - p_j)^{n - x_j}}$$

$$= \frac{(n - x_j)!}{x_i!\, (n - x_j - x_i)!} \left(\frac{p_i}{1 - p_j}\right)^{x_i} \left(\frac{1 - p_j - p_i}{1 - p_j}\right)^{n - x_j - x_i}$$

$$\Rightarrow\ X_i \mid X_j = x_j \sim \mathrm{Bin}\left(n - x_j,\ \frac{p_i}{1 - p_j}\right).$$

6. *Let's say we want the conditional distribution* $X_i \mid X_j + X_i = t \sim$ *BIN*$(t, p_i/(p_i + p_j))$

$$X_i \mid (X_j + X_i = t) \sim \text{Bin}\left(t, \frac{p_i}{p_i + p_j}\right)$$

Verify using definition of conditional distribution:

$$\Pr(X_i = x_i \mid X_j + X_i = t) = \frac{\Pr(X_i = x_i,\, X_j + X_i = t)}{\Pr(X_j + X_i = t)} = \frac{\Pr(X_i = x_i,\, X_j = t - x_i)}{\Pr(X_j + X_i = t)}$$

$$= \frac{\dfrac{n!}{x_i!\,(t-x_i)!\,(n-t)!}\, p_i^{x_i} p_j^{t-x_i} (1 - p_i - p_j)^{\,n-t}}{\dfrac{n!}{t!\,(n-t)!}\,(p_i + p_j)^t (1 - p_i - p_j)^{\,n-t}}$$

$$= \frac{t!}{x_i!\,(t-x_i)!} \left(\frac{p_i}{p_i + p_j}\right)^{x_i} \left(\frac{p_j}{p_i + p_j}\right)^{t-x_i}.$$

Remark 12.2. Let's consider as an example where, let a success be considered as when we get a ball from box i or j for $i \neq j$, and a failure be defined as anything otherwise. Then, $T = X_i + X_j$ can be interpreted as the number of successes out of the n draws where $p = p_i + p_j$ is the total probability, so $T \sim \text{BIN}(n, p_i + p_j)$.

12.1 Bivariate Normal Distribution

Definition 12.2. Bivariate Normal Distribution
Suppose that we have a random variable matrix $X^\top(X_1, X_2)$ which follows a bivariate normal distribution if its joint pdf is, for $-\infty < x_1, x_2 < \infty$, we have,

$$f(x_1, x_2) = \frac{1}{2\pi|\Sigma|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x} - \mu)^\top \Sigma^{-1} (\mathbf{x} - \mu)\right\}$$

Remark 12.3. **Notation for BVN**

Notation: $X \sim \text{BVN}(\mu, \Sigma)$

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}, \quad |\Sigma| : \text{determinant of } \Sigma$$

Σ is positive definite $\iff \forall\, y \neq 0,\ y^\top \Sigma y > 0$

$|\Sigma| = (1 - \rho^2)\,\sigma_1^2\sigma_2^2 > 0 \ \Rightarrow\ |\rho| < 1.$

Remark 12.4. The BVN can be expanded as,

$$f(x_1, x_2) = \frac{1}{2\pi\,\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)}\left\{\frac{(x_1 - \mu_1)^2}{\sigma_1^2} + \frac{(x_2 - \mu_2)^2}{\sigma_2^2} - \frac{2\rho\,(x_1 - \mu_1)(x_2 - \mu_2)}{\sigma_1\sigma_2}\right\}\right].$$

1. **Joint mgf**

$$\begin{aligned} M_{(t_1,t_2)} &= \exp\big(\mu^\top t + \tfrac{1}{2} t^\top \Sigma t\big) \quad (\, t = [t_1\ t_2]^\top \in \mathbb{R}^2\,) \\ &= \exp\big(t_1\mu_1 + t_2\mu_2 + \tfrac{1}{2}t_1^2\sigma_1^2 + \tfrac{1}{2}t_2^2\sigma_2^2 + t_1t_2\rho\sigma_1\sigma_2\big) \end{aligned}$$

2. **Marginal dist of X_1 or X_2**

$$M_{X_1}(t_1) = M_{(t_1,0)} = \exp\big(t_1\mu_1 + \tfrac{1}{2}t_1^2\sigma_1^2\big) \quad \text{(mgf of univariate normal)} \Rightarrow \quad X_1 \sim N(\mu_1, \sigma_1^2)$$

Similarly, $X_2 \sim N(\mu_2, \sigma_2^2)$

$$\mu_1 = \mathbb{E}[X_1], \qquad \sigma_1^2 = \mathrm{Var}(X_1)$$
$$\mu_2 = \mathbb{E}[X_2], \qquad \sigma_2^2 = \mathrm{Var}(X_2)$$

3. **Conditional distribution** *use def. of conditional dist.*

$$X_2 \mid X_1 = x_1 \sim N\left(\mu_2 + \rho\sigma_2\frac{x_1-\mu_1}{\sigma_1}, (1-\rho^2)\sigma_2^2\right)$$
$$X_1 \mid X_2 = x_2 \sim N\left(\mu_1 + \rho\sigma_1\frac{x_2-\mu_2}{\sigma_2}, (1-\rho^2)\sigma_1^2\right)$$

calculate: $f(x_2 \mid x_1) = \dfrac{f(x_1,x_2)}{f(x_1)}$ and $f(x_1 \mid x_2) = \dfrac{f(x_1,x_2)}{f(x_2)}$

4. **Covariance**

$$\begin{aligned} \mathrm{Cov}(X_1,X_2) &= \mathbb{E}[X_1X_2] - \mathbb{E}[X_1]\mathbb{E}[X_2] \\ \mathbb{E}[X_1X_2] &= \mathbb{E}\big[\mathbb{E}[X_1X_2 \mid X_1]\big] \quad \text{(Double expectation)} \\ &= \mathbb{E}\big[X_1\,\mathbb{E}[X_2 \mid X_1]\big] \\ &= \mathbb{E}\big[X_1(\mu_2 + \rho\sigma_2\tfrac{X_1-\mu_1}{\sigma_1})\big] \\ &= \mu_2\mathbb{E}[X_1] + \rho\tfrac{\sigma_2}{\sigma_1}\mathbb{E}[X_1^2] - \rho\mu_1\tfrac{\sigma_2}{\sigma_1}\mathbb{E}[X_1] \\ &= \mu_1\mu_2 + \rho\sigma_1\sigma_2 \quad \Rightarrow \quad \mathrm{Cov}(X_1,X_2) = \rho\sigma_1\sigma_2 \end{aligned}$$

Meaning of the parameters:

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathrm{BVN}(\mu,\Sigma)\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

$$\mu = \mathbb{E}[X], \quad \mu_1 = \mathbb{E}[X_1], \quad \mu_2 = \mathbb{E}[X_2] \implies \Sigma = \begin{pmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1,X_2) \\ \mathrm{Cov}(X_1,X_2) & \mathrm{Var}(X_2) \end{pmatrix} \quad : \text{covariance matrix.}$$

$$\sigma_1^2 = \mathrm{Var}(X_1), \quad \sigma_2^2 = \mathrm{Var}(X_2) \implies \rho = \frac{\mathrm{Cov}(X_1,X_2)}{\sqrt{\mathrm{Var}(X_1)\,\mathrm{Var}(X_2)}} = \rho(X_1,X_2) \quad : \text{correlation coefficient.}$$

1. Independence. Recall that generally, for two random variables $X_1 \perp X_2$, we have that

$$X_1 \perp X_2 \implies \text{Cov}(X_1, X_2) = \rho(X_1, X_2) = 0$$

However, if X_1 and X_2 are normally distribution then,

$$\text{Cov}(X_1, X_2) = \rho(X_1, X_2) = 0 \iff X_1 \perp X_2$$

Solution. Method 1
If we have that $\rho(X_1, X_2) \iff f(x_1, x_2)$ and we know that,

$$\begin{aligned} f(x_1, x_2) &= \frac{1}{2\pi\sigma_1\sigma_2} \exp\left\{-\frac{1}{2}\frac{(x_1-\mu_1)^2}{\sigma_1^2} - \frac{1}{2}\frac{(x_2-\mu_2)^2}{\sigma_2^2}\right\} \\ &= \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left\{-\frac{1}{2}\frac{(x_1-\mu_1)^2}{\sigma_1^2}\right\} \cdot \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left\{-\frac{1}{2}\frac{(x_2-\mu_2)^2}{\sigma_2^2}\right\} \\ &= f(x_1)f(x_2) \implies X_1 \perp X_2 \end{aligned}$$

■

Solution. Method 2
Since we know that $\rho = 0 \iff M(t_1, t_2)$, then we can use the MGFs,

$$\begin{aligned} M(t_1, t_2) &= \exp\left(t_1\mu_1 + t_2\mu_2 + \frac{1}{2}t_1^2\sigma_1^2 + \frac{1}{2}t_2\sigma_2^2\right) \\ &= \exp\left(t_1\mu_1 + \frac{1}{2}t_1^2\sigma_1^2\right) \cdot \exp\left(t_2\mu_2 + \frac{1}{2}t_2^2\sigma_2^2\right) \\ &= M_{X_1}(t_1) \cdot M_{X_2}(t_2) \iff X_1 \perp X_2 \end{aligned}$$

■

2. Linear Transformations. Suppose that we have $X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim \text{BVN}(\mu, \Sigma)$

(a) $c = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$ then we know that $c^\top X = c_1X_1 + c_2X_2 \sim N(c^\top\mu, c^\top\Sigma)$

$$\mathbb{E}[c^\top X] = c^\top\mathbb{E}[X] = c^\top\mu, \qquad \text{Var}(c^\top X) = c^\top\text{Var}(X) \cdot c = c^\top\text{Cov}(X)c = c^\top\Sigma c$$

(b) Given an $A_{2\times2}$ matrix, and a column vector $b_{2\times1}$, then we know that,

$$\begin{aligned} &AX + b \sim \text{BVN}(A\mu + b, A\Sigma A^\top) \\ \implies &\mathbb{E}[AX + b] = A\mathbb{E}[X] + b = A\mu + b \\ \implies &\text{Var}[AX + b] = A\text{Var}(X)A^\top = A\Sigma A^\top \end{aligned}$$

3. Given that $Z \sim N(0,1)$, find the distribution of Z^2. From STAT 231, we know that $Z^2 \sim \chi^2_1$. However, to be more rigorous, we provide a more explicit proof using the MGFs.

   Proof.

$$\begin{aligned}
M_{Z^2}(t) &= \mathbb{E}[e^{tz^2}] \\
&= \int_{-\infty}^{\infty} e^{tz^2} \cdot \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{z^2}{2}} dz, \qquad 1-2t>0 \implies t < \frac{1}{2} \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(1-2t)z^2} dz \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2} \cdot \frac{z^2}{(1-2t)^{-1}}\right\} dz, \qquad \text{Let } \sigma^2 = (1-2t)^{-1} \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp \underbrace{\left\{-\frac{z^2}{2\sigma^2}\right\}}_{\text{part of pdf of } N(0,\sigma^2)} dz, \qquad \text{Gamma}(\alpha,\beta) : (1-\beta t)^{-2} \\
&= \frac{\sqrt{2\pi\sigma^2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{z^2}{2\sigma^2}\right\} dz \\
&= \sigma = (1-2t)^{-1/2}
\end{aligned}$$

   Thus by Gamma distribution, we see that $Z^2 \sim \text{Gamma}(\alpha,\beta) = \chi^2_1$, as required.

4. Z_1 and Z_2 are independent standard normal variables. Find the distribution of $Z_1^2 + Z_2^2$ (i.e., the sum of 2 independent χ^2 distribution). Note that, $Z_1 \perp Z_2 \implies Z_1^2 \perp Z_2^2$, i.e., the square of two chi-squared random variables are also independent.

$$\begin{aligned}
M_{Z_1^2} + M_{Z_2^2}(t) &= \mathbb{E}[e^{tZ_1^2 + tZ_2^2}] \\
&= \mathbb{E}[e^{tZ_1^2}]\mathbb{E}[e^{tZ_2^2}] \\
&= (1-2t)^{-1/2} \cdot (1-2t)^{-1/2} \\
&= (1-2t)^{-1}
\end{aligned}$$

   Thus, we have that $Z_1^2 + Z_2^2 \sim \text{Gamma}(1,2) = \chi^2_2$ with 2 degrees of freedom.

   Remark 12.5. Note that we can also extend this to the multivariate case where for $Z_1, \ldots, Z_n \sim N(0,1)$ (iid random variables), we have that, $Z_1^2 + \cdots + Z_n^2$ are also distributed by a $\text{Gamma}(n/2, 2) = \chi^2_n$ because $M_{Z_1^2 + \cdots Z_n^2}(t) = (1-2t)^{-n/2}$.

5. Suppose that $X \sim \text{BVN}(\mu, \Sigma)$. Find the distribution of $(X-\mu)^\top \Sigma^{-1}(X-\mu)$. Note that if X is a scalar then $X \sim N(\mu, \sigma^2)$, and thus we have,

$$(X-\mu)\frac{1}{\sigma^2}(X-\mu) = \left(\frac{X-\mu}{\sigma}\right)^2 \sim \chi_2 \qquad \frac{X-\mu}{\sigma} \sim N(0,1)$$

   If X is bivariate, similarly, we know that $(X-\mu)^\top \Sigma^{-1}(X-\mu) \sim \chi^2_2$. In order to prove this we require the help of two lemmas.

   Remark 12.6. If Σ is positive definite, then there exists a matrix A which is non-singular such that $\Sigma = AA^\top$

   Remark 12.7. If $X \sim \text{BVN}(\mu, \Sigma)$ then by the first lemme we know that $\Sigma = AA^\top$ for some matrix A and thus X and $AZ + \mu$ have the same distribution where $Z = [Z_1 \quad Z_2] = \text{BVN}([0 \quad 0], \Sigma)$ where $\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

$$\begin{aligned}
(X-\mu)^\top \Sigma^{-1}(X-\mu) &\sim (AZ+\mu-\mu)^\top \Sigma^{-1}(AZ+\mu-\mu) \\
&= (AZ)^\top (AA^\top)^{-1}(AZ) \\
&= Z^\top A^\top (A^\top)^{-1} A^{-1} AZ \\
&= Z^\top Z = [Z_1 \quad Z_2] \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} = Z_1^2 + Z_2^2 \sim \chi^2_2
\end{aligned}$$

12.2 Function of Random Variables

Given $X_1, \ldots, X_n$ which are usually continuous random variables, we would like to determine the distribution of $Y = h(X_1, \ldots, X_n)$. We will discuss three methods,

- CDF Method
- One-to-One transformation
- MGF Method

12.2.1 CDF Method

- Find the cdf of Y
- Find the pdf ot Y by

$$g(y) = G'(y)$$

Example 12.3. X and Y are random variables with joint pdf given by,

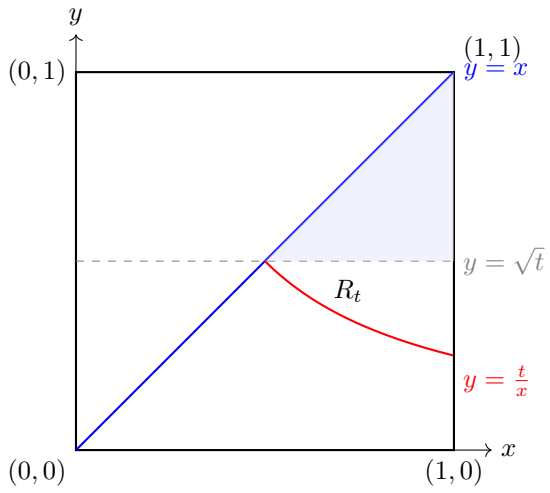
$$f(x, y) = 3y, \quad 0x \leq y < 1$$

Find the pdf of $T = XY$.

Solution. From the above, note that $T = XY \implies Y = T/X$ so we have that,

$$\begin{aligned} G_T(t) &= P(T \leq t) = P(XY \leq t) \\ &= 1 - P(XY > t) \\ &= 1 - \int_{\sqrt{t}}^{1} \int_{t/y}^{y} 3y dx dy \\ &= 1 - \int_{\sqrt{t}}^{1} (3y^2 - 3t) dy = 3t - 2t^{3/2}, \quad t \in [0, 1] \\ \implies g_T(t) &= G_T'(t) = 3 - 3t^{1/2}, \quad t \in [0, 1] \end{aligned}$$

■

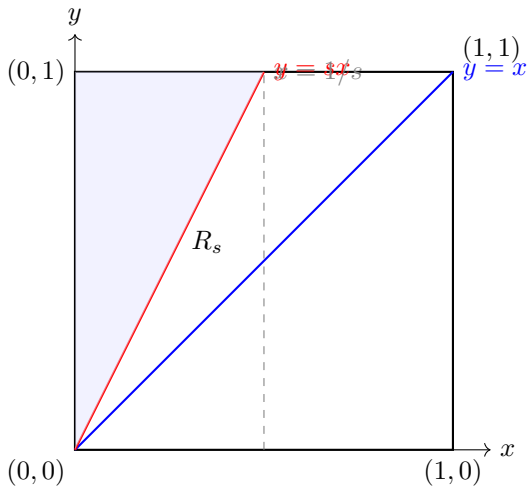


Example 12.4. Suppose that now we have that $S = \frac{Y}{X}$. Find the pdf.

Solution.

$$\begin{aligned}
G_S(s) &= P\left(\frac{Y}{X} \le s\right) = \iint_{\{0<x<y<1,\, y\le sx\}} 3y\,dx\,dy \\
&= \int_0^{1/s}\int_x^{sx} 3y\,dy\,dx + \int_{1/s}^1\int_x^1 3y\,dy\,dx \\
&= \int_0^{1/s} \frac{3}{2}\left((sx)^2 - x^2\right)dx + \int_{1/s}^1 \frac{3}{2}\left(1-x^2\right)dx \\
&= \frac{1}{2}\frac{s^2-1}{s^3} + \left(1 - \frac{3}{2s} + \frac{1}{2s^3}\right) \\
&= 1 - \frac{1}{s}, \quad s \in [1,\infty) \\
\implies g_S(s) &= G_S'(s) = \frac{1}{s^2}, \quad s \in [1,\infty)
\end{aligned}$$

■



Example 12.5. Suppose that we have the order statistics $X_1, \ldots, X_n$ such that $X_{(n)} = \max\{X_1, \ldots, X_n\}$ and $X_{(1)} = \min\{X_1, \ldots, X_n\}$. Find the pdf of $X_{(n)}$ and $X_{(1)}$ such that $X_1, \ldots, X_n$ are iid Uniformly distributed around 0, θ for $\theta > 0$.

Solution.

$$\begin{aligned}
G_{X_{(n)}}(y) &= P(X_{(n)} \le y) = P\{X_{(n)} \le y\} = P\{X_1 \le y,\, X_2 \le y,\, \cdots,\, X_n \le y\} \\
&= P(X_1 \le y,\, X_2 \le y,\, \cdots,\, X_n \le y) \\
&= \prod_{i=1}^n P(X_i \le y) \\
&= \prod_{i=1}^n \left(\frac{y}{\theta}\right) = \left(\frac{y}{\theta}\right)^n, \quad y \in [0,\theta] \\
\implies g_{X_{(n)}}(y) &= G_{X_{(n)}}'(y) = n\frac{y^{n-1}}{\theta^n}, \quad y \in [0,\theta]
\end{aligned}$$

$$\begin{aligned}
G_{X_{(1)}}(y) &= P(X_{(1)} \le y) \\
&= 1 - P(X_{(1)} > y) = 1 - P\{X_{(1)} > y\} = 1 - P\{X_1 > y,\, X_2 > y,\, \cdots,\, X_n > y\} \\
&= 1 - \prod_{i=1}^{n} P(X_i > y) \\
&= 1 - \left(1 - \frac{y}{\theta}\right)^n, \quad y \in [0, \theta] \\
\implies g_{X_{(1)}}(y) &= G'_{X_{(1)}}(y) = \frac{n}{\theta}\left(1 - \frac{y}{\theta}\right)^{n-1}, \quad y \in [0, \theta]
\end{aligned}$$

■

Example 12.6. Suppose that $X_1, \ldots, X_n \sim \text{EXP}(1)$ iid random variables.
Find the pdf of $X_{(n)}$ and $X_{(1)}$.

Solution.

$$\begin{aligned}
G_{X_{(n)}}(y) &= P(X_{(n)} \le y) = P(X_1 \le y,\, X_2 \le y,\, \cdots,\, X_n \le y) \\
&= \prod_{i=1}^{n} P(X_i \le y) = \left(1 - e^{-y}\right)^n, \quad y \in [0, \infty) \\
\implies g_{X_{(n)}}(y) &= G'_{X_{(n)}}(y) = n\left(1 - e^{-y}\right)^{n-1} e^{-y}, \quad y \in [0, \infty)
\end{aligned}$$

$$\begin{aligned}
G_{X_{(1)}}(y) &= P(X_{(1)} \le y) = 1 - P(X_{(1)} > y) = 1 - P(X_1 > y,\, X_2 > y,\, \cdots,\, X_n > y) \\
&= 1 - \prod_{i=1}^{n} P(X_i > y) = 1 - \left(e^{-y}\right)^n = 1 - e^{-ny}, \quad y \in [0, \infty) \\
\implies g_{X_{(1)}}(y) &= G'_{X_{(1)}}(y) = ne^{-ny}, \quad y \in [0, \infty)
\end{aligned}$$

■

12.3 CDF Method

Example 12.7. **Demonstrations of the CDF Method**

1. Suppose that $Z \sim \mathcal{N}(0, 1)$. Find the pdf of $Y = Z^2$.
   Solution.

$$\begin{aligned}
G_Y(y) &= P(Y \le y) = P(Z^2 \le y) \\
&= P(-\sqrt{y} \le Z \le \sqrt{y}) \\
&= F_Z(\sqrt{y}) - F_Z(-\sqrt{y}) \\
g_Y(y) &= G'_Y(y) \\
&= F'_Z(\sqrt{y}) \cdot \frac{1}{2} y^{-\frac{1}{2}} + F'_Z(-\sqrt{y}) \cdot \frac{1}{2} y^{-\frac{1}{2}} \\
&= [f_Z(\sqrt{y}) + f_Z(-\sqrt{y})] \cdot \frac{1}{2} y^{-\frac{1}{2}} \\
&= \frac{1}{\sqrt{2\pi}} e^{-\frac{y}{2}} \cdot 2 \cdot \frac{1}{2} y^{-\frac{1}{2}} \\
&= -\frac{1}{\sqrt{2\pi}} e^{-\frac{y}{2}} \cdot 2 \cdot \frac{1}{2} \cdot y^{-\frac{1}{2}} \\
&= \frac{1}{\sqrt{2\pi}} y^{-\frac{1}{2}} \cdot e^{-\frac{1}{2}y}, \qquad y > 0 \\
\implies Y &\sim \text{Gamma}(1/2, 2) = \chi^2_1
\end{aligned}$$

■

2. Suppose that we have a random variable X with pdf given by,

$$f(x) = \frac{\theta}{x^{\theta+1}}, \qquad x \geq 1, \theta > 0$$

Find the pdf of $Y = \log X$ for $x \geq 1$ and $\theta > 0$.

Solution. Note that the support of Y is $y \in [0, +\infty)$ so we have that,

$$\begin{aligned} G_Y(y) &= P(Y \leq y) = P(\log X \leq y) = P(X \leq e^y), \qquad y > 0 \\ &= \int_1^{e^y} \frac{\theta}{x^{\theta+1}} dx \\ &= -x^{-\theta}]_1^{e^y} = 1 - e^{-\theta y} \\ g_Y(y) &= G_Y'(y) \\ &= \theta e^{-\theta y} \qquad y \geq 0 \\ \implies & Y \sim \text{EXP}(1/\theta) \end{aligned}$$

■

12.4 One-to-one Transformation Method

Suppose that X is a continuous random variable with pdf $f(x)$ and $Y = h(X)$ with $h(\cdot)$ is a continuous one to one function (i.e., monotonic one-to-one function). Then we can write $X = h^{-1}(Y)$. Find the pdf of Y. Note that the pdf of Y can be written using $g(y)$ as,

$$g(y) = f(x) \left| \frac{dx}{dy} \right|$$

where $f(x) = f(h^{-1}(y))$ is a function of y and,

$$\left| \frac{dx}{dy} \right| := \text{absolute value of } \frac{dx}{dy} \implies \left| \frac{dx}{dy} \right| = \frac{dh^{-1}(y)}{dy}$$

where the 2nd line is a function of y. Note that we can also write $g(y)$ as,

$$g(y) = f(x) \left| \frac{dx}{dy} \right| = f(h^{-1}(y)) \left| \frac{dh^{-1}(y)}{dy} \right|$$

is a function of y. We will now prove this.

Proof. For this proof, note that there are two cases: (1) if $h(\cdot)$ is increasing and (2) if $h(\cdot)$ is decreasing. Note that we are dealing with monotonically increasing/decreasing functions here so these 2 cases suffice.

1. If $h(\cdot)$ is increasing, we have that

$$\begin{aligned} & G_Y(y) = P(Y \leq y) = P(h(X) \leq y) = P(x \leq h^{-1}(y)) = F_X(h^{-1}(y)) \\ \implies & g_Y(y) = G_Y'(y) = F_X'(h^{-1}(y))(h^{-1}(y))' \\ & = f(h^{-1}(y)) \cdot \frac{dh^{-1}(y)}{dy} = f(x) \cdot \frac{dx}{dy} \end{aligned}$$

2. If $h(\cdot)$ is decreasing, we have that

$$\begin{aligned} G_Y(y) &= P(Y \leq y) = P(h(X) \leq y) \\ &= P(X \geq h^{-1}(y)) = 1 - P(X < h'(Y)) \\ &= 1 - F_X(h^{-1}(y)) = 1 - F_X(h^{-1}(y)) \\ \implies g_Y(y) &= G_Y'(y) = -F_X'(h^{-1}(y))(h^{-1}(y))' \\ &= -f(h^{-1}(y)) \cdot \frac{dh^{-1}(y)}{dy} = f(x) \cdot \left(-\frac{dx}{dy} \right) = f(x) \cdot \left| \frac{dx}{dy} \right| \end{aligned}$$

Remark 12.8. Recall that from calculus, if we have that $h(h^{-1})(y) = y$, then this can be re-written as,

$$h'(h^{-1}(y))(h^{-1}(y))' = 1 \implies (h^{-1}(y))' = \frac{1}{h'(h^{-1}(y))}$$

Example 12.8. Suppose that we are given a cdf $F(\cdot)$ and we are told to generate an random variable that has the cdf $F(\cdot)$ on our computer? Then we can generate $U \sim U(0,1)$ first then get $F^{-1}(U)$.

1. **Probability Integral Transformation**
   If X is a continuous random variable with pdf $f(x)$ and cdf $F(x)$, find the distribution of $Y = F(X)$ which can also be written $X = F^{-1}(Y)$. Then the support of Y is $y \in [0,1]$ so,

$$\begin{aligned} g(y) &= f(x)\left|\frac{dx}{dy}\right| \\ \left|\frac{dx}{dy}\right| &= \left|\frac{dF^{-1}(y)}{dy}\right| = \left|\frac{1}{F'(F^{-1}(y))}\right| \\ &= \left|\frac{1}{f(F^{-1}(y))}\right| = \left|\frac{1}{f(X)}\right| = \left|\frac{1}{f(X)}\right| = \frac{1}{f(X)} \\ g(y) &= f(x)\cdot\frac{1}{f(x)} = 1, \qquad y \in [0,1] \end{aligned}$$

   So we see that $Y \sim U(0,1)$ regardless of the distribution of X.

2. However, when $\frac{dx}{dy}$ is difficult to get, we see that

$$\begin{aligned} 1 &= \frac{dx}{dy}\cdot\frac{dx}{dy} \\ g(y) &= f(x)\left|\frac{dx}{dy}\right| = f(x)\cdot\left|\frac{1}{\frac{dy}{dx}}\right| = f(x)\left|\frac{1}{f(x)}\right| = 1 \end{aligned}$$

3. Suppose that $X \sim U(0,1)$. Find the pdf of $Y = -\log X$. Note that the support of Y is given by $[0,+\infty)$ so we get that,

$$\begin{aligned} g(y) &= f(x)\left|\frac{dx}{dy}\right| = 1\cdot\left|\frac{de^{-y}}{dy}\right| \\ &= e^{-y} = 1\cdot\left|\frac{1}{\frac{dy}{dx}}\right| = 1\cdot\left|\frac{1}{-\frac{1}{\lambda}}\right| = x = e^{-y}, \qquad y \geq 0 \end{aligned}$$

12.5 One-to-One Bivariate Transformation Method

Suppose that X, Y is a joint pdf with $f(x,y)$. Then, there is a one-to-one transformation given by,

$$\begin{cases} U = h_1(X,Y), \\ Y = h_2(X,Y) \end{cases}$$

where $h_1(\cdot)$ and $h_2(\cdot)$ are known functions. Thus, we wish to find the joint pdf of (U,V) denoted as $g(U,V)$.

1. Find the inverse of the transformation

$$\begin{cases} U = h_1(X,Y), \\ V = h_2(X,Y) \end{cases} \iff \exists g_1, g_2, \text{ s.t. } \begin{cases} X = g_1(U,V), \\ Y = g_2(U,V) \end{cases}$$

   i.e., we solve for (x,y) from the cases on the left.

2. Get the Jacobian. Recall that the Jacobian is the partial derivative matrix given as,

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix} \iff \left|\frac{\partial(u,v)}{\partial(x,y)}\right|, \qquad |\cdot| \text{ is the determinant.}$$

   Then we can use $J_1 = \left|\frac{\partial(u,v)}{\partial(x,y)}\right|$ denotes the Jacobian to get,

$$\begin{cases} \frac{\partial(x,y)}{\partial(u,v)} = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} \\ J_2 = \left|\frac{\partial(x,y)}{\partial(u,v)}\right| \end{cases}$$

   Note that if J_1 is difficult to get then we can use the fact that $J_1 \cdot J_2 = 1$.

3. Using the formula, we can get

$$\begin{aligned} g(u,v) &= f(x,y)\,|J_2|\,, \quad |\cdot| \text{ is the determinant.} \\ &= f(g_1(u,v), g_2(u,v)) \cdot |J_2| \\ &= \underbrace{f(g_1(u,v), g_2(u,v))}_{\text{functions of } u,\, v} \cdot \frac{1}{|\underbrace{\quad J_1 \quad}_{\text{function of } u,\, v}|} \end{aligned}$$

Example 12.9. Suppose that X and Y are continuous random variable with joint pdf $f(x,y) = \exp(-x-y)$ for $x \in (0,\infty)$ and $y \in (0,\infty)$. Let $U = X + Y$ and $V = X$. Find the joint pdf of U and V and the marginal pdf of U.

Solution.

$$\begin{cases} U = X + Y \\ V = X \end{cases} \implies \begin{cases} X = V \\ Y = U - X = U - V \end{cases}$$

Note that we can also use the Jacobians J_1 and J_2 corresponding to (u,v) and (x,y) to get,

$$\begin{aligned} J_1 = \left|\frac{\partial(u,v)}{\partial(x,y)}\right| &= \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = 1 \cdot 0 - 1 \cdot 1 = -1 \\ J_2 = \left|\frac{\partial(x,y)}{\partial(u,v)}\right| &= \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = 0 \cdot (-1) - 1 \cdot 1 = -1 \\ g(u,v) &= f(x,y)|J_2| \\ &= e^{-x-y} = e^{-v-(u-v)} = e^{-u} \end{aligned}$$

Note that since $x = v > 0$ and $y = u - v > 0$ then we can conclude that $0 < v < u$ is our support. Then, integrating we get $g(v)$ written to be as,

$$g(u) = \int_{-\infty}^{\infty} g(u,v)dv = \int_0^u e^{-u}dv = ue^{-u}, \quad u > 0$$

■

Example 12.10. X and Y are independent standard normal random variables. $U = X + Y$ and $V = X - Y$ find the joint pdf of (U,V). Hint: use the one-to-one bivariate transformation.

Solution.

$$\begin{cases} U = X + Y \\ V = X - Y \end{cases} \implies \begin{cases} X = \frac{1}{2}(U+V) \\ Y = \frac{1}{2}(U-V) \end{cases}$$

$$J_1 = \left|\frac{\partial(u,v)}{\partial(x,y)}\right| = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = 1(-1) - 1(1) = -2.$$

$$J_2 = \left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix} = \left(\tfrac{1}{2}\right)\left(-\tfrac{1}{2}\right) - \left(\tfrac{1}{2}\right)\left(\tfrac{1}{2}\right) = -\tfrac{1}{2}.$$

Note that $J_1 \cdot J_2 = 1$. Now we can compute $g(u,v)$ using the formula,

$$\begin{aligned} g(u,v) = f(x,y)\,|J_2| \quad \text{or} \quad f(x,y) &= \frac{g(u,v)}{|J_2|} \\ &= \tfrac{1}{2} f(x,y) \\ &= \tfrac{1}{2} f\big(\tfrac{1}{2}(u+v), \tfrac{1}{2}(u-v)\big) \\ &= \tfrac{1}{2} \cdot \tfrac{1}{2\pi} \exp\big\{-\tfrac{1}{2}\left[\tfrac{1}{4}(u+v)^2 + \tfrac{1}{4}(u-v)^2\right]\big\} \\ &= \tfrac{1}{4\pi} \exp\big\{-\tfrac{1}{4}(u^2+v^2)\big\}, \quad u,v \in \mathbb{R}. \end{aligned}$$

Alternatively, we can use the properties of the normal random variables,

$$\begin{bmatrix} U \\ V \end{bmatrix} = \begin{bmatrix} X+Y \\ X-Y \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = AZ$$

$$A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad Z = \begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathrm{BVN}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) = \mathrm{BVN}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

$$\begin{pmatrix} U \\ V \end{pmatrix} = AZ \sim \mathrm{BVN}(A\boldsymbol{\mu},\, A\boldsymbol{\Sigma}A^\top) = \mathrm{BVN}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \right).$$

■

12.6 Non-one-to-one Transformations

1. If we can partition the support of (X,Y) into $A = A_1 \cup \cdots \cup A_K$, where $A_i \cap A_j = \varnothing$ for $i \neq j$, and on each A_i, the transformation is one-to-one.
2. Then, the pdf of (U,V) is
$$g(u,v) = \sum_{i=1}^{K} f(x,y)\,|J_{2i}| \quad \text{or} \quad \sum_{i=1}^{K} f(x,y)\,\frac{1}{|J_{2i}|}.$$
The inverse transformations on $A_1, \ldots, A_K$ are different.

Example 12.11.

$$X_1 \sim N(0,1),\ X_2 \sim N(0,1),\ X_1 \perp X_2$$

$$\begin{cases} Y_1 = X_1^2 + X_2^2 \in [0, +\infty) \\ Y_2 = \dfrac{X_1}{\sqrt{X_1^2 + X_2^2}} \in [-1,1] \end{cases}$$

$$X_1 = Y_2\sqrt{Y_1}, \quad X_2^2 = Y_1 - X_1^2 = Y_1(1-Y_2^2) \Rightarrow X_2 = \pm\sqrt{Y_1(1-Y_2^2)}$$

Support for (X_1, X_2) : $A = \mathbb{R} \times \mathbb{R}$

$$A = A_1 \cup A_2, \quad A_1 = \mathbb{R} \times [0, +\infty), \quad A_2 = \mathbb{R} \times (-\infty, 0)$$

Solution. Step 1 (on A_1):

$$X_1 = \sqrt{Y_1}\,Y_2, \quad X_2 = \sqrt{Y_1(1-Y_2^2)}$$

Step 2:

$$J_{12} = \left| \frac{\partial(x_1,x_2)}{\partial(y_1,y_2)} \right| = \begin{vmatrix} \frac{1}{2}y_1^{-1/2}y_2 & y_1^{1/2} \\ \frac{1}{2}y_1^{-1/2}(1-y_2^2)^{1/2} & -y_1^{1/2}y_2(1-y_2^2)^{-1/2} \end{vmatrix} = -\tfrac{1}{2}(1-y_2^2)^{-1/2}$$

$$J_{11} = \frac{1}{J_{12}} = 2(1-y_2^2)^{1/2}$$

Step 3:

$$f(x_1,x_2)|J_{12}| = f(\sqrt{y_1}y_2, \sqrt{y_1(1-y_2^2)}) \cdot \tfrac{1}{2}(1-y_2^2)^{-1/2}$$

Step 1 (on A_2):

$$X_1 = \sqrt{Y_1}\,Y_2, \quad X_2 = -\sqrt{Y_1(1-Y_2^2)}$$

Step 2:

$$J_{22} = \left|\frac{\partial(x_1,x_2)}{\partial(y_1,y_2)}\right| = \begin{vmatrix} \frac{1}{2}y_1^{-1/2}y_2 & y_1^{1/2} \\ -\frac{1}{2}y_1^{-1/2}(1-y_2^2)^{1/2} & y_1^{1/2}y_2(1-y_2^2)^{-1/2} \end{vmatrix} = \tfrac{1}{2}(1-y_2^2)^{-1/2}$$

$$J_{21} = \frac{1}{J_{22}} = 2(1-y_2^2)^{1/2}$$

Step 3:

$$f(x_1,x_2)|J_{22}| = f(\sqrt{y_1}y_2, -\sqrt{y_1(1-y_2^2)}) \cdot \tfrac{1}{2}(1-y_2^2)^{-1/2}$$

$$\begin{aligned} \Rightarrow\ g(y_1,y_2) &= \tfrac{1}{2}(1-y_2^2)^{-\frac{1}{2}}\Big[f(\sqrt{y_1}y_2, \sqrt{y_1(1-y_2^2)}) + f(\sqrt{y_1}y_2, -\sqrt{y_1(1-y_2^2)})\Big] \\ &= \tfrac{1}{2\pi}\cdot\tfrac{1}{\sqrt{1-y_2^2}}\, e^{-\frac{1}{2}y_1}, \quad (y_1,y_2) \in [0,+\infty)\times[-1,1]. \end{aligned}$$

■

12.7 MGF Technique

1. Find the MGF of a Random Variables
2. Use the Uniqueness of the MGF to find the distribution of the new random variable

Remark 12.9. Suppose that $X_1,\ldots,X_n$ are independent, then $T = \sum_{i=1}^n X_i$ has the MGF given by,

$$\begin{aligned} M_T(t) &= \mathbb{E}\left[\exp\left(t\sum_{i=1}^n X_i\right)\right] \\ &= \mathbb{E}\left[\prod_{i=1}^n \exp(tX_i)\right] \\ &= \prod_{i=1}^n \mathbb{E}[\exp(tX_i)] = \prod_{i=1}^n M_{X_i}(t) \end{aligned}$$

Further more, if $X_1,\ldots,X_n$ are i.i.d., then, $T = \sum_{i=1}^n X_i$ has the MGF,

$$M_{X_1}(t_1) = \cdots = M_{X_n}(t) \implies M_T(t) = \prod_{i=1}^n M_{X_i}(t) = \{M_{X_1}(t)\}^n.$$

12.8 Normal Distribution

1. If $X \sim N(\mu,\sigma^2)$ then $aX+b \sim N(a\mu+b, a^2\sigma^2)$

$$\begin{aligned} M_Y(t) &= \mathbb{E}\big[e^{tY}\big] = \mathbb{E}\Big[e^{t(aX+b)}\Big] \\ &= e^{bt}\,\mathbb{E}\big[e^{atX}\big] \\ &= e^{bt}\,M_X(at) \\ &= \exp\big\{\tfrac{1}{2}a^2t^2\sigma^2 + (a\mu+b)t\big\} \\ \implies Y &\sim N(a\mu+b, a^2\sigma^2) \quad M_X(t) = \exp\left(\frac{1}{2}\sigma^2t^2 + \mu t\right) \end{aligned}$$

2. If $X_i \sim N(\mu_i, \sigma_i^2)$ for $i = 1, \ldots, n$ and $X_1, \ldots, X_n$ are independent, then, $\sum_{i=1}^{n} a_i X_i \sim N\left(\sum_{i=1}^{n} a_i \mu_i, \sum_{i=1}^{n} a_i^2 \sigma_i^2\right)$

$$\begin{aligned} Y &= \sum_{i=1}^{n} a_i X_i \\ M_Y(t) &= \prod_{i=1}^{n} M_{a_i X_i}(t) = \prod_{i=1}^{n} M_{X_i}(a_i t) \\ &= \prod_{i=1}^{n} \exp\left(\tfrac{1}{2} a_i^2 \sigma_i^2 t^2 + a_i \mu_i t\right) \\ &= \exp\left\{ \tfrac{1}{2} \left(\sum_{i=1}^{n} a_i^2 \sigma_i^2 \right) t^2 + \left(\sum_{i=1}^{n} a_i \mu_i \right) t \right\} \\ \implies Y &\sim N\left(\sum_{i=1}^{n} a_i \mu_i,\ \sum_{i=1}^{n} a_i^2 \sigma_i^2 \right) \end{aligned}$$

3. If $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ are i.i.d. Then, for $Y = \sum_{i=1}^{n} X_i$

$$\begin{aligned} M_Y(t) &= \{M_{X_1}(t)\}^n \\ &= \exp\left(\tfrac{1}{2} n \sigma^2 t^2 + n\mu t\right) \implies Y \sim N(n\mu,\, n\sigma^2) \\ \overline{X}_n &= n^{-1} \sum_{i=1}^{n} X_i = n^{-1} Y \implies \overline{X}_n \sim N\left(\mu,\, \tfrac{\sigma^2}{n}\right) \end{aligned}$$

12.9 Some Important Distributions

1. $Z \sim N(0, 1)$ then the distribution of Z^2,

$$Z^2 \sim \chi_1^2 \iff \text{Gamma}(\frac{1}{2}, 2)$$

Proved via method 1 (mgf method) or method 2 (cdf method). Alternatively we can show that,

$$Z_1^2 + \cdots + Z_n^2 \sim \chi_n^2 \iff \text{Gamma}(n/2, 2)$$

where $Z_1, \ldots, Z_n \sim N(0, 1)$

2. If $Y_i \chi_k^2$ and $Y_1, \ldots, Y_n$ are independent. The distribution $\sum_{i=1}^{n} Y_i$ can written as,

$$\sum_{i=1}^{n} Y_i \sim \chi^2_{k_1 + k_2 + \cdots + k_n}$$

which is the sum of independent random variables.
We can use the MGF method to prove this result.

3. If $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ are iid then,

$$\sum_{i=1}^{n} \left(\frac{X_i - \mu}{\sigma} \right)^2 \sim \chi_n^2 \iff \frac{X_i - \mu}{\sigma} \chi N(0, 1) \iff \left(\frac{X_i - \mu}{\sigma} \right)^2 \sim \chi_1^2$$

12.10 t distribution

Z, Y are r.v.s

$$\begin{cases} (1)\ Z \sim N(0,1) \\ (2)\ Y \sim \chi^2_n \\ (3)\ Z \perp Y \end{cases} \implies T = \frac{Z}{\sqrt{Y/n}} \sim t_n$$

t distribution with df = n

12.10.1 How to find the pdf ot T?

$$\begin{cases} T = \dfrac{Z}{\sqrt{Y/n}} \\ V = Y \end{cases} \implies \begin{cases} Z = T\sqrt{V/n} \\ Y = V \end{cases}$$

1. Use one-to-one transformation to get the pdf $f_{T,V}(t, v)$.
2. Get $f_T(t)$ by integrating: $f_T(t) = \int f_{T,V}(t, v)\, dv$.

12.11 F Distribution

X, Y are r.v.s

$$\begin{cases} (1)\ X \sim \chi^2_n \\ (2)\ Y \sim \chi^2_m \\ (3)\ X \perp Y \end{cases} \implies \frac{X/n}{Y/m} \sim F_{n,m}$$

F distribution with numerator df$= n$ and denominator df$= m$.

Question: If $T \sim t_m$, what is the distribution of T^2?

$$T^2 \sim F_{1,m} \quad \text{because}$$

$$T = \frac{Z}{\sqrt{Y/m}}, \quad \text{where} \begin{cases} Z \sim N(0,1) \\ Y \sim \chi^2_m \\ Z \perp Y \end{cases}$$

$$T^2 = \frac{Z^2}{Y/m} = \frac{Z^2/1}{Y/m}, \quad \text{where} \begin{cases} Z^2 \sim \chi^2_1 \\ Y \sim \chi^2_m \\ Z^2 \perp Y \end{cases} \implies T^2 \sim F_{1,m}.$$

12.12 Distributional Results from Normal Samples

Consider the iid sampled sample $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$

12.13 Sample Mean and Sample Variance

1. Definitions
   (a) Sample mean
   $$\bar{X_n} = \frac{1}{n}\sum_{i=1}^{n} X_i$$
   (b) Sample Variance
   $$S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(\bar{X_i} - \bar{X})^2$$
   Note that both of the above are Random Variables.
2. Random or Non-random
   (a) Sample mean and sample variance are random variables
   (b) Mean and Variance are deterministic non-random values
3. Estimator/Estimate/Parameter
   (a) $X_1, \ldots, X_n$ for $\bar{X_n}$ is an Estimator which is random
   (b) $X_1 = 1, X_2 = -1, X_3 = 3, \ldots$ with
   $$\bar{X}_n = \frac{1 + (-1) + 3}{n} = 2.2$$
   is an Estimate which is a realized estimator and is non-random.
   (c) $\mu = E[X]$ is a capital X which is a parameter and is fixed, non-random.
4. Distributional Results

- $\bar{X}_n \perp S_n^2$
- $\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \sim N(0,1)$ because $\bar{X}_n \sim N(\mu, \sigma^2/n)$
- $\frac{(n-1)S_n^2}{\sigma^2} \sim \chi^2_{n-1}$

12.14 Important Distributional Results for Asymptotics

Recall that from STAT231 we have a couple of distributional results. Moreover, recall that $S_n = \sqrt{S_n^2}$ is an estimator for σ. With this, we can derive the following,

1. $\dfrac{\bar{X}_n - \mu}{S_n/\sqrt{n}} \approx \dfrac{(\bar{X}_n - \mu)/(\sigma\sqrt{n})}{S_n/\sigma} \sim t_{n-1}$ which follows because

$$\begin{aligned} \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} &\sim N(0,1) \\ \frac{S_n}{\sigma} = \sqrt{\frac{S_n^2}{\sigma^2}} &= \sqrt{\frac{1}{n-1} \cdot \frac{(n-1)S_n^2}{\sigma^2}} \sim \sqrt{\frac{\chi^2_{n-1}}{n-1}} \\ \frac{\bar{X}_n - \mu}{S_n/\sqrt{n}} &\perp \frac{S_n}{\sigma} \\ \implies \frac{\bar{X}_n - \mu}{S_n/\sqrt{n}} &\sim t_{n-1} \end{aligned}$$

   which is the null hypothesis $H_0 : \mu = \mu_0$ which is known under

$$H_0 : \frac{\bar{X}_n - \mu_0}{S_n/\sqrt{n}} \sim t_{n-1}$$

2. $X_1, \ldots, X_n \sim N(\mu_1, \sigma_1^2)$ and $Y_1, \ldots, Y_m \sim N(\mu_2, \sigma_2^2)$ are independent.

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} = \frac{\frac{\sum_{i=1}^n (X_i - \bar{X}_m)}{\sigma_1^2}/n-1}{\frac{\sum_{i=1}^m (Y_i - \bar{Y}_m)}{\sigma_2^2}/m-1} \sim \frac{\chi^2_{n-1}/n-1}{\chi^2_{m-1}/m-1} \sim F_{n-1,m-1}$$

   under the null hypothesis that $H_0 = \dfrac{S_1^2}{S_2^2} = \dfrac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{n-1,m-1}$.

13 Introduction to Asymptotics

So far our focus is on how to describe the behaviours of random variables.

1. PDF, PMF and CDF describe the random variables behaviours from a population level (i.e., the underlying truth)
2. Parameters, including the mean, variance, and correlation coefficient summarize the distribution.

We almost always assume that we have access to the true distributions of the random variables. But this is not the case in real-world applications.

1. In real-world applications, we are given a dataset
   but we do not know the true distribution and true parameters
2. We treat the dataset as realizations $X = x$ of some random variables X
3. We use functions $h(x)$ of the data to estimate parameters
4. If we replace x with in estimate $h(x)$ with random variable X then we call $h(X)$ an estimator.
5. $h(X)$ is a random variable so $h(X)$ has a distribution and we want to get this distribution so that we can perform statistical inference.
6. In some cases, the behaviour of $h(X)$ is easy to describe. For example, if we assume that $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$, we have that,

$$\bar{X}_n \sim N(\mu, \sigma^2/n).$$

7. In some cases, the finite sample behaviour of the estimator is hard to get. For example, if we have that $X_1, \dots, X_n \sim \text{Uniform}(0, \theta)$ it is complicated to get $\bar{X}_n \sim$? However, its limiting distribution is easy to get when the sample size $n \to \infty$. More notably, $\bar{X}_n$ will get closer to a normal distribution as $n \to \infty$ by CLT. We call this normal distribution the asymptotic distribution.

8. Asymptotic statistics aims to answer the question of what the distribution of estimators is when $n \to \infty$ both through probability and distribution.

Remark 13.1. Statistical inference relies on the distribution of $h(X)$

- Hypothesis tests
- Confidence intervals

13.1 Convergence in Distribution

Definition 13.1. Convergence in Distribution. Suppose we have a sequence of random variables $X_1, \dots, X_n$ and the cdf of X_i is denoted $F_i(x)$. Let X be a random variable with cdf $F(x)$, then if,

$$\lim_{n\to\infty} F_n(x) = F(x)$$

where $F(x)$ is a valid cdf, for all x at which $F(x)$ is continuous, then we say that the sequence $\{X_i\}$ converges to X in distribution. We denote the convergence in distribution as,

$$X_n \xrightarrow{d} X.$$

Remark 13.2.

1. Sequence of Random Variables

$$Y_1, \dots, Y_n \sim N(\mu, \sigma^2) \iff n = 1, \bar{X}_1 = Y_1; \quad n = 2, \bar{X}_2 = \frac{Y_2 + Y_2}{2}; \quad n = 3, X_3 = \frac{Y_1 + Y_2 + Y_3}{3} \cdots$$

   We call the above the sample mean of an increasing sample.

2. $F(\cdot)$ is called the *Limiting Distribution* (limit) or asymptotic distribution of $\{X_i\}$.

3. For all x at which $F(x)$ is continuous implies that we must have,

$$F(X) = \begin{cases} 1 & x \geq 0 \\ 0 & x < a \end{cases} \iff F_n(X) = \begin{cases} 1 & x > a + \frac{1}{n} \\ n(x-a) & a < x \leq a + \frac{1}{n} \\ 0 & x \leq a \end{cases} \iff \lim_{n\to\infty} F_n(X) = \begin{cases} 1 & x > a \\ 0 & x \leq a \end{cases}.$$

   We also call $\xrightarrow{d}$ *Weak Convergence* because it is only about the distribution and not about the random variable. Moreover, we can also impose a couple remarks on the limit for $F_n(X)$ namely,

   (i) $\lim_{n\to\infty} F_n(X)$ is not a CDF

   (ii) $\lim_{n\to\infty} F_n(X)$ except for $x = a$ where $F(X)$ is not continuous

   (iii) We can still say $X_n \xrightarrow{d} X$

4. The definition of $\xrightarrow{d}$ is applicable for both continuous and discrete random variables.

   Example 13.1. Suppose that we have random variables $X, Y \sim N(0, 1)$ where $X_1 = Y, X_2 = -Y, X_3 = Y, \dots$. Then we have that $X_n \xrightarrow{d} X$ because $F_n(X) = F(X)$ for all $n \in \mathbb{N}$. But, observe that X_i and X can be very different which is the rationale for calling it weak convergence.

Example 13.2. Suppose that we have a sequence of random variables $X_1, \ldots, X_n \sim \text{Uniform}(0,1)$. Let $X_{(1)}$ and $X_{(n)}$ be the minimal and maximum of $X_1, \ldots, X_n$. Find the limiting distribution of,

1. $nX_{(1)}$ and $n(1 - X_{(n)})$

   Solution. We have that $nX_{(1)} \in (0, n)$ and,

$$F_n(X) = P(nX_{(1)} \le x) = \begin{cases} 1 & x \ge n \\ P(X_{(1)} \le x/n) & 0 < x < n \\ 0 & x \le 0 \end{cases}$$

$$\begin{aligned} P\left(X_{(1)} \le \frac{x}{n}\right) &= 1 - P\left(X_{(1)} > \frac{x}{n}\right) \\ &= 1 - \left\{P(X_1 > \frac{x}{n})\right\}^n \\ &= 1 - \left(1 - \frac{x}{n}\right)^n \\ \lim_{n\to\infty} P\left(X_{(1)} \le \frac{x}{n}\right) &= 1 - \lim_{n\to\infty}\left(1 - \frac{x}{n}\right)^n = 1 - e^{-x}, \\ \implies \lim_{n\to\infty} F_n(X) &= \begin{cases} 1 - e^{-x} & x \ge 0 \\ 0 & x \le 0 \end{cases} \end{aligned}$$

   Then for $X \sim \text{EXP}(1)$ then $F(x)$ equals the line above so $\lim_{n\to\infty} F_n(X) = F(X)$ for all $x \in \mathbb{R}$ so $nX_{(1)} \xrightarrow{d} X$ where $X \sim \text{EXP}(1)$. ■

2. $X_{(1)}$ and $X_{(n)}$

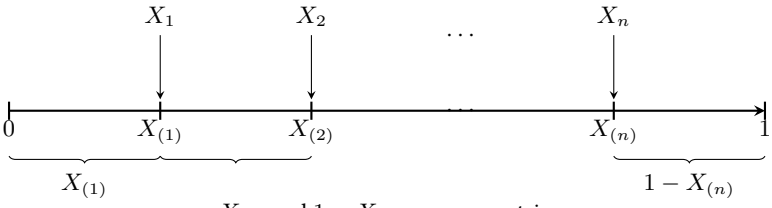
   Solution.

$$\begin{aligned} F_n(X) &= P\left(n(1 - X_{(n)} \le x\right) \\ &= \begin{cases} 1 & x \ge n \\ P\left(X_{(n)} \ge 1 - \frac{x}{n}\right) & 0 < x < n \\ 0 & x \le 0 \end{cases} \\ P\left(X_{(n)} \ge 1 - \frac{x}{n}\right) &= 1 - P\left(X_{(n)} < 1 - \frac{x}{n}\right) \\ &= 1 - \left\{P\left(X_1 < 1 - \frac{x}{n}\right)\right\}^n \\ &= 1 - \left(1 - \frac{x}{n}\right)^n \implies 1 - e^{-x} \quad \text{as} \quad n \to \infty. \end{aligned}$$

   Therefore, we have that,

$$\lim_{n\to\infty} F_n(X) = \begin{cases} 1 - e^{-x} & x > 0 \\ 0 & x \le 0 \end{cases} \implies n(1 - X_{(n)}) \xrightarrow{d} X$$

   where $X \sim \text{EXP}(1)$. ■



$X_{(1)}$ and $1 - X_{(n)}$ are symmetric.

3. $X_{(1)} \in (0,1)$. We first find the cdf F_n,

$$F_n(X) = P(X_{(1)} \leq x) = \begin{cases} 1 & x \geq 0 \\ 1-(1-x)^n & 0 < x < 1 \\ 0 & x \leq 0 \end{cases}$$

Then, we get the limit,

$$\lim_{n\to\infty} F_n(X) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases} \iff \lim_{n\to\infty} (1-x)^n = 0 \quad \text{for } x \in (0,1).$$

Clearly, we can see that F_n is not a cdf but $F(X) = 1$ for $x \geq 0$ and 0 otherwise is a cdf. Thus we can write that,

$$\lim_{n\to\infty} F_n(X) = F(X) \implies X_{(1)} \xrightarrow{d} X$$

except for $x = 0$ and where $X \sim F(X)$. Thus we say that $P(X = 0) = 1$ or $X = 0$ almost surely.

4. $X_n \in (0,1)$

Solution. We can again, first find the cdf F_n,

$$F_n(X) = P(X_{(n)} \leq x) = \begin{cases} 1 & x \geq 0 \\ x^n & 0 < x < 1 \\ 0 & x \leq 0 \end{cases}.$$

Then, we can take the limit and get,

$$\lim_{n\to\infty} F_n(X) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases} \implies \lim_{n\to\infty} x^n = 0 \quad x \in (0,1)$$

So we can conclude that,

$$F(X) = \begin{cases} 1 & x \geq 1 \\ 0 & x < 1 \end{cases} \implies \lim_{n\to\infty} F_n(X) = F(X) \quad x \in \mathbb{R} \iff X_{(n)} \xrightarrow{d} X, \quad X \sim F(X).$$

so therefore $P(X = 1) = 1$ or $X = 1$ almost surely. ■

Example 13.3 (Convergence in Distribution (Cont.)). Continuous sequence of random variables X_n with range $\mathbb{R}$ and cdf

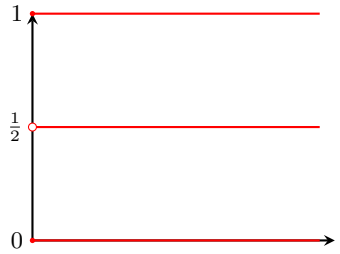
$$F_n(x) = \frac{\exp(nx)}{1+\exp(nx)}, \quad x \in \mathbb{R}.$$

Find the limiting distribution.

$$F_n(x) = \frac{1}{1+\exp(-nx)}, \quad x \in \mathbb{R}.$$

$$\lim_{n\to\infty} F_n(x) = \begin{cases} 1, & x > 0 \\ \frac{1}{2}, & x = 0 \\ 0, & x < 0 \end{cases}$$

Not a cdf, not a problem.



Let

$$F(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

be a valid cdf.

$$\lim_{n\to\infty} F_n(x) = F(x) \quad \text{except for } x = 0,$$

where $F(x)$ is not continuous.

$$\implies X_n \xrightarrow{d} X, \quad \text{where } X \sim F(x),$$

so that $P(X = 0) = 1$ or $X = 0$ almost surely.

Convergence in distribution only imposes requirements on the cdfs. It **does not directly address the random variable** (which is also a function).

Example 13.4. Two random variables may be different but have the same cdf. Let $X \sim \text{Uniform}(0,1)$ and define $Y = 1 - X$.

$$X \neq Y \text{ a.s.} \quad \left(P(X \neq Y) = P(X \neq \tfrac{1}{2}) = 1\right)$$

$$F_X(t) = t, \quad t \in (0,1)$$

$$F_Y(t) = P(Y \leq t) = P(1 - X \leq t) = P(X \geq 1 - t) = t, \quad t \in (0,1)$$

Thus, X and Y have the same distribution.

$F(\cdot)$ is a cdf,	X is a random variable, a function
$F(t) = P(X \leq t)$,	$X(\omega)$, ω is an outcome
$F : \mathbb{R} \to [0,1]$.	$X : \Omega \to \mathbb{R}$, Ω is the sample space.

Convergence in distribution only addresses convergence at the cdf level. The previous example shows that two random variables can be very different while having the same cdf. **Convergence in probability** directly addresses convergence at the random variable level.

13.2 Convergence in Probability

Definition 13.2. Let $X_1, \ldots, X_n$ be a sequence of random variables, and let X be a random variable. We say that the sequence $X_1, \ldots, X_n$ converges to X in probability if

$$\lim_{n\to\infty} P(|X_n - X| \geq \varepsilon) = 0$$

for any $\varepsilon > 0$.

Remark 13.3.

$$X_n \xrightarrow{p} X \quad \text{as } n \to \infty.$$

Remark 13.4.

1. $$\{|X_n - X| \geq \varepsilon\} = \{\omega : |X_n(\omega) - X(\omega)| \geq \varepsilon\}$$
   Event of "X_n and X are very different"
   $$|X_n(\omega) - X(\omega)| \geq \varepsilon$$
2. "In probability" means that
   $$P(X_n \text{ and } X \text{ are very different}) \to 0 \quad \text{as } n \to \infty.$$
   The threshold ε can be very small.

13.3 Convergence to a Constant

Theorem 13.1.

1. $X_n \xrightarrow{p} X \implies X_n \xrightarrow{d} X.$
2. $X_n \xrightarrow{d} X \not\Rightarrow X_n \xrightarrow{p} X.$

Example 13.5. Let $X \sim N(0,1)$ and $Y = -X \sim N(0,1)$. Define $Z_1 = Y, Z_2 = Y, \ldots, Z_n = Y, \ldots$.
Then $Z_n \xrightarrow{d} X$.

$$\begin{aligned} P(|Z_n - X| \geq \varepsilon) &= P(|Y - X| \geq \varepsilon) \\ &= P(|2X| \geq \varepsilon) \\ &= P(|X| \geq \tfrac{\varepsilon}{2}) \not\to 0 \quad \text{as } n \to \infty, \end{aligned}$$

where $\frac{\varepsilon}{2}$ is a positive constant.

$$Z_n \xrightarrow{p} X.$$

Proof & Implication.

1. $X_n \xrightarrow{p} b \implies X_n \xrightarrow{d} b$ (Use previous results)
2. $X_n \xrightarrow{d} b \implies X_n \xrightarrow{p} b$

$$\begin{aligned} P(|X_n - b| \geq \varepsilon) &= P(X_n \geq b + \varepsilon) + P(X_n \leq b - \varepsilon) \\ &= 1 - P(X_n < b + \varepsilon) + P(X_n \leq b - \varepsilon) \\ &\leq 1 - P(X_n < b + \tfrac{\varepsilon}{2}) + P(X_n \leq b - \varepsilon) \\ &= 1 - F_n(b + \tfrac{\varepsilon}{2}) + F_n(b - \varepsilon) \to 0, \quad n \to \infty. \end{aligned}$$

$$A = \{X_n < b + \varepsilon\}, \quad B = \{X_n \leq b + \tfrac{\varepsilon}{2}\}.$$

If $\omega \in B$, then $\omega \in A$.

$$\Rightarrow B \subseteq A \quad \Rightarrow P(B) \leq P(A).$$

$$\lim_{n\to\infty} F_n(b + \tfrac{\varepsilon}{2}) = 1, \qquad \lim_{n\to\infty} F_n(b - \varepsilon) = 0$$

$$\implies 0 \le \lim_{n\to\infty} P(|X_n - b| \ge \varepsilon) \le 0 \implies \lim_{n\to\infty} P(|X_n - b| \ge \varepsilon) = 0.$$

Example 13.6. Suppose there is an i.i.d. sample $X_1, \dots, X_n \sim \text{Uniform}(0,1)$. Let $X_{(1)} = \min\{X_1, \dots, X_n\}$ and $X_{(n)} = \max\{X_1, \dots, X_n\}$. Prove that $X_{(1)} \xrightarrow{p} 0$ and $X_{(n)} \xrightarrow{p} 1$.

$$X_{(1)} \xrightarrow{d} X, \text{ where } X = 0 \text{ a.s.} \iff X_{(1)} \xrightarrow{p} 0$$

$$X_{(n)} \xrightarrow{d} X, \text{ where } X = 1 \text{ a.s.} \iff X_{(n)} \xrightarrow{p} 1$$

Now, let $X_1, \dots, X_n$ be i.i.d. with pdf

$$f(x) = \exp\{-(x-\theta)\}, \quad x > \theta,$$

and cdf

$$F(x) = 1 - \exp\{-(x-\theta)\}, \quad x > \theta.$$

Prove that $X_{(1)} \xrightarrow{p} \theta$.

Method 1: Use definition of $\xrightarrow{p}$.

$$\begin{aligned} P(|X_{(1)} - \theta| \ge \varepsilon) &= P(X_{(1)} \ge \theta + \varepsilon) + P(X_{(1)} \le \theta - \varepsilon) \\ &= P(X_{(1)} \ge \theta + \varepsilon) \quad = 0 \text{ because } X_{(1)} > \theta. \\ &= P(X_1 \ge \theta + \varepsilon, \dots, X_n \ge \theta + \varepsilon) \\ &= \{P(X_1 \ge \theta + \varepsilon)\}^n \\ &= (e^{-\varepsilon})^n = e^{-n\varepsilon} \to 0, \quad n \to \infty. \end{aligned}$$

Therefore,

$$\lim_{n\to\infty} P(|X_{(1)} - \theta| \ge \varepsilon) = 0.$$

$$\implies X_{(1)} \xrightarrow{p} \theta.$$

Method 2: Since θ is constant,

$$\xrightarrow{d} \Rightarrow \xrightarrow{p}.$$

$$F_n(x) = P(X_{(1)} \le x) = \begin{cases} 1 - P(X_{(1)} > x), & x > \theta \\ 0, & x \le \theta \end{cases} \iff \left(P(X_{(1)} > x) = \exp\{-n(x-\theta)\}\right)$$

$$F_n(x) = \begin{cases} 1 - \exp\{-n(x-\theta)\}, & x > \theta \\ 0, & x \le \theta \end{cases} \implies \lim_{n\to\infty} F_n(x) = \begin{cases} 1, & x > \theta \\ 0, & x \le \theta \end{cases}$$

Let

$$F(x) = \begin{cases} 1, & x \ge \theta \\ 0, & x < \theta \end{cases}$$

Then

$$\lim_{n\to\infty} F_n(x) = F(x) \text{ except for } x = \theta.$$

$$X_n \xrightarrow{d} X, \quad X \sim F(x).$$

$$X = \theta \text{ a.s. or } P(X = \theta) = 1 \implies X_n \xrightarrow{d} \theta \Rightarrow X_n \xrightarrow{p} \theta.$$

13.4 Markov Inequality

Theorem 13.2 (Markov Inequality)**.**

1. *Suppose $X \geq 0$. Then,*

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}, \quad a > 0.$$

Proof:

$$\begin{aligned}\mathbb{E}[X] &= \int_0^\infty x f(x)\,dx \geq \int_a^\infty x f(x)\,dx \\ &\geq \int_a^\infty a f(x)\,dx = a\int_a^\infty f(x)\,dx = aP(X \geq a)\end{aligned}$$

$$\implies P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

2. *Suppose $X \geq 0$. Then, for any $k = 1, 2, \ldots,$*

$$P(X \geq a) \leq \frac{\mathbb{E}[X^k]}{a^k}, \quad a > 0.$$

Proof:

$$P(X \geq a) = P(X^k \geq a^k) \leq \frac{\mathbb{E}[X^k]}{a^k}$$

use (1)

3. *For a random variable $X \in \mathbb{R}$,*

$$P(|X| \geq a) \leq \frac{\mathbb{E}[|X|^k]}{a^k}, \quad a > 0,\ k = 1, 2, \ldots$$

Proof: *use (2)*

Useful in proving convergence in probability because:

$$P(|X_n - X| \geq \varepsilon) \leq \frac{\mathbb{E}[|X_n - X|^k]}{\varepsilon^k}.$$

Convergence in probability holds as long as you can show

$$\lim_{n\to\infty} \mathbb{E}[|X_n - X|^k] = 0.$$

4. *For a random variable $X \in \mathbb{R}$,*

$$P(|X - \mu| \geq a) \leq \frac{\mathbb{E}[(X-\mu)^2]}{a^2} = \frac{\mathrm{Var}(X)}{a^2},$$

where $\mu = \mathbb{E}[X]$.

Provides an upper bound on the probability of deviation of a random variable from its expectation.

13.5 Chernoff Bounds

Let $X \geq 0$ be a random variable and $a > 0$. Then for all $t > 0$,

$$P(X \geq a) = P(e^{tX} \geq e^{ta}) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}} = \frac{M_X(t)}{e^{ta}},$$

$$\implies P(X \geq a) \leq \inf_{t>0} \frac{M_X(t)}{e^{ta}} \quad \text{(Bound on the right tail)}.$$

Similarly, for all $t < 0$,

$$P(X \leq a) = P(e^{tX} \geq e^{ta}) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}} = \frac{M_X(t)}{e^{ta}},$$

$$\implies P(X \leq a) \leq \inf_{t<0} \frac{M_X(t)}{e^{ta}} \quad \text{(Bound on the left tail)}.$$

Upper bound on the tail of a random variable based on its MGF.

Example 13.7. Let $Y \sim \text{Bin}(n, p)$ and $\mu = \mathbb{E}[Y] = np$. Then,

$$P(Y \geq (1+\delta)\mu) \leq \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu}.$$

Proof.

$$P(Y \geq (1+\delta)\mu) \leq M_Y(t) \cdot e^{-t(1+\delta)\mu}, \quad \forall t > 0.$$

$$M_Y(t) = (pe^t + 1 - p)^n = \left\{1 + p(e^t - 1)\right\}^n \leq e^{p(e^t-1)n} = \exp\{\mu(e^t - 1)\}.$$

$$\implies P(Y \geq (1+\delta)\mu) \leq \exp\{\mu(e^t - 1) - t(1+\delta)\mu\}, \quad \forall t > 0.$$

$$\inf_{t>0} \exp\{\mu(e^t - 1 - t(1+\delta))\} = \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu} \quad \text{when } t = \log(1+\delta).$$

$f(t) = e^t - (1+\delta)t, \quad f'(t) = e^t - (1+\delta) = 0 \implies t = \log(1+\delta).$

13.6 Markov Inequality - WLLN

1. Suppose $X_1, \ldots, X_n$ are independent, $\mathbb{E}[X_i] = \mu$, and $\text{Var}(X_i) = \sigma^2 < \infty$. Then,

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{p} \mu.$$

Proof:

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sum_{i=1}^n \text{Var}(X_i)}{n^2\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \to 0 \quad \text{as } n \to \infty.$$

$$\implies \lim_{n\to\infty} P(|\bar{X}_n - \mu| \geq \varepsilon) = 0.$$

Hence, $\bar{X}_n \xrightarrow{p} \mu$.

2. (*Other version*) Suppose $X_1, \ldots, X_n$ are independent with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma_i^2 \leq c$ for some constant $c < \infty$. Then, the Weak Law of Large Numbers (WLLN) still holds:

$$P(|\bar{X}_n - \mu| \geq \varepsilon) = \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sum_{i=1}^n \text{Var}(X_i)}{n^2\varepsilon^2} \leq \frac{nc}{n^2\varepsilon^2} = \frac{c}{n\varepsilon^2} \to 0.$$

Therefore, $\bar{X}_n \xrightarrow{p} \mu$.

Example 13.8.

1.
$$X_1, \dots, X_n \text{ i.i.d. } \chi^2(1) \implies \bar{X}_n \xrightarrow{p} \mathbb{E}[X_i] = 1.$$
$$Y_n = \sum_{i=1}^{n} X_i^2 \implies \frac{Y_n}{n} \xrightarrow{p} 1.$$

   Because
$$Y_n = X_1^2 + \cdots + X_n^2, \quad X_i \text{ i.i.d. } \chi^2(1),$$
   we have
$$\frac{Y_n}{n} = \bar{X}_n \xrightarrow{p} 1 \implies \frac{Y_n}{n} \xrightarrow{p} 1.$$

2. If $X_1, \dots, X_n \sim \text{Poisson}(\mu)$, then $\bar{X}_n \xrightarrow{p} \mu$ and
$$\frac{X_1^2 + \cdots + X_n^2}{n} \xrightarrow{p} \mathbb{E}[X_i^2] = \mu + \mu^2.$$
   (Make sure $\text{Var}(X_i^2) < \infty$.)

   The fourth moment of X_i must exist; need to show $\text{Var}(X_i^2)$ exists.

3. $X_1, \dots, X_n$ are i.i.d. random variables with cdf
$$F(x) = \begin{cases} 1 - \exp(-\mu x^2), & x \geq 0,\ \mu > 0, \\ 0, & x < 0. \end{cases}$$
   Find a such that
$$\lim_{n\to\infty} P\left(\left| \frac{1}{n} \sum_{i=1}^{n} X_i^2 - a \right| \geq \varepsilon \right) = 0 \quad \forall \varepsilon > 0$$

   Solution. By the WLLN,
$$\frac{1}{n} \sum_{i=1}^{n} X_i^2 \xrightarrow{p} a = \mathbb{E}[X_i^2].$$
   The pdf is
$$f(x) = 2\mu x e^{-\mu x^2}, \quad x > 0.$$
   Then
$$\mathbb{E}[X^4] = \int_0^\infty x^4 f(x)\, dx = \int_0^\infty 2\mu x^5 e^{-\mu x^2}\, dx < \infty.$$
   Substituting $y = \mu x^2$, we have
$$\mathbb{E}[X^2] = \int_0^\infty 2\mu x^3 e^{-\mu x^2}\, dx = \frac{1}{\mu}.$$
   Therefore, $a = \frac{1}{\mu}$. ■

4. **(Cauchy sample mean)**
$$X_1, \dots, X_n \text{ i.i.d. Cauchy}(0,1), \quad f(x) = \frac{1}{\pi(1+x^2)}, \quad x \in \mathbb{R}.$$
$$\bar{X}_n \xrightarrow{p} 0 \text{ ?!}$$
   $\mathbb{E}[X_i]$ and $\text{Var}(X_i)$ do not exist. The WLLN is not applicable. In fact, $\bar{X}_n \xrightarrow{d}$ Cauchy$(0,1)$.

13.7 Central Limit Theorem

Theorem 13.3 (Central Limit Theorem). *Let $X_1, \ldots, X_n$ be a sequence of i.i.d. random variables having mean 0 and variance σ^2. Let $Y_n = \sum_{i=1}^n X_i$. Then,*

$$\lim_{n\to\infty} P\left(\frac{Y_n}{\sigma\sqrt{n}} \le x\right) = \Phi(x), \quad -\infty < x < \infty,$$

where $\Phi(\cdot)$ is the cdf of the standard normal distribution.

$$Z \sim N(0,1).$$

1. *Define*

   $$Z_n = \frac{Y_n}{\sigma\sqrt{n}} = \frac{\sqrt{n}\,\bar{X}_n}{\sigma}.$$

   Then

   $$F_n(x) = P\left(\frac{Y_n}{\sigma\sqrt{n}} \le x\right)$$

   is the cdf of Z_n. Hence,

   $$\lim_{n\to\infty} F_n(x) = \Phi(x), \quad x \in \mathbb{R}.$$

   Therefore,

   $$Z_n \xrightarrow{d} Z, \quad \text{where } Z \sim N(0,1).$$

2. *If $\mathbb{E}[X_i] = \mu$, $i = 1, 2, \ldots, n$, use $X_i - \mu$ to replace the original X_i so that $\mathbb{E}[X_i - \mu] = 0$. This gives*

   $$\frac{Y_n - n\mu}{\sigma\sqrt{n}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0,1).$$

3. *From the **WLLN**:*

   $$\bar{X}_n - \mu \xrightarrow{p} 0, \quad \mu = \mathbb{E}[X_i].$$

   *From the **CLT**:*

   $$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2).$$

   $n(\bar{X}_n - \mu)$ diverges.

CLT Proof. Let $X_1, \ldots, X_n$ be a sequence of i.i.d. random variables having mean 0 and variance σ^2. Let $Y_n = \sum_{i=1}^n X_i$. Then,

$$\lim_{n\to\infty} P\left(\frac{Y_n}{\sigma\sqrt{n}} \le x\right) = \Phi(x), \quad -\infty < x < \infty,$$

where $\Phi(\cdot)$ is the cdf of the standard normal distribution. mgf and cdf are one-to-one.
Strategy: prove the limit of mgf of Z_n is the mgf of $N(0,1)$. Let,

$$Z_n = \frac{Y_n}{\sigma\sqrt{n}}, \quad Y_n = \sum_{i=1}^n X_i, \quad \mathbb{E}[X_i] = 0, \quad \mathrm{Var}(X_i) = \mathbb{E}[X_i^2] = \sigma^2 < \infty.$$

Let $M_X(t)$ be the mgf of X_i. Then,

$$M_{Y_n}(t) = [M_X(t)]^n, \quad M_{Z_n}(t) = [M_X(t/\sigma\sqrt{n})]^n.$$

$$M_X\left(\frac{t}{\sigma\sqrt{n}}\right) = M_X(0) + M_X'(0)\frac{t}{\sigma\sqrt{n}} + \frac{1}{2}M_X''(0)\left(\frac{t}{\sigma\sqrt{n}}\right)^2 + \varepsilon_n, \quad n\varepsilon_n \to 0.$$

Since $M_X(0) = 1$, $M_X'(0) = 0$, and $M_X''(0) = \sigma^2$, we have

$$M_X\left(\frac{t}{\sigma\sqrt{n}}\right) = 1 + \frac{t^2}{2n} + \varepsilon_n.$$

Hence,

$$M_{Z_n}(t) = \left(1 + \frac{t^2}{2n} + \varepsilon_n\right)^n.$$

As $n \to \infty$,

$$\lim_{n\to\infty} M_{Z_n}(t) = \lim_{n\to\infty} \left(1 + \frac{t^2}{2n}\right)^n = e^{\frac{t^2}{2}},$$

since $\lim_{n\to\infty}(1 + \frac{a}{n})^n = e^a,\ a \in \mathbb{R}$.

$$e^{\frac{t^2}{2}} \text{ is the mgf of } N(0,1).$$
$$\implies \text{Limit of the cdf of } Z_n \text{ is the cdf of } N(0,1).$$
$$\therefore Z_n \xrightarrow{d} Z, \quad \text{where } Z \sim N(0,1).$$

Example 13.9. Let $X_1, \ldots, X_n$ be an i.i.d. sample from a χ^2 distribution, and define

$$Y_n = \sum_{i=1}^n X_i.$$

Find the limiting distribution of

$$\frac{Y_n - n}{\sqrt{2n}}.$$

We know $\mathbb{E}[X_i] = 1$ and $\mathrm{Var}(X_i) = 2$. By the CLT,

$$\frac{Y_n - n\mathbb{E}[X_i]}{\sqrt{n\,\mathrm{Var}(X_i)}} = \frac{Y_n - n}{\sqrt{2n}} \xrightarrow{d} N(0,1).$$

Example 13.10. **Example 2:** Let $X_1, \ldots, X_n$ be an i.i.d. sample from Poisson(μ), and define

$$Y_n = \sum_{i=1}^n X_i.$$

Find the limiting distribution of

$$\frac{Y_n - n\mu}{\sqrt{n\mu}}.$$

We have $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \mu$. By the CLT,

$$\frac{Y_n - n\mathbb{E}[X_i]}{\sqrt{n\,\mathrm{Var}(X_i)}} = \frac{Y_n - n\mu}{\sqrt{n\mu}} \xrightarrow{d} N(0,1).$$

To use CLT: Identify $\mathbb{E}[X_i]$ and $\mathrm{Var}(X_i)$, then

$$\frac{Y_n - n\mathbb{E}[X_i]}{\sqrt{n\,\mathrm{Var}(X_i)}} = \frac{\sqrt{n}(\bar{X}_n - \mathbb{E}[X_i])}{\sqrt{\mathrm{Var}(X_i)}} \xrightarrow{d} N(0,1).$$

13.8 Continuous Mapping Theorem

The CMT answers the question: *What is the asymptotic behavior of a sequence of random variables after transformation?*

Theorem 13.4 (Continuous Mapping Theorem)**.** *Let $g : \mathbb{R} \to \mathbb{R}$ be a continuous function.*

1. *If $X_n \xrightarrow{d} X$, then $g(X_n) \xrightarrow{d} g(X)$.*
2. *If $X_n \xrightarrow{p} X$, then $g(X_n) \xrightarrow{p} g(X)$.*

The CMT also holds for general $g : \mathbb{R}^d \to \mathbb{R}^b$.

Asymptotic behavior is preserved under continuous transformations.

Example 13.11. Define

$$S_n = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X}_n)^2}{n-1}},$$

where $X_1, \ldots, X_n$ is an i.i.d. sample from Norm(μ, σ^2). Find the constant c such that $S_n \xrightarrow{p} c$.

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

From previous lectures, we know

$$\frac{(n-1)S_n^2}{\sigma^2} \sim \chi^2_{n-1}.$$

Hence,

$$\frac{S_n^2}{\sigma^2} \sim \frac{1}{n-1}\chi^2_{n-1} \quad \text{(scaled Chi-squared distribution).}$$

We can write

$$\frac{1}{n-1}\chi^2_{n-1} = \frac{1}{n-1}(Z_1^2 + \cdots + Z_{n-1}^2), \quad Z_i \text{ i.i.d. } N(0,1).$$

By the WLLN,

$$\frac{1}{n-1} \sum_{i=1}^{n-1} Z_i^2 \xrightarrow{p} \mathbb{E}[Z_i^2] = 1.$$

Therefore,

$$\frac{S_n^2}{\sigma^2} \xrightarrow{p} 1 \quad \Longrightarrow \quad S_n^2 \xrightarrow{p} \sigma^2 \quad \text{by CMT with } g(x) = \sigma^2 x.$$

Finally,

$$S_n \xrightarrow{p} \sigma \quad \text{by CMT with } g(x) = \sqrt{x}.$$

13.9 Slutsky's Theorem

Theorem 13.5 (Slutsky's Theorem)**.** *(Implied by the CMT) If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} b$, where X is a random variable and b is a constant, then*

$$\begin{pmatrix} X_n \\ Y_n \end{pmatrix} \xrightarrow{d} \begin{pmatrix} X \\ b \end{pmatrix}.$$

(Joint distribution convergence.) If $g(\cdot) : \mathbb{R}^2 \to \mathbb{R}$ is continuous at (x, b) for any $x \in \mathbb{R}$, then

$$g(X_n, Y_n) \xrightarrow{d} g(X, b).$$

Proof: *By the Continuous Mapping Theorem (CMT).*

In particular:

1. $g(x, y) = x + y \quad \Rightarrow \quad X_n + Y_n \xrightarrow{d} X + b.$
2. $g(x, y) = xy \quad \Rightarrow \quad X_n Y_n \xrightarrow{d} bX.$
3. $g(x, y) = \dfrac{x}{y} \quad \Rightarrow \quad \frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{b}, \quad$ given that $b \neq 0,\ Y_n \neq 0.$

Example 13.12.

1. If $X_n \xrightarrow{p} a$, then $X_n^2 \xrightarrow{p} a^2$ by CMT. If $a \geq 0$ and $X_n \geq 0$, then $\sqrt{X_n} \xrightarrow{p} \sqrt{a}$ by CMT.
2. Suppose $X_n \xrightarrow{d} Z$, where $Z \sim N(0, 1)$.
$$\begin{aligned} &\Rightarrow\ 2X_n \xrightarrow{d} 2Z \sim N(0, 4) \\ &\Rightarrow\ 2X_n + 1 \xrightarrow{d} 2Z + 1 \sim N(1, 4) \quad \text{by CMT.} \\ &\Rightarrow\ X_n^2 \xrightarrow{d} Z^2 \sim \chi_1^2. \end{aligned}$$
3. Suppose $X_n \xrightarrow{d} Z \sim N(0, 1)$ and $Y_n \xrightarrow{p} b \neq 0$.
$$\begin{cases} X_n + Y_n \xrightarrow{d} b + Z \sim N(b, 1), \\ X_n Y_n \xrightarrow{d} bZ \sim N(0, b^2), \end{cases}$$
by Slutsky's Theorem.
$$\Rightarrow \frac{X_n}{Y_n} \xrightarrow{d} \frac{Z}{b} \sim N\left(0, \frac{1}{b^2}\right).$$

Example 13.13. Let $X_1, \ldots, X_n$ be an i.i.d. sample from Poisson(μ). Find the limiting distribution of

$$Z_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sqrt{\bar{X}_n}} \quad \text{and} \quad U_n = \sqrt{n}(\bar{X}_n - \mu),$$

$$Z_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sqrt{\bar{X}_n}} = \underbrace{\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sqrt{\mu}}}_{\text{CLT term}} \cdot \underbrace{\frac{\sqrt{\mu}}{\sqrt{\bar{X}_n}}}_{\text{CMT term}}.$$

1. By the CLT,
$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sqrt{\mu}} \xrightarrow{d} Z, \quad Z \sim N(0, 1).$$
2. By the WLLN, $\bar{X}_n \xrightarrow{p} \mu$. Hence, by the CMT with $g(x) = \frac{\sqrt{\mu}}{\sqrt{x}}$,
$$\frac{\sqrt{\mu}}{\sqrt{\bar{X}_n}} \xrightarrow{p} 1.$$

Combining (1) and (2) using Slutsky's Theorem,

$$Z_n \xrightarrow{d} Z \sim N(0,1).$$

$$U_n = \sqrt{n}(\bar{X}_n - \mu) = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sqrt{\bar{X}_n}} \cdot \sqrt{\bar{X}_n} \xrightarrow{d} \sqrt{\mu}\, Z \sim N(0, \mu)$$

by CMT or Slutsky's Theorem.

Example 13.14. Let $X_1, \dots, X_n$ be an i.i.d. Unif$(0,1)$ sample. Define $U_1 = X_{(1)}$ and $U_n = X_{(n)}$. From previous lecture results, we know that

$$n(1 - U_n) \xrightarrow{d} \text{Exp}(1), \quad U_1 \xrightarrow{p} 0, \quad U_n \xrightarrow{p} 1.$$

1. $U_n^n \xrightarrow{p} e^{-1}$ by CMT with $g(x) = e^{-x}$.
2. $\sin(U_1) \xrightarrow{p} \sin(0) = 0$ by CMT with $g(x) = \sin x$.
3. Find the limiting distribution of $\exp\{-n(1 - U_n)\}$. Let $g(x) = e^{-x}$. By CMT,

   $$\exp\{-n(1 - U_n)\} \xrightarrow{d} \exp(-X), \quad X \sim \text{Exp}(1).$$

   Using the cdf method, let $Y = e^{-X}$ where $X \in (0, \infty)$.

   $$F_Y(y) = P(Y \le y) = P(X \ge -\log y) = 1 - F_X(-\log y) = y, \quad y \in (0,1).$$

   Hence $f_Y(y) = 1$ for $y \in (0,1)$, so

   $$Y \sim \text{Unif}(0,1) \quad \Rightarrow \quad \exp\{-n(1 - U_n)\} \xrightarrow{d} \text{Unif}(0,1).$$

4. $(U_n + 1)^2\, n(1 - U_n) \xrightarrow{d} 4X$, where $X \sim \text{Exp}(1)$, by Slutsky's Theorem.

   Since $(U_n + 1)^2 \xrightarrow{p} 2^2 = 4$ by CMT ($g(x) = (x+1)^2$) and $n(1 - U_n) \xrightarrow{d} X$ with $X \sim \text{Exp}(1)$, the result follows.

Example 13.15. Let $X_1, \dots, X_n$ be an i.i.d. sample from Norm(μ, σ^2). The sample variance is given by

$$S_n^2 = \frac{\sum_{i=1}^n (X_i - \bar{X}_n)^2}{n-1}.$$

We know that

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{S_n} \sim t_{n-1}.$$

Find its limiting distribution (exact distribution known).

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{S_n} = \underbrace{\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}}_{\xrightarrow{d} N(0,1) \text{ by CLT}} \cdot \underbrace{\frac{\sigma}{S_n}}_{\xrightarrow{p} 1 \text{ by CMT}}.$$

By Slutsky's Theorem,

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{S_n} \xrightarrow{d} N(0,1).$$

In other words,

$$t_n \xrightarrow{d} N(0,1) \quad \text{as } n \to \infty.$$

Theorem 13.6 (Delta Method)**.** *The **Delta Method** finds the limiting distribution of a transformation $g(Y_n)$ of a sequence of random variables Y_n, which has a normal limit distribution after centering and scaling. If*

1. $\sqrt{n}(Y_n - \theta) \xrightarrow{d} N(0, \sigma^2)$,
2. *$g(\cdot)$ is differentiable at θ with $g'(\theta) \neq 0$,*

then

$$\sqrt{n}\left(g(Y_n) - g(\theta)\right) \xrightarrow{d} N\left(0, [g'(\theta)]^2 \sigma^2\right).$$

Proof. We start with

$$\sqrt{n}\left(g(Y_n)-g(\theta)\right)=\sqrt{n}(Y_n-\theta)\,\frac{g(Y_n)-g(\theta)}{Y_n-\theta}.$$

Rewrite as

$$\sqrt{n}\left(g(Y_n)-g(\theta)\right)=\sqrt{n}(Y_n-\theta)\,g'(\theta)+\sqrt{n}(Y_n-\theta)\left\{\frac{g(Y_n)-g(\theta)}{Y_n-\theta}-g'(\theta)\right\}.$$

The first term (underlined in notes as desired) satisfies

$$\sqrt{n}(Y_n-\theta)\,g'(\theta)\xrightarrow{d}N\big(0,[g'(\theta)]^2\sigma^2\big)\quad\text{by the CMT.}$$

For the second term (underlined in notes as $\xrightarrow{p}0$), define

$$u(y)=\begin{cases}\dfrac{g(y)-g(\theta)}{y-\theta}-g'(\theta), & y\neq\theta,\\ 0, & y=\theta.\end{cases}$$

Since $\lim_{y\to\theta}u(y)=u(\theta)=0$, $u(y)$ is continuous at θ. Moreover, $Y_n-\theta=\dfrac{1}{\sqrt{n}}\sqrt{n}(Y_n-\theta)\xrightarrow{p}0$ and $\sqrt{n}(Y_n-\theta)\xrightarrow{d}N(0,\sigma^2)$. Hence, by CMT and Slutsky's Theorem,

$$u(Y_n)\xrightarrow{p}u(\theta)=0\quad\text{and}\quad\sqrt{n}(Y_n-\theta)\,u(Y_n)\xrightarrow{d}N(0,\sigma^2)\cdot 0=0.$$

Therefore, combining both terms,

$$\sqrt{n}\left(g(Y_n)-g(\theta)\right)\xrightarrow{d}N\big(0,[g'(\theta)]^2\sigma^2\big)\quad\text{by Slutsky's Theorem.}$$

Remark 13.5 (More General Form). If $a_n(Y_n-\theta)\xrightarrow{d}X$, where X is a random variable and $a_n\to\infty$, then under the same differentiability requirements for $g(\cdot)$,

$$a_n\big(g(Y_n)-g(\theta)\big)\xrightarrow{d}g'(\theta)X.$$

For example, $a_n=n^{1/3},n^{1/4},\dots$ Note that X can have a different distribution from $N(\cdot,\cdot)$ — the result does not require normality.

14 (Cont') Delta Method

14.1 Big $O_p(\cdot)$ Small $o_p(\cdot)$ Notations:

If X_n is a sequence of random variables, and a_n is a sequence of constants.

1. We say $X_n=o_p(a_n)$ if
$$\frac{X_n}{a_n}\xrightarrow{p}0.$$
2. We say $X_n=O_p(a_n)$ if X_n/a_n is *stochastically bounded*: for any $\varepsilon>0$, there exists a finite constant M_ε and a finite constant N_ε such that
$$\Pr\left(\left|\frac{X_n}{a_n}\right|>M_\varepsilon\right)<\varepsilon\quad\text{or equivalently}\quad\Pr\left(\left|\frac{X_n}{a_n}\right|\le M_\varepsilon\right)\ge 1-\varepsilon$$
for any $n>N_\varepsilon$.

   m_ε and N_ε depend on ε.

   Interpretation: As long as n is large enough, we can find a finite number (M_ε) such that
$$\left|\frac{X_n}{a_n}\right|$$
is bounded by M_ε with a probability that is essentially 1.

For example the sample mean of an iid sample from a distribution with variance σ^2 and mean μ satisfies

$$\bar{X}_n - \mu \xrightarrow{p} 0, \quad \text{by WLLN}$$

which implies that

$$\bar{X}_n - \mu = o_p(1).$$

It also satisfies

$$Z_n = \sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2) \quad \text{by CLT}$$

which implies that

$$Z_n = O_p(1) \quad \text{or} \quad \bar{X}_n - \mu = O_p\left(\frac{1}{\sqrt{n}}\right).$$

If $Y_n \xrightarrow{d} Y$ where Y is a continuous r.v., then $Y_n = O_p(1)$.

14.2 Combining Big/small O:

- $o_p(1) + o_p(1) = o_p(1)$
- $O_p(1) + o_p(1) = O_p(1)$
- $O_p(1)o_p(1) = o_p(1)$

14.3 Alternative Proof

We provide an alternative proof of the Delta method.

1. If we know Y_n satisfies the following
$$\sqrt{n}(Y_n - \theta) \xrightarrow{d} N(0, \sigma^2),$$
then we have
$$O_p\left(\frac{1}{\sqrt{n}}\right) = O_p(1) \cdot \frac{1}{\sqrt{n}} = o_p(1).$$
$$Y_n - \theta = O_p\left(\frac{1}{\sqrt{n}}\right) \Rightarrow Y_n - \theta = o_p(1).$$
We have proved this in Lec 17 so $Y_n \xrightarrow{p} \theta$.
2. Use a Taylor series expansion of $g(Y_n)$ around θ gives
$$g(Y_n) = g(\theta) + g'(\theta)(Y_n - \theta) + \frac{1}{2}g''(\theta_n^*)(Y_n - \theta)^2$$
   (a) Assume $g(\cdot)$ is sufficiently smooth.
   (b) θ_n^* is between Y_n and θ.
3. Thus, we have
$$\sqrt{n}\{g(Y_n) - g(\theta)\} = g'(\theta)\sqrt{n}(Y_n - \theta) + \frac{\sqrt{n}}{2}g''(\theta_n^*)(Y_n - \theta)^2.$$
   (a) $g'(\theta)\sqrt{n}(Y_n - \theta) \xrightarrow{d} N(0, (g'(\theta))^2\sigma^2)$.
   (b)
$$\frac{\sqrt{n}}{2}g''(\theta_n^*)(Y_n - \theta)^2 = \frac{1}{2}g''(\theta_n^*) \cdot \sqrt{n}(Y_n - \theta) \cdot (Y_n - \theta) = O_p(1) \cdot O_p(1) \cdot o_p(1) = o_p(1).$$
$$\Rightarrow \sqrt{n}\{g(Y_n) - g(\theta)\} \xrightarrow{d} N(0, (g'(\theta))^2\sigma^2).$$
   Any term $o_p(1)$ or smaller can be ignored when considering the asymptotic or limiting distribution.

14.4 Examples

1. $X_1, \dots, X_n \overset{iid}{\sim} \text{Poi}(\mu)$. Find the limiting distribution of

$$Z_n = \sqrt{n}(\sqrt{\bar{X}_n} - \sqrt{\mu}).$$

$\bar{X}_n$ using CLT $\Rightarrow$

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sqrt{\mu}} \xrightarrow{d} N(0,1) \Rightarrow \sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \mu).$$

Delta method: let $g(x) = \sqrt{x}$ and so $g'(\mu) = \frac{1}{2\sqrt{\mu}}$.

$$\Rightarrow \sqrt{n}(\sqrt{\bar{X}_n} - \sqrt{\mu}) \xrightarrow{d} N\left(0, \frac{1}{4}\right)$$

since $(g'(\mu))^2\mu = \frac{1}{4}$.

2. $X_1, \dots, X_n \overset{iid}{\sim} \text{Exp}\left(\frac{1}{\theta}\right)$ with mean θ.

   (a) $\bar{X}_n$
   (b) $Z_n = \sqrt{n}(\bar{X}_n - \theta)$
   (c) $U_n = \sqrt{n}\frac{\bar{X}_n - \theta}{\bar{X}_n}$
   (d) $V_n = \sqrt{n}[\log \bar{X}_n - \log \theta]$

   (a) by WLLN we have $\bar{X}_n \xrightarrow{p} \theta \Rightarrow \bar{X}_n \xrightarrow{d} \theta$.
   (b) by CLT

$$\frac{\sqrt{n}(\bar{X}_n - \theta)}{\theta} \xrightarrow{d} N(0,1) \Rightarrow Z_n \xrightarrow{d} N(0, \theta^2).$$

   (c)

$$U_n = \frac{\sqrt{n}(\bar{X}_n - \theta)}{\theta} \cdot \frac{\theta}{\bar{X}_n} \xrightarrow{d} N(0,1) \cdot 1 = N(0,1)$$

   by Slutsky's Theorem.
   (d) By Delta method

$$V_n = \sqrt{n}[\log \bar{X}_n - \log \theta] \xrightarrow{d} N\left(0, \frac{1}{\theta^2} \cdot \theta^2\right) = N(0,1).$$

   Here $g(\theta) = \log \theta$ and $g'(\theta) = \frac{1}{\theta} > 0$.

15 Point Estimation

Setting: We have an iid sample $X_1, \dots, X_n$ from a distribution $f(x; \theta)$ (we assume iid for simplicity). We usually say that the distribution is indexed by θ.

The parameter θ could be a scalar, or it could be a vector $\theta^T = (\theta_1, \theta_2)$.

Model: The parameter(s) θ is usually unknown. Thus, we provide a family of models by assuming "a" model $f(x; \theta)$. A model is a data-generating mechanism, or you can think of this as a pdf/cdf etc.

Parameter Space: Describe the family of models you choose. You believe the true data-generating mechanism lives in the chosen family, which is equivalent to the true value of θ lives in the parameter space.

Estimation: Based on the sample generated using the unknown true value of the parameter, what is your "guess" of the true value of the parameter?

Statistic: A function of the random sample that does not depend on any parameters.

$$T(X_1, \dots, X_n) = \bar{X}_n, \quad T(X_1, \dots, X_n) = S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2.$$

$\frac{\bar{X}_n - \mu}{\sigma}$, $\frac{(n-1)S_n^2}{\sigma^2}$ are NOT statistics.

15.1 Estimator & Estimate

Estimator: a statistic that is used to do estimation, i.e. guess what the true parameter is.

Estimators are random variables.

θ is the parameter. $\hat{\theta}_n = \hat{\theta}(X_1, \dots, X_n)$ is an estimator for θ.

$\bar{X}_n$ is an estimator for μ.

$\frac{n}{n+1}\bar{X}_n$ is also an estimator for μ.

S_n^2 is an estimator for σ^2.

15.2 How to evaluate an estimator?

1. **Unbiasedness**

   If $E[\hat{\theta}_n] = \theta$ we say $\hat{\theta}_n$ is unbiased.

   If $\lim_{n\to\infty} E[\hat{\theta}_n] = \theta$ we say $\hat{\theta}_n$ is asymptotically unbiased.

2. **Consistency**

   If $\hat{\theta}_n \xrightarrow{p} \theta$ we say $\hat{\theta}_n$ is consistent.

3. **Metric for comparison: Mean Squared Error (MSE)**

   $$\text{MSE} = E[(\hat{\theta}_n - \theta)^2].$$

   Smaller MSE is preferred.

   Decomposition:

   $$\begin{aligned}\text{MSE} &= E[(\hat{\theta}_n - \mu_n + \mu_n - \theta)^2] \\ &= E[(\hat{\theta}_n - \mu_n)^2] + E[(\mu_n - \theta)^2] + 2E[(\hat{\theta}_n - \mu_n)(\mu_n - \theta)] \\ &= \text{Var}(\hat{\theta}_n) + (\mu_n - \theta)^2 \\ &= \text{Var}(\hat{\theta}_n) + (\text{Bias})^2.\end{aligned}$$

15.3 Method of Moments

Setting: Suppose $X_1, \dots, X_n$ are iid with pdf/pmf $f(x;\theta)$. We need to estimate

$$\theta^T = (\theta_1, \dots, \theta_k).$$

Method: Method of Moments (MM) estimator.

1. Find the moments of X_i using θ.

   $$\mu_1 = E[X_1], \dots, \mu_k = E[X^k].$$

   $(\mu_1, \dots, \mu_k)$ is a function of $\theta^T = (\theta_1, \dots, \theta_k)$.

2. Find the estimators for the moments (sample moments)

   $$\hat{\mu}_1 = \bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i, \quad \hat{\mu}_k = \frac{1}{n}\sum_{i=1}^n X_i^k.$$

3. From Step 1 find the inverse

   $$\theta_1 = \theta_1(\mu_1, \dots, \mu_k), \dots, \theta_k = \theta_k(\mu_1, \dots, \mu_k).$$

4. Replace $\mu_1, \dots, \mu_k$ with $\hat{\mu}_1, \dots, \hat{\mu}_k$.

   $$\hat{\theta}_1 = \theta_1(\hat{\mu}_1, \dots, \hat{\mu}_k), \dots, \hat{\theta}_k = \theta_k(\hat{\mu}_1, \dots, \hat{\mu}_k).$$

   These are MOM estimators.

If the parameter of interest is $\tau(\theta_1, \ldots, \theta_k)$, the MOM estimator is $\tau(\hat{\theta}_1, \ldots, \hat{\theta}_k)$.

15.4 Examples

1. $X_1, \ldots, X_n \overset{iid}{\sim} \text{Poi}(\theta)$. Find the MOM estimator for θ.

   $$\mu_1 = E[X] = \theta, \quad \hat{\mu}_1 = \bar{X}_n, \quad \hat{\theta} = \bar{X}_n.$$

   Find the MOM estimator for θ^2:

   $$\hat{\theta}^2 = (\bar{X}_n)^2.$$

2. Same context:

   (a) $\text{Exp}\left(\frac{1}{\theta}\right)$
   (b) $\text{Unif}(0, \theta)$
   (c) $f(x;\theta) = \theta x^{\theta-1}, \quad x \in (0,1),\ \theta > 0$

   (a) $\mu_1 = E[X] = \theta \Rightarrow \hat{\theta} = \bar{X}_n$.
   (b) $\mu_1 = E[X] = \frac{\theta}{2} \Rightarrow \hat{\theta} = 2\bar{X}_n$.
   (c) $\mu_1 = E[X] = \frac{\theta}{\theta+1} \Rightarrow \hat{\theta} = \frac{\bar{X}_n}{1-\bar{X}_n}$.

Maximum Likelihood Method

15.5 Likelihood Function

1. The <u>likelihood function</u> is the joint pdf/pmf evaluated at the realized sample.
   $X_1, \ldots, X_n$ random variables
   $f(x_1, \ldots, x_n; \theta)$ joint pdf/pmf
   $x_1 = x_1, \ldots, x_n = x_n$ realized sample
   Plug realized sample into function, into the joint pdf/pmf.
   $f(x_1, \ldots, x_n; \theta)$
   consider θ fixed (i.e. likelihood func).

2. Suppose $X_1, \ldots, X_n$ are iid from $f(x;\theta)$, where θ is unknown. Then, the likelihood function is

   $$f(x_1, \ldots, x_n; \theta) = \prod_{i=1}^{n} f(x_i; \theta)$$

   (specified distribution)

3. We use $L(\theta; x_1, \ldots, x_n)$ or $L(\theta)$ to denote the likelihood function.

   $$L(\theta) = L(\theta; x_1, \ldots, x_n) = \prod_{i=1}^{n} f(x_i; \theta)$$

   (dep on realised samples)
   $\{x_1, \ldots, x_n\}$ is the realised sample.

15.6 Comments:

1. The likelihood function measures how likely we observe the observed data given θ.
   $\{x_1, \ldots, x_n\}$
2. Smaller $L(\theta)$ means θ is less likely to generate the observed data.
3. Larger $L(\theta)$ means θ is more likely to generate the observed data.

15.7 Example

Suppose we have two types of coins; the first is a fair coin, and the second is an unfair coin with the probability of heads being 80%. We were given a coin (not knowing it was fair or unfair), tossed it five times, and got five heads and 0 tails.

"unfair coin"

"toss the coin 5 times"

"got 5 heads"

$\theta \in \Theta = \{0.5, 0.8\}$ parameter space

$X_1, \ldots, X_5$ random variables coin tossing

$x_1, \ldots, x_5 = 1$ realized sample

$L(\theta) = \theta^y (1-\theta)^{5-y}$ where $y = \#$ heads in total

$y = \sum_{i=1}^{5} x_i = 5$

$\theta = 0.5 \Rightarrow L(\theta) = 0.5^5$

$\theta = 0.8 \Rightarrow L(\theta) = 0.8^5$

Thus our estimate is that this coin is unfair $\Rightarrow \hat{\theta} = 0.8$.

15.8 Maximum Likelihood Method

Choose the θ to maximize the likelihood function $L(\theta)$.

1. ML estimate:
$$\hat{\theta}_n = \arg\max_{\theta \in \Theta} L(\theta) = \hat{\theta}_n(x_1, \ldots, x_n)$$
(realised sample, fixed!)

2. ML estimator:
Replace $x_1, \ldots, x_n$ with $X_1, \ldots, X_n$ to obtain
$$\hat{\theta}_n = \hat{\theta}_n(X_1, \ldots, X_n)$$
the ML estimator which is an r.v.
(random sample)

3. Log-likelihood function:
$L(\theta)$ usually has the form $\prod_{i=1}^{n} f(x_i; \theta)$.
$\ell(\theta)$ "log likelihood" function
$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log f(x_i; \theta).$$
$\log(\cdot)$ is monotonic, does NOT change maximum.
$$\hat{\theta}_n = \arg\max_{\theta \in \Theta} L(\theta) = \arg\max_{\theta \in \Theta} \ell(\theta).$$

4. If $\hat{\theta}$ is the ML estimator for θ, then $\tau(\hat{\theta})$ is the ML estimator for $\tau(\theta)$, where τ is a known function.
If $\tau(\theta)$ ("transformed θ") is the parameter of interest
$\hat{\theta}_n$ is the ML estimator for θ.
then $\tau(\hat{\theta}_n)$ is the ML estimator for $\tau(\theta)$.

15.9 Examples

$X_1, \ldots, X_n$ is an iid sample from $f(x;\theta)$. Find the ML estimator for θ.

1. Poisson distribution: $f(x;\theta) = \frac{\theta^x}{x!}e^{-\theta}$

$$L(\theta) = \prod_{i=1}^{n} f(x_i;\theta) = e^{-n\theta} \cdot \exp\left(\sum_{i=1}^{n} x_i\right) \Big/ \prod_{i=1}^{n} x_i!$$

$$\ell(\theta) = -n\theta + \sum_{i=1}^{n} x_i \log\theta - \log\left(\prod_{i=1}^{n} x_i!\right)$$

(can be skipped)

$$\frac{d\ell(\theta)}{d\theta} = -n + \frac{\sum_{i=1}^{n} x_i}{\theta} = 0 \Rightarrow \hat{\theta}_n = \frac{1}{n}\sum_{i=1}^{n} x_i$$

(replacement is trivial)

2. $f(x;\theta) = \theta x^{\theta-1},\ 0 < x < 1,\ \theta > 0.$

$$L(\theta) = \prod_{i=1}^{n} f(x_i;\theta) = \theta^n \left(\prod_{i=1}^{n} x_i\right)^{\theta-1}$$

$$\ell(\theta) = n\log\theta + (\theta-1)\sum_{i=1}^{n}\log x_i$$

$$\frac{d\ell(\theta)}{d\theta} = \frac{n}{\theta} + \sum_{i=1}^{n}\log x_i = 0 \Rightarrow \hat{\theta}_n = -\frac{n}{\sum_{i=1}^{n}\log x_i}$$

3. Normal distribution $N(\mu,\sigma^2)$ $\quad \theta^T = (\mu,\sigma^2)$

$$L(\theta) = \prod_{i=1}^{n} f(x_i;\theta)$$

$$\ell(\theta) = -\frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2 + \text{const.}$$

$$\frac{d\ell(\theta)}{d\mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu) = 0$$

$$\frac{d\ell(\theta)}{d\sigma} = \frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i-\mu)^2 = 0$$

$$\Rightarrow \hat{\mu}_n = \bar{X}_n \quad \text{and} \quad \hat{\sigma}_n^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x}_n)^2$$

$\Rightarrow$ ML estimator of σ^2 use invariant property

$$\hat{\sigma}_n = \sqrt{\hat{\sigma}_n^2} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x}_n)^2}$$

$$\hat{\sigma}_n^2 = \frac{n}{n-1}S_n^2 \quad \text{and} \quad E[\hat{\sigma}_n^2] = \sigma^2 \quad \text{(unbiased estimator)}$$

$$E[\hat{\sigma}_n^2] = \frac{n-1}{n}\sigma^2 \quad \text{(asymptotically unbiased)}$$

$$\hat{\sigma}_n^2 \xrightarrow{p} \sigma^2 \quad \text{by SLLN's Theorem}$$

4. Unif$(0, \theta)$

$$f(x) = \begin{cases} \frac{1}{\theta} & x \in [0, \theta] \\ 0 & \text{otherwise} \end{cases}$$

$$L(\theta) = \prod_{i=1}^{n} f(x_i; \theta) = \theta^{-n} \cdot 1\{x_1 \le \theta\} \cdots 1\{x_n \le \theta\}$$

$\Rightarrow$ the ML estimator for θ is

$$\hat{\theta}_n = X_{(n)}.$$

any $\theta < x_{(n)} \Rightarrow 0$ (not useful).

smaller $\theta \Rightarrow$ larger $L(\theta)$.

If the support of $X_1, \ldots, X_n$ depends on θ, we need to be careful. Usually no need to take the derivative.

15.10 Inverse Gaussian

Let $X_1, \ldots, X_n$ be an iid random sample from the inverse Gaussian distribution with pdf

$$f(x; \mu, \lambda) = \left(\frac{\lambda}{2\pi x^3}\right)^{1/2} \exp\left\{-\frac{\lambda(x-\mu)^2}{2\mu^2 x}\right\}, \quad 0 < x < \infty.$$

The log-likelihood is

$$\ell(\mu, \lambda) = \frac{n}{2}\log\lambda + n\lambda/\mu - \frac{n}{2}\log(2\pi) - \frac{3}{2}\sum_{i=1}^{n}\log x_i - \frac{\lambda}{2\mu^2}\sum_{i=1}^{n} x_i - \frac{\lambda}{2}\sum_{i=1}^{n} x_i^{-1}.$$

$$\frac{\partial \ell}{\partial \lambda} = \frac{n}{2\lambda} + \frac{n}{\mu} - \frac{1}{2\mu^2}\sum_{i=1}^{n} x_i - \frac{1}{2}\sum_{i=1}^{n} x_i^{-1} = 0$$

$$\frac{\partial \ell}{\partial \mu} = -\frac{n\lambda}{\mu^2} + \frac{\lambda}{\mu^3}\sum_{i=1}^{n} x_i = 0 \quad (2)$$

From (2) $\hat{\mu}_n = \bar{X}_n$.

Plug $\hat{\mu}_n = \bar{X}_n$ into (1)

$$\frac{n}{2\lambda} + \frac{n}{\bar{X}_n} - \frac{1}{2\bar{X}_n^2}\sum_{i=1}^{n} x_i - \frac{1}{2}\sum_{i=1}^{n} x_i^{-1} = 0$$

$$\Rightarrow \hat{\lambda}_n = \left(\frac{1}{n}\sum_{i=1}^{n} x_i^{-1} - \frac{1}{\bar{X}_n}\right)^{-1}.$$

15.11 Properties of ML Estimator

(commonly used estimator)

1. We consider a scalar θ.
2. The ML estimator of θ, denoted by $\hat{\theta}_n$, is a random variable.
3. We only consider the case where the support of $X_1, \ldots, X_n$ does not depend on θ.

15.12 Preparations

1. Score function:

$$S(\theta) = \frac{d\ell(\theta)}{d\theta} = \frac{d}{d\theta} \log L(\theta) = \frac{L'(\theta)}{L(\theta)}.$$

$$S(\theta) = S(\theta; x_1, \ldots, x_n) \quad \text{(realised sample)}$$

$$S(\hat{\theta}_n) = 0 \quad \text{where } \hat{\theta}_n \text{ is the ML estimator.}$$

2. Information function:

$$I(\theta) = -\frac{dS(\theta)}{d\theta} = -\frac{d^2\ell(\theta)}{d\theta^2} = -\frac{d^2}{d\theta^2} \log L(\theta)$$

$$I(\theta) = I(\theta; x_1, \ldots, x_n) \quad \text{(realised sample)}$$

3. Fisher Information / Expected Information

$$I(\theta) = I(\theta; X_1, \ldots, X_n)$$

replace $x_1, \ldots, x_n$ with $X_1, \ldots, X_n$ (random sample)
$I(\theta; X_1, \ldots, X_n)$ is an r.v.

$$I(\theta) = E[I(\theta; X_1, \ldots, X_n)]$$

is the Fisher information.
If $X_1, \ldots, X_n$ iid $f(x; \theta)$,

$$I(\theta) = E\left[\sum_{i=1}^{n} \left(-\frac{d^2}{d\theta^2} \log f(X_i; \theta)\right)\right] = -\sum_{i=1}^{n} E\left[\frac{d^2}{d\theta^2} \log f(X_i; \theta)\right].$$

16 (Cont') Fisher Information

1. $X_1, \ldots, X_n$ are iid with $f(x;\theta)$
   $J(\theta) = nI(\theta)$ where $I(\theta)$ is Fisher Information of X_i.

$$I(\theta) = -E\left[\frac{d^2 \log f(X;\theta)}{d\theta^2}\right]$$

2. Alternative

$$I(\theta) = E\left[\left(\frac{d \log f(X;\theta)}{d\theta}\right)^2\right]$$

$$\frac{d^2 \log f}{d\theta^2} = \frac{f''}{f} - \left(\frac{f'}{f}\right)^2$$

(a)

$$\frac{f''}{f} = E\left[\frac{d^2 f(X;\theta)}{d\theta^2} \Big/ f(X;\theta)\right] = \int \frac{d^2 f(x;\theta)}{d\theta^2}\, dx = \frac{d^2}{d\theta^2} \int f(x;\theta)\, dx = 0$$

(b)

$$\left(\frac{f'}{f}\right)^2 = E\left[\left(\frac{f'}{f}\right)^2\right] = E\left[(\log f)'^2\right]$$

16.1 Examples:

(1) Poi(θ): $f(x;\theta) = e^{-\theta}\theta^x / x!$

$$\log f(x;\theta) = x \log \theta - \theta - \log x!$$

$$S(\theta) = \frac{x}{\theta} - 1 \quad \text{and} \quad I(\theta) = \frac{1}{\theta}$$

$$J(\theta) = E[I(\theta; X)] = \frac{1}{\theta}$$

$$J(\theta) = nI(\theta) = \frac{n}{\theta}$$

$$S(\theta) = \sum_{i=1}^{n} \frac{x_i}{\theta} - n = 0 \Rightarrow \hat{\theta} = \bar{X}_n$$

$$\mathrm{Var}(\hat{\theta}) = \mathrm{Var}(\bar{X}_n) = \frac{\theta}{n} = \frac{1}{J(\theta)}$$

(2) Exp with mean θ.
$f(x;\theta) = \frac{1}{\theta} e^{-x/\theta}$

$$\log f(x;\theta) = -\frac{x}{\theta} - \log \theta$$

$$S(\theta) = \frac{x}{\theta^2} - \frac{1}{\theta}$$

$$I(\theta) = \frac{1}{\theta^2}$$

$$J(\theta) = E[I(\theta; X)] = \frac{1}{\theta^2}$$

$$J(\theta) = nI(\theta) = \frac{n}{\theta^2}$$

$$S(\theta) = \sum_{i=1}^{n} \frac{x_i}{\theta^2} - \frac{n}{\theta} = 0 \Rightarrow \hat{\theta}_n = \bar{X}_n$$

$$\Rightarrow \mathrm{Var}(\hat{\theta}_n) = \mathrm{Var}(\bar{X}_n) = \frac{\theta^2}{n} = \frac{1}{J(\theta)}$$

(3) $N(\mu, \sigma^2)$ assume σ^2 is known.

$$f(x; \mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2\sigma^2}(x-\mu)^2\right]$$

$$\log f(x; \mu) = -\frac{1}{2\sigma^2}(x-\mu)^2$$

$$S(\mu) = \frac{x-\mu}{\sigma^2} \quad I(\mu) = \frac{1}{\sigma^2}$$

$$J(\mu) = \frac{n}{\sigma^2}$$

$$\hat{\mu}_n = \bar{X}_n \Rightarrow \mathrm{Var}(\hat{\mu}_n) = \mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n} = \frac{1}{J(\mu)}$$

16.2 The Cramer-Rao Lower Bound

The Cramer-Rao bound says that for any unbiased estimator $T = T(X_1, \ldots, X_n)$ for $\tau(\theta)$, the lower bound of its variance is given by

$$\mathrm{Var}(T) \geq \frac{(\tau'(\theta))^2}{J(\theta)}.$$

$\tau(\theta)$ is a known function.

16.3 Comments:

Theoretical minimum variance achievable by an unbiased estimator.

- Conditions:

  support of $X_1, \ldots, X_n$ does NOT depend on θ.

- If $\tau(\theta) = \theta$:

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{J(\theta)}.$$

- If iid data:

$$J(\theta) = nI(\theta) \quad \text{where } I(\theta) \text{ is the Fisher info of } X_i.$$

$$\frac{(\tau'(\theta))^2}{J(\theta)} = \frac{(\tau'(\theta))^2}{nI(\theta)}.$$

- Attainable?

1. CR Lower Bound is NOT always attainable.
2. If $\hat{\theta}$ is unbiased for θ and $\text{Var}(\hat{\theta}) = \frac{1}{J(\theta)}$, we say $\hat{\theta}$ is a uniformly minimal variance unbiased estimator (UMVUE).

16.4 Examples:

The previous examples

$$\text{Var}(\hat{\theta}_{MLE}) = \frac{1}{J(\theta)}, \quad \hat{\theta}_{MLE} \text{ is the UMVUE.}$$

These are special cases.

Generally $\hat{\theta}_{MLE}$ is NOT.

Proof of Cramer-Rao Lower Bound

$$\tau(\theta) = E[T(X)] = \int_A T(x) f(x;\theta)\, dx.$$

$X = (x_1, \ldots, x_n)$ random sample where $f(x;\theta)$ is the joint pdf of $(x_1, \ldots, x_n)$.

$$\tau'(\theta) = \frac{d}{d\theta} \int_A T(x) f(x;\theta)\, dx$$

$$= \int_A T(x) \frac{d}{d\theta} f(x;\theta)\, dx = \int_A T(x) \frac{f'(x;\theta)}{f(x;\theta)} f(x;\theta)\, dx$$

$$= E[T(X)\ell'(\theta;X)] \quad \text{since } \ell(\theta;X) = \log f(X;\theta).$$

Because,

$$E[\ell'(\theta;X)] = \int_A \frac{f'(x;\theta)}{f(x;\theta)} f(x;\theta)\, dx = \int_A f'(x;\theta)\, dx = \frac{d}{d\theta} \int_A f(x;\theta)\, dx = 0.$$

$$\tau'(\theta) = E[(T(X) - \tau(\theta))\ell'(\theta;X)].$$

$$\Rightarrow (\tau'(\theta))^2 = \left(E[(T(X) - \tau(\theta))\ell'(\theta;X)]\right)^2$$

$$\leq E[(T(X) - \tau(\theta))^2]\, E[(\ell'(\theta;X))^2] \quad \text{by Cauchy-Schwarz Inequality}$$

$$= \text{Var}(T(X)) \cdot J(\theta).$$

$$\Rightarrow \text{Var}(T(X)) \geq \frac{(\tau'(\theta))^2}{J(\theta)} \quad \text{as required.}$$

16.5 ML Estimator Property

Under regularity conditions (e.g., the support does not depend on θ), with iid $X_1, \ldots, X_n$, the ML estimator for θ satisfies

1. $\hat{\theta}_n \xrightarrow{p} \theta$ as $n \to \infty$.
2. $$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N\left(0, \frac{1}{J_1(\theta)}\right).$$
3. $$\sqrt{n}(\tau(\hat{\theta}_n) - \tau(\theta)) \xrightarrow{d} N\left(0, \frac{(\tau'(\theta))^2}{J_1(\theta)}\right).$$

Asymptotical Normality of MLE.

Comments:

1. $$\sqrt{n}(\hat{\theta}_n - \theta) \approx N\left(0, \frac{1}{J(\theta)}\right)$$
   $$\hat{\theta}_n - \theta \approx N\left(0, \frac{1}{nJ(\theta)}\right)$$
   $$\frac{1}{nJ(\theta)} = \frac{1}{J(\theta)} \quad \text{is the CR lower bound.}$$
   $$\hat{\theta}_n \approx N\left(\theta, \frac{1}{J(\theta)}\right).$$
   The ML estimator has an asymptotic variance that attains the CR lower bound.
2. From (2) using the Delta method $g(\theta) = \tau(\theta)$.

16.6 Example

$X_1, \ldots, X_n$ are iid Poi(θ).

1. Find the ML estimator of θ, denoted by $\hat{\theta}$.
   $$\hat{\theta} = \bar{X}_n.$$
2. Find the limiting distribution of $\sqrt{n}(\hat{\theta} - \theta) = \sqrt{n}(\bar{X}_n - \theta)$.
   method 1: CLT
   $$\sqrt{n}(\bar{X}_n - \theta) \xrightarrow{d} N(0, \theta).$$
   method 2: $\hat{\theta}$ is the ML estimator.
   $$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N\left(0, \frac{1}{J(\theta)}\right) = N(0, \theta).$$
   $$J(\theta) = \frac{1}{\theta}.$$
3. Find the ML estimator of $\psi = \Pr(X_1 = 0)$, denoted by $\hat{\psi}$.
   $$\psi = \Pr(X_1 = 0) = f(0; \theta) = e^{-\theta}.$$
   By invariance property of ML estimator,
   The ML estimator of ψ is
   $$\hat{\psi} = e^{-\hat{\theta}} = e^{-\bar{X}_n}.$$

4. Find the limiting distribution of $\sqrt{n}(\hat{\psi} - \psi)$.

   method 1: CLT + Delta method

   $g(\theta) = e^{-\theta}$, $g'(\theta) = -e^{-\theta}$.

$$\sqrt{n}(\hat{\psi} - \psi) = \sqrt{n}(g(\hat{\theta}) - g(\theta)) \xrightarrow{d} N(0, g'(\theta)^2\theta) = N(0, \theta e^{-2\theta}).$$

   method 2: use property of ML estimator

$$\tau(\theta) = e^{-\theta}.$$

$$\sqrt{n}(\hat{\psi} - \psi) \xrightarrow{d} N\left(0, \frac{\tau'(\theta)^2}{J(\theta)}\right) = N(0, \theta e^{-2\theta}).$$

5. Find the mean of $\hat{\theta}$ and $\hat{\psi}$.

$$E[\hat{\theta}] = E[\bar{X}_n] = \theta. \quad \hat{\theta} \text{ is unbiased for } \theta.$$

$$E[\hat{\psi}] = E[e^{-\bar{X}_n}] = E[e^{-\frac{1}{n}\sum_{i=1}^n X_i}]$$

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i \sim \text{Poi}(n\theta).$$

$$\Rightarrow E[\hat{\psi}] = \exp\Big\{\theta n(e^{-1/n} - 1)\Big\} \neq e^{-\theta}.$$

   $\hat{\psi}$ is biased for $\psi = e^{-\theta}$.

$$\lim_{n\to\infty} E[\hat{\psi}] = \lim_{n\to\infty} \exp\{\theta n(e^{-1/n} - 1)\} = e^{-\theta}.$$

$$n(e^{-1/n} - 1) \to -1 \Rightarrow (e^{-x} - 1)/x \to -1.$$

$$\Rightarrow \lim_{n\to\infty} E[\hat{\psi}] = e^{-\theta}.$$

   $\hat{\psi}$ is asymptotically unbiased for ψ.

17 Frequentist vs. Bayesian Statistics

17.1 Two notations: $f(x;\theta)$ vs. $f(x \mid \theta)$

In frequentist inference, we use the notation $f(x;\theta)$:

e.g.

$$f(x;\theta) = \binom{n}{x}\theta^x(1-\theta)^{n-x}.$$

1. We treat θ as a constant.
2. We use a statistic as an estimator $\hat{\theta}$ for θ.
   We are interested in the dist. of $\hat{\theta}$.

In Bayesian inference, we use $f(x \mid \theta)$:

e.g.

$$f(x \mid \theta) = \binom{n}{x}\theta^x(1-\theta)^{n-x}.$$

1. We treat θ as an r.v.
2. Likelihood becomes a conditional distribution.

18 Bayesian inference

Prior distribution: $\pi(\theta)$

$G(\theta)$: dist response on θ

$L(\theta)$: beliefs / knowledge about θ without seeing the data x.

Likelihood: $L(\theta;x) = f(x \mid \theta)$

Conditional dist of x given θ.

Posterior distribution: $p(\theta \mid x)$

updated beliefs / knowledge about θ after observing the data.

goal.

$\pi(\theta)$ and $f(x \mid \theta)$ are given (known).

$$f(x,\theta) = f(x \mid \theta)\pi(\theta).$$

$$p(\theta \mid x) = \frac{f(x,\theta)}{f(x)} = \frac{f(x \mid \theta)\pi(\theta)}{\int f(x \mid \theta)\pi(\theta)\,d\theta}.$$

$p(\theta \mid x)$ is the pdf.

$f(x)$ is a normalizing constant.

$$p(\theta \mid x) \propto f(x \mid \theta)\pi(\theta).$$

18.1 Bayes Estimator

If we have $p(\theta \mid x)$, a natural estimator is the posterior mean.

$$\hat{\theta} = E[\theta \mid X].$$

Estimator $\hat{\theta} = E[\theta \mid X = x] = \int \theta p(\theta \mid x)\,d\theta$.

18.2 Example 1

Suppose $Y \sim \text{Bin}(n, \theta)$, where $\theta \in (0,1)$ is the unknown probability of success. We place a Beta prior on θ, $\theta \sim \text{Beta}(\alpha, \beta)$, with the pdf given by

$$\pi(\theta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha,\beta)}, \quad \theta \in (0,1),$$

where $B(\alpha, \beta)$ is the Beta function. Find the Bayes estimator.

$$f(y \mid \theta) = \binom{n}{y}\theta^y(1-\theta)^{n-y}, \quad y \in \{0,1,\dots,n\}.$$

$$p(\theta \mid y) \propto f(y \mid \theta)\pi(\theta)$$

$$\propto \theta^y(1-\theta)^{n-y} \cdot \theta^{\alpha-1}(1-\theta)^{\beta-1}$$

$$= \theta^{y+\alpha-1}(1-\theta)^{n-y+\beta-1}.$$

(kernel of Beta pdf)

$$\theta \mid Y = y \sim \text{Beta}(y+\alpha, n-y+\beta).$$

The Bayes estimator

$$\hat{\theta} = \frac{y+\alpha}{n+\alpha+\beta}$$

(posterior mean).

If $Z \sim \text{Beta}(\alpha, \beta)$,

$$E[Z] = \frac{\alpha}{\alpha+\beta}.$$

Example 2

Suppose we can observe a single random variable $Y \mid \theta \sim N(\theta, \sigma^2)$, where θ is unknown and σ^2 is known. We place a normal prior on θ,

$$\pi(\theta) = \frac{1}{\sqrt{2\pi\tau^2}}\exp\left\{-\frac{(\theta-\mu_0)^2}{2\tau^2}\right\}, \quad \theta \sim N(\mu_0, \tau^2).$$

μ_0, τ^2 are known.

Find the Bayes estimator.

$$f(y \mid \theta) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(y-\theta)^2}{2\sigma^2}\right\}.$$

$$p(\theta \mid y) \propto f(y \mid \theta)\pi(\theta)$$

$$\propto \exp\left\{-\frac{(y-\theta)^2}{2\sigma^2}\right\} \cdot \exp\left\{-\frac{(\theta-\mu_0)^2}{2\tau^2}\right\}.$$

$$\propto \exp\left\{-\frac{1}{2}\left(\frac{1}{\tau^2}+\frac{1}{\sigma^2}\right)\theta^2 + \left(\frac{y}{\sigma^2}+\frac{\mu_0}{\tau^2}\right)\theta\right\}.$$

$$= \exp\left\{-\frac{1}{2\tau_1^2}(\theta-\mu_1)^2\right\},$$

where

$$\tau_1^2 = \left(\frac{1}{\tau^2}+\frac{1}{\sigma^2}\right)^{-1} \quad \text{and} \quad \mu_1 = \frac{y}{\sigma^2}+\frac{\mu_0}{\tau^2}.$$

Normal pdf Kernel.

$$\theta \mid Y = y \sim N(\mu_1, \tau_1^2).$$

The Bayes estimator is

$$\hat{\theta} = \mu_1 = \frac{y/\sigma^2+\mu_0/\tau^2}{1/\sigma^2+1/\tau^2}.$$

$\tau^2 \to 0$, τ^2 fixed (accurate data).

μ_0 is $\tau^2 \to 0$, σ^2 fixed (strong beliefs).

(weighted average of Y & μ_0)

We say $\pi(\theta)$ is the conjugate prior for $f(y \mid \theta)$. $p(\theta \mid x)$ and $\pi(\theta)$ have the same type of distribution.

18.3 Example 3: Improper Prior

The prior distribution does not have to be a proper distribution.

Let $Y_1, \ldots, Y_n \sim N(\theta, \sigma^2)$, where θ is the unknown mean, σ^2 is known, and $n \geq 1$. We use the improper prior $\pi(\theta) \propto 1$ for $\theta \in \mathbb{R}$.

(flat prior)

uniform over $\mathbb{R}$.

$$f(y_1, \ldots, y_n \mid \theta) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(y_i-\theta)^2}{2\sigma^2}\right\}.$$

$$p(\theta \mid y_1, \ldots, y_n) \propto f(y_1, \ldots, y_n \mid \theta)\pi(\theta)$$

$$\propto \exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-\theta)^2\right\}$$

$$\propto \exp\left\{-\frac{n}{2\sigma^2}\theta^2 + \frac{\sum y_i}{\sigma^2}\theta\right\}$$

$$\propto \exp\left\{-\frac{n}{2\sigma^2}\left(\theta - \frac{1}{n}\sum_{i=1}^{n} y_i\right)^2\right\}.$$

$$\Rightarrow \theta \mid Y_1 = y_1, \ldots, Y_n = y_n \sim N\left(\frac{1}{n}\sum_{i=1}^{n} y_i, \frac{\sigma^2}{n}\right).$$

Bayes estimator:

$$\hat{\theta} = \bar{X}_n.$$

18.4 How to choose prior distributions

- Subjective prior: Beta, Normal, …
- Weakly informative prior: $N(\mu_0, \sigma^2)$ with large σ^2
- Objective prior: flat prior, Jeffreys Prior (likelihood)

Jeffreys Prior

The most famous objective prior. When θ is a scalar parameter, the Jeffreys prior is defined as

$$\pi(\theta) \propto \sqrt{J(\theta)},$$

where $J(\theta)$ is the Fisher information.

18.5 Example 1

Suppose X has a binomial distribution $\text{Bin}(n, \theta)$. Find the Jeffreys prior for θ.

$$f(x \mid \theta) = \binom{n}{x} \theta^x (1-\theta)^{n-x}.$$

$$\ell = \log f(x \mid \theta) = x \log \theta + (n-x) \log(1-\theta) + \log \binom{n}{x}.$$

$$\frac{d\ell}{d\theta} = \frac{x}{\theta} - \frac{n-x}{1-\theta}, \quad \frac{d^2\ell}{d\theta^2} = -\frac{x}{\theta^2} - \frac{n-x}{(1-\theta)^2}.$$

$$J(\theta) = -E\left[\frac{d^2\ell}{d\theta^2}\right] = \frac{n}{\theta(1-\theta)}.$$

The Jeffreys Prior:

$$\pi(\theta) \propto \sqrt{J(\theta)} = \pi n^{1/2} \theta^{-1/2} (1-\theta)^{-1/2}$$

$$\propto \theta^{-1/2}(1-\theta)^{-1/2} \sim \text{Beta}\left(\frac{1}{2}, \frac{1}{2}\right).$$

18.6 Example 2

Suppose X has a normal distribution $N(\mu, \sigma^2)$, where σ^2 is known. Find the Jeffreys prior for μ.

From previous lecture.

$$J(\mu) = \frac{1}{\sigma^2}.$$

$$\pi(\mu) \propto \sqrt{J(\mu)} = \frac{1}{\sigma} \propto 1.$$

(flat prior)

18.7 Invariance

The Jeffreys prior is invariant to monotone transformation.

18.8 Meaning of Invariance

1. parameter: θ, Jeffreys Prior $\pi_\theta(\theta)$.
2. Transformation: $\eta = h(\theta)$, $h(\cdot)$ is monotone since dist of η:

$$\pi_\eta(\eta) = \pi_\theta(\theta) \left| \frac{d\theta}{d\eta} \right|$$

   one-to-one transform.

Then the distribution of η, $\pi_\eta(\eta)$, is also the Jeffreys Prior for η.

18.9 Proof

We know

$$\pi_\eta(\eta) = \pi_\theta(\theta) \left| \frac{d\theta}{d\eta} \right|.$$

We need to show $\pi_\eta(\eta) \propto \sqrt{J_\eta(\eta)}$.

$$J_\eta(\eta) = E\left[\left(\frac{d\ell}{d\eta} \right)^2 \right], \quad \frac{d\ell}{d\eta} = \frac{d\ell}{d\theta} \frac{d\theta}{d\eta}.$$

$$= E\left[\left(\frac{d\ell}{d\theta} \right)^2 \left(\frac{d\theta}{d\eta} \right)^2 \right] = \left(\frac{d\theta}{d\eta} \right)^2 J_\theta(\theta).$$

$$\sqrt{J_\eta(\eta)} = \sqrt{J_\theta(\theta)} \left| \frac{d\theta}{d\eta} \right| \propto \pi_\theta(\theta) \left| \frac{d\theta}{d\eta} \right|.$$

18.10 Example

$X \sim \text{Bin}(n, \theta)$, $\pi(\theta) \sim \text{Beta}\left(\frac{1}{2}, \frac{1}{2}\right)$.

Find the Jeffreys prior for

$$\eta = \log \frac{\theta}{1-\theta}.$$

$\pi(\theta)$ is the Jeffreys Prior.

Use the invariance property.

$$\pi_\eta(\eta) = \pi_\theta(\theta) \left| \frac{d\theta}{d\eta} \right| = \pi_\theta(\theta) \frac{1}{\left| \frac{d\eta}{d\theta} \right|}$$

$$\propto \theta^{-1/2}(1-\theta)^{-1/2} \cdot \theta(1-\theta) = \theta^{1/2}(1-\theta)^{1/2}.$$

$$\theta = \frac{1}{1+e^{-\eta}}.$$

$$= \frac{e^{\eta/2}}{1+e^{\eta}}.$$